Article

Accurate predictions of disordered protein ensembles with STARLING

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Intrinsically disordered proteins and regions (collectively IDRs) are found across all kingdoms of life and have critical roles in virtually every eukaryotic cellular process[1]. IDRs exist in a broad ensemble of structurally distinct conformations. This structural plasticity facilitates diverse molecular recognition and function[2–4]. Here we combine advances in physics-based force fields with the power of multi-modal generative deep learning to develop STARLING, a framework for rapid generation of accurate IDR ensembles and ensemble-aware representations from sequence. STARLING supports environmental conditioning across ionic strengths and demonstrates proof of concept for the interpolative ability of generative models beyond their training domain. Moreover, we enable ensemble refinement under experimental constraints using a Bayesian maximum-entropy reweighting scheme. Beyond ensemble characterization, STARLING sequence representations can be used in multiple ways. We showcase two examples: first, STARLING lets us perform ensemble-based search for 'biophysical look-alikes'. Second, we demonstrate how these latent representations can be used to accelerate ensemble-first sequence design from weeks or hours per candidate to seconds, enabling library-scale designs. Together, STARLING dramatically lowers the barrier to the computational interrogation of IDR function through the lens of emergent biophysical properties, complementing bioinformatic protein sequence analysis. We evaluate the accuracy of STARLING against extant experimental data and offer a series of vignettes illustrating how STARLING can enable rapid hypothesis generation for IDR function and aid the interpretation of experimental data.

IDRs are structurally heterogeneous protein regions that are estimated to constitute approximately 30% of eukaryotic proteomes[1]. Despite lacking a fixed structure, IDRs have key roles in essential cellular processes such as transcription, translation and cell signalling[1]. Owing to their broad structural heterogeneity, IDRs must be described by a conformational ensemble: a large collection of structurally distinct and interchanging conformations[1]. However, although IDRs cannot be represented as a single 3D structure, they do still possess sequence-encoded conformational biases, and ensembles can have essential roles in IDR function and may be perturbed in disease[2–4]. Just as structural biology has been instrumental in understanding the molecular basis for folded domain function, there is a growing appreciation that the characterization of IDR ensembles may be important for understanding IDR function[5,6].

Various experimental techniques have been applied to interrogate sequence–ensemble relationships[7–9]. Although these report on specific aspects of IDR ensembles, they fall short of providing a holistic description of the distribution of conformers (that is, the 3D coordinates of all residues in the protein across many different conformations, referred to here as a 'full structural ensemble'). To achieve this, the integration of computational modelling with experimental data has proven an effective route[10–13].

Conceptually, computational models and experiments can be combined in two different ways. One approach involves using physics-based models and reweighting or biasing towards experimental observables[14,15]. Another involves using experimental data to parameterize transferable force fields, which, in principle, do not require additional reweighting[16]. Although both approaches have been effective in providing insight into IDR behaviour, they require deep technical expertise to ensure reliable conclusions are drawn, and can also be computationally expensive.

Although recent advances in coarse-grained simulations[17–19] offer a faster alternative (and both modalities described above can be applied here), even coarse-grained simulations can still take hours to obtain sufficient sampling and still require a relatively high level of technical expertise to set up, run and analyse. Recent deep learning predictors trained on coarse-grained simulations have enabled proteome-scale predictions for ensemble average values, but are limited to specific observables for which a predictor was trained (for example, radius of gyration (R_g) or radius of end-to-end distances (R_e))[20,21].

Deep learning approaches have transformed protein structure prediction, significantly reducing the barrier to large-scale exploration of sequence–structure relationships[22] (Fig. 1a). However, these

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文章

利用 Starling 对无序蛋白质集合进行精确预测

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内在无序蛋白质及区域（统称为 IDR）广泛存在于所有生命界中，在几乎每一个真核细胞过程中都发挥着至关重要的作用[1]。IDR 存在于一个由多种结构迥异的构象组成的庞大集合中；这种结构可塑性促进了多样化的分子识别与功能实现[2–4]。本文结合基于物理的力场研究进展与多模态生成式深度学习的强大能力，开发了 Starling——一个用于从序列快速生成精确的 IDR 集合以及构建集合作用感知表征的框架。Starling 支持在不同离子强度条件下进行环境条件化，并验证了生成式模型在超出其训练域时所具有的插值能力。此外，我们利用贝叶斯最大熵重加权方案，在实验约束条件下实现集合作用的优化。除集合作用表征外，Starling 的序列表征还可通过多种方式加以利用。我们展示了两个示例：首先，Starling 可支持基于集合作用的“生物物理相似物”搜索。其次，我们展示了如何利用这些潜在表征将“集成优先”序列设计的耗时从每候选序列数周或数小时缩短至数秒，从而支持大规模库规模的序列设计。总体而言，Starling 通过从涌现的生物物理特性视角出发，显著降低了对 IDR 功能进行计算探究的门槛，与基于生物信息学的蛋白质序列分析方法形成互补。我们通过现有实验数据对 Starling 的准确性进行了评估，并通过一系列示例演示了 Starling 如何助力针对 IDR 功能的快速假说生成，以及如何辅助实验数据的解读。

IDRs 是经估算出的结构异质性蛋白质区域。
约占真核生物蛋白质组的 30%[1]。尽管如此
由于缺乏固定结构，IDRs 在细胞必需的促生化过程中发挥着关键作用。
诸如转录、翻译及细胞信号转导等过程[1]。
鉴于IDRs具有广泛的结构异质性，必须对其进行描述。
构象集合：一个包含大量结构各异的分子集合
溶液态与互换构象[1]。然而，尽管 IDRs
虽无法表示为单一的三维结构，但它们仍具有
序列编码的构象偏好性，以及系综可能具有
在 IDR 功能中发挥着至关重要的作用，且在疾病状态下可能受到干扰[2–4]。
正如结构生物学在理解折叠结构域功能的分子基础方面发挥了重要作用
一样，目前人们对这一领域的关注度也日益增加。
研究表明，对 IDR 系统集合进行特征描述可能具有重要意义。
用于理解 IDR 函数[5,6]。
多种实验技术已被应用于对相关现象进行探究。
序列与集成模型之间的关系[7–9]。尽管这些研究涉及特定内容
尽管涉及 IDR 簇的相关方面，但这些方法尚无法对构象异构体的分布
（即所有残基的三维坐标）提供全面的描述。
该蛋白质在多种不同构象中存在，此处称为
“完整结构系综”）。为实现这一目标，需将计算方法进行整合——
基于实验数据的理论建模已被证明是一种有效的研究途径[10–13]。
从概念上讲，计算模型与实验可以相
可通过两种不同的方式实现；其中一种方法涉及采用基于物理原理的方法。
模型及对实验观测值进行重新加权或引入偏差
bles [14,15]；另一种方法则是利用实验数据对参数进行标定
可转移力场——原则上无需额外计算资源。
重新加权[16]。尽管这两种方法在提供信息方面均取得了良好效果——
要深入理解 IDR 的行为，需要具备深厚的技术专长。
确保得出可靠的结论，并可通过计算手段实现。
昂贵的.
尽管近期粗粒化模拟领域的进展[17–19]为
更快的替代方案（上述两种方法均可用于此场景），即便进行粗粒度
模拟，仍可能需要数小时才能完成。
采样量充足，但仍需较高的技术水平。
专业能力，涵盖系统搭建、运行与数据分析；近期推出的深度学习预测模型
基于粗粒度模拟训练的方法已实现蛋白质组规模分析
用于集合平均值的预测，但仅适用于特定情况
用于训练预测模型的可观测量（例如，回转半径（R_g）或端到端距离半径（R_e））[20,21]。
深度学习方法彻底改变了蛋白质结构研究。
预测技术显著降低了大规模探索的门槛——
序列与结构关系的分析[22]（图1a）。然而，这些

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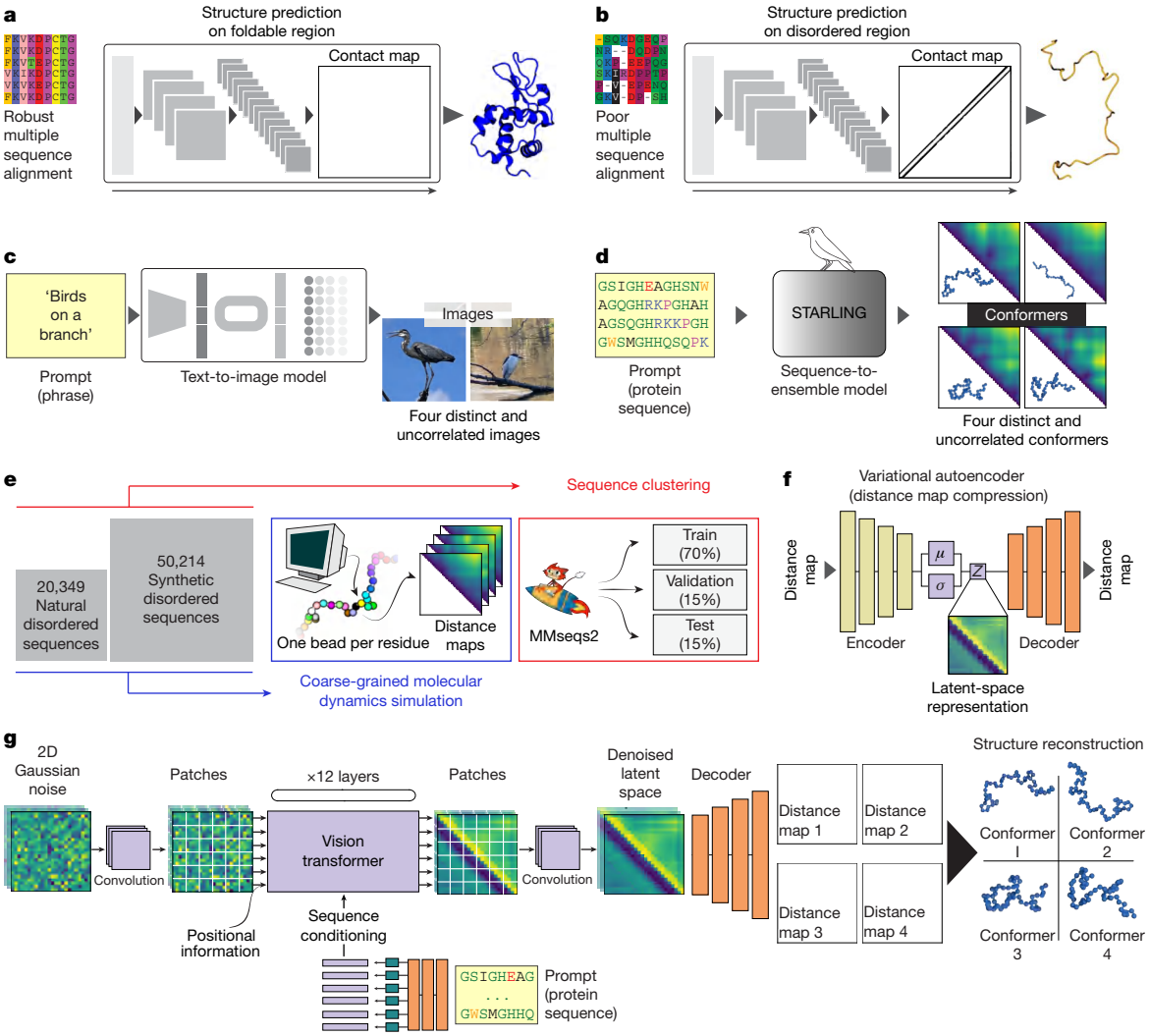

Fig. 1 | STARLING approach and model architecture. a, Deep learning has revolutionized protein structure prediction, with major advances being facilitated by large-scale evolutionary information. **b**, Structure prediction methods for folded domains are poorly suited for predicting the behaviour of IDRs. These limitations stem from the absence of a native-state structure and because evolutionary information is often poorly captured in multiple sequence alignments (MSAs) of IDRs. **c**, Generative text-to-image models enable the creation of many unique and independent images consistent with a single input prompt. **d**, IDR ensemble generation shares many similarities with text-to-image generation; we required many distinct, uncorrelated conformers, all of which are consistent with an input prompt (an amino acid sequence). **e**, STARLING was trained on approximately 50,000 amino acid sequences at 150 mM ionic strength and approximately 14,000 sequences at 20 mM and 300 mM ionic strength. The sequences performed at 20 mM and 300 mM ionic strength are a subset of those simulated at 150 mM ionic strength. For each sequence, hundreds of distinct conformers were generated using coarse-grained molecular dynamics simulations, and each conformer was converted into a distance map (an image). Sequences were split into training, testing and validation sets. The MARV cartoon was reproduced from ref. 61, Martin Steinegger. **f**, STARLING makes use of a VAE to compress distance maps to a latent space, allowing a denoising diffusion model to work in this latent space ('latent diffusion'). **g**, The overall architecture of the STARLING model combines a latent-space probabilistic denoising diffusion model with a vision transformer architecture using both convolutions and transformer blocks. Latent-space maps were decoded to real space via the VAE decoder. Finally, distance maps can be reconstructed into 3D coordinates via a parallelized multidimensional scaling approach.

methods are poorly suited to investigate IDRs[23] owing to a reduction in alignment-based conservation in disordered proteins, a paucity of appropriate experimental training data and optimization for an inappropriate objective (predicting a single best structure from sequence, whereas IDRs should be described by a large, conformationally heterogeneous ensemble; Fig. 1b). In short, although we now possess easy-to-use tools for accurately predicting the 3D structure of folded proteins, equivalent tools for rapid and accurate IDR ensemble prediction are lacking.

Here we addressed these challenges by developing a fast and accurate approach for predicting full coarse-grained disordered protein structural ensembles directly from the amino acid sequence. Our approach leveraged advances in generative modelling, a deep learning technique capable of creating new and original data. However, developing a generative model poses a key challenge: the need for large training datasets. To address this, we performed large-scale coarse-grained simulations to generate full structural ensembles across tens of thousands of natural and synthetic IDRs. The resulting method – STARLING – allowed us to generate structural ensembles directly from sequence in seconds. A major goal in developing STARLING was to avoid hardware barriers. Although STARLING is fast (approximately 35 conformers per second) on GPUs, it can still generate ensembles in minutes on Intel/AMD CPUs and seconds on Apple CPUs.

STARLING-generated ensembles show good agreement with experimental data, enabling de novo exploration of uncharacterized IDRs or aiding in the biophysical interpretation of experimental data. Moreover, here we show proof of concept that STARLING can be used to

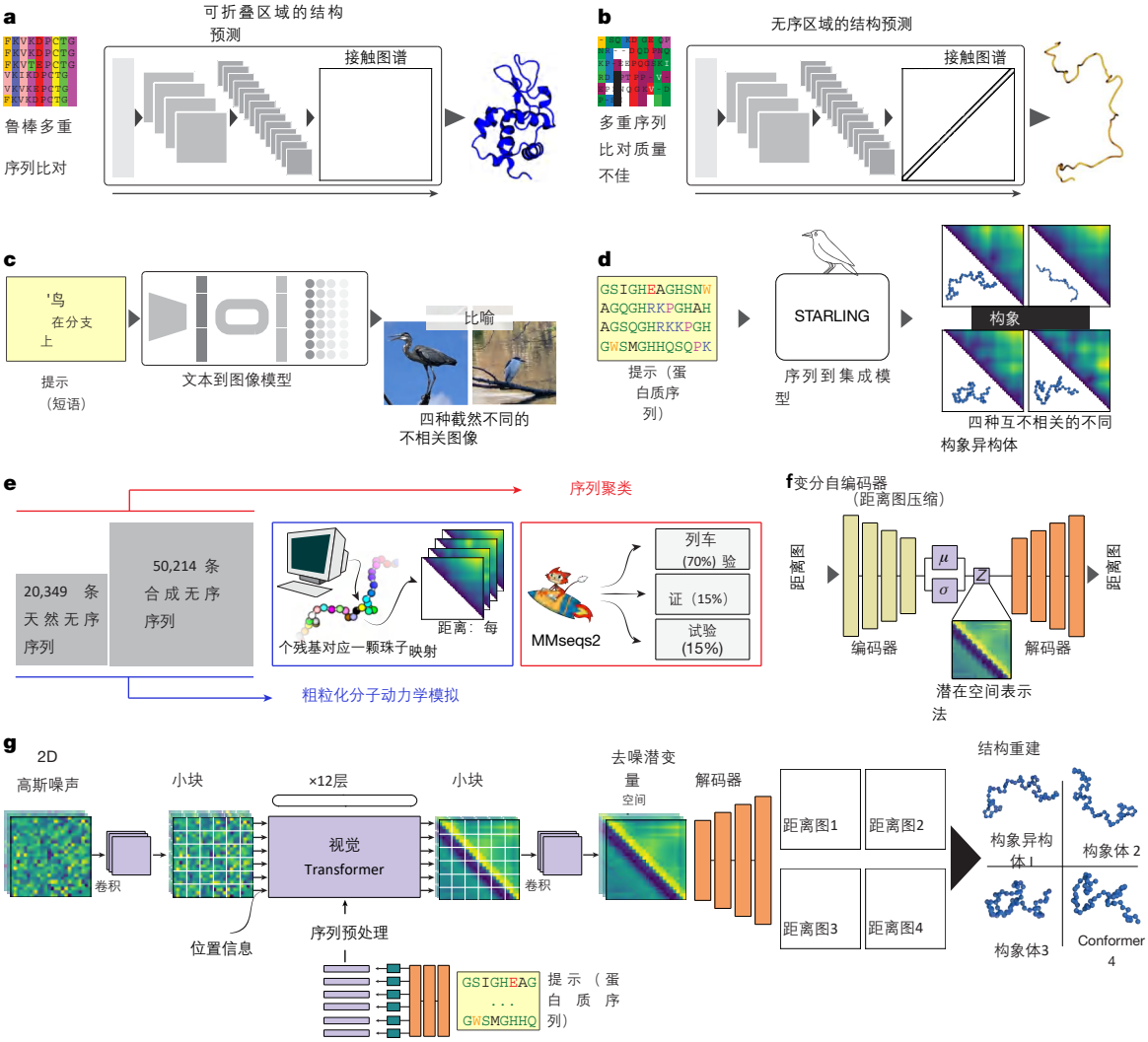

图 1 | Starling 模型及模型架构。a：深度学习彻底改变了蛋白质结构预测领域，其中大规模进化信息为重大进展提供了助力。**b**：针对折叠结构域的结构预测方法并不适用于预测内在无序区域（IDRs）的行为。这些局限性源于缺乏天然状态结构，且进化信息在 IDR 的多序列比对（MSAs）中往往难以准确捕捉。**c**：生成式文本到图像模型能够根据单个输入提示生成大量独特且独立的图像。**d**：IDR 集成生成法与文本到图像生成法具有诸多相似之处：我们需要生成许多互不相关且各不相同的构象异构体，且所有这些构象均与输入提示（氨基酸序列）相符。**e**：Starling 模型在离子强度为 150 mM 的条件下基于约 50,000 条氨基酸序列进行训练，在离子强度为 20 mM 和 300 mM 的条件下则基于约 14,000 条氨基酸序列进行训练。在 20 mM 和 300 mM 离子强度下运行的序列，属于在 150 mM 离子强度下模拟所得序列的子集。对于每个序列，通过粗粒度分子动力学模拟生成了数百个不同的构象异构体，并将每个构象异构体转换为一张距离图（一张图像）。这些序列被划分为训练集、测试集和验证集。MARV 简图参考自文献61（Martin Steinegger）。**f**，starling 系统利用 aVAE 将距离图压缩到潜在空间，从而使去噪扩散模型能够在该潜在空间中运行（“潜在扩散”）。

g：Starling 模型的整体架构将潜在空间概率性去噪扩散模型与采用卷积层及 Transformer 模块的视觉Transformer 架构相结合。潜在空间映射通过 VAE 解码器解码为真实空间映射；最后，距离映射可通过并行化多维标度法重建为三维坐标。

这些方法并不十分适用于研究无序区（IDRs）[23]，原因在于：在无序蛋白质中，基于序列比对的保守性分析效果较差；缺乏合适的实验训练数据；且优化目标并不恰当（即从序列中预测单一最优结构，而IDRs应通过一个规模较大、构象异质性的结构集合来描述；图1b）。简而言之，尽管目前我们已拥有易于使用且能准确预测折叠蛋白质三维结构的工具，但尚缺乏能够实现快速、准确的 IDR 集合预测的相应工具。

针对上述挑战，我们开发了一种快速且准确的方法，可直接根据氨基酸序列预测完整的粗粒度无序蛋白质结构集合。该方法充分利用了生成式建模技术的最新进展——这是一种能够生成全新且原创数据的深度学习技术。然而，构建生成式模型面临一个关键挑战：即需要大规模的训练数据集。为解决这一问题，我们开展了大规模的粗粒度模拟计算，从而针对数万种天然及合成的IDR序列生成完整的结构集合。由此开发出的 Starling 方法使我们能够在数秒内直接从氨基酸序列生成结构集合。开发 Starling 的一个主要目标是规避硬件性能限制：尽管 Starling 在 GPU 上运行速度较快（约每秒生成 35 个构象），但在 Intel/AMD CPU 上仍可在数分钟内生成结构集合，在 Apple CPU 上则可在数秒内完成。

Starling生成的集合与实验数据吻合良好，这使得对尚未表征的内在无序区域（IDRs）进行从头探索或辅助对实验数据进行生物物理层面的阐释成为可能。此外，本文还验证了Starling可用于

(1) investigate sequence–ensemble relationships for disordered proteins; (2) explore bound-state conformational ensembles for binary disordered protein complexes; and (3) provide conformation-aware latent representations for IDR characterization, search and design. Together, we propose that the ease of use and speed of STARLING make it a powerful tool for democratizing large-scale exploration of sequence–ensemble relationships in IDRs.

Generative artificial intelligence (AI) has been transformative for text-to-image generation[24,25]. In text-to-image generative AI, an image is generated by passing a prompt (a short phrase describing the desired image) to a pre-trained deep learning model. The model then generates an image consistent with the prompt, a process referred to as inference. Deep learning models capable of inference must first be trained. For models used in modern text-to-image generative AI, training is not simply memorization but also learning the relationship between prompts and the features in the associated images. As a result, once a model is trained, if the same prompt is reused many times, it will generate many independent images entirely distinct from any of the individual instances observed in training. Of note, despite being different from one another, each generated image should be consistent with the prompt (Fig. 1c).

The mapping of a single text prompt to a collection of different images – each of which is consistent with the input prompt – is precisely the problem that we wished to solve for IDR ensemble generation. In IDR ensemble generation, we wanted to take a text prompt (amino acid sequence) and generate a collection of many distinct and uncorrelated IDR conformations consistent with that prompt (Fig. 1d). Moreover, we wanted this generation process to be fast (seconds) and possible on commodity hardware (laptops and desktops). To achieve this goal, we combined a variational autoencoder (VAE) with a discrete-time denoising-diffusion probabilistic model (DDPM) to create a latent diffusion model[25]. The resulting method (STARLING) enabled the accurate and rapid prediction of coarse-grained conformational ensembles of IDRs.

STARLING was trained on approximately 50,000 unique rationally designed and naturally occurring IDR sequences (10–384 residues) simulated at 150 mM ionic strength (Supplementary Table 1). Among these, approximately 14,000 were additionally simulated at 20 mM and 300 mM ionic strength (Fig. 1e, Extended Data Fig. 1a,b and Supplementary Table 2). Rationally designed sequences were designed using GOOSE[26] (see Methods; Extended Data Fig. 1c–f). Although naturally occurring IDRs provide training data focused on the most relevant manifold of sequences, the inclusion of a large number of rationally designed sequences allowed us to construct training data that systematically titrates across sequence space, yielding a well-rounded training set that captures the extremes of sequence composition and patterning.

A necessary decision in deep learning models is defining the limits of your training data. We focused on sequences up to 384 amino acids in length for several reasons. First, almost 95% of naturally occurring IDRs across common model organisms are shorter than 384 amino acids (Extended Data Fig. 1h,i). Second, this length enables well-sampled coarse-grained simulations to be completed on a reasonable timescale. Third, the underlying architecture of the model requires a fixed upper limit (see Methods). Finally, experimental characterization of disordered protein sequences longer than approximately 350 residues is lacking, making it difficult to assess the validity of model predictions relative to experiments for very long disordered proteins.

Molecular dynamics simulations were performed using the Mpipi-GG force field. Mpipi-GG[20] – a variant of the original Mpipi force field[17] – is a one-bead-per-residue coarse-grained model developed for disordered regions. Once run, conformations from simulations of a sequence of length n were converted into distance maps, $n \times n$ matrices, where each element describes the distance between the i-th and j-th residue for a specific conformation. This converts each IDR conformation into an 'image', allowing us to directly leverage innovations developed for conditional image generation. Overall, our training dataset comprised nearly 12 million distance maps.

Two core limitations of DDPM models are their substantial memory requirements and the slow generative process during inference. These limitations arise from performing the reverse denoising process in a complex, high-dimensional space (for example, images of size 384 × 384 pixels). To mitigate this, we developed a VAE to compress each distance map into a lower-resolution latent space (24 × 24 pixels)[27] (Fig. 1f). The denoising diffusion process can then occur in this latent space (latent diffusion), significantly reducing memory requirements and inference time. Given this, STARLING was trained in two independent stages.

We first trained a highly accurate VAE that enables the compression of full-resolution distance maps into latent space (Extended Data Fig. 2a). Our VAE uses a ResNet18 architecture and learns parameters for an encoder (full resolution to latent space) and a decoder (latent space to full resolution)[28]. We assessed the accuracy of VAE by encoding and decoding distance maps derived from sequences substantially different from those used in the training and validation sets. We evaluated our model on a held-out test set comprising approximately 16,000 simulations, totalling nearly 2.5 million distance maps. The model achieved a root-mean-square reconstruction error (RMSE) of 1.16 Å (Extended Data Fig. 2b). Furthermore, our model accurately reconstructed bond lengths, with a root-mean-square reconstruction error of 0.18 Å (Extended Data Fig. 2c), crucial for modelling protein conformations.

In the second stage of training, we developed a DDPM (Fig. 1d and Extended Data Fig. 2d–f). In brief, this model was trained to learn parameters that map random noise to individual latent-space conformer distance maps, conditioned on the associated amino acid sequence and ionic strength (see Methods). The training data came from nearly 12 million distance maps across almost 78,000 simulations. Each map was first compressed into a latent space, and then the fixed forward diffusion process was used to add noise to each latent-space distance map. The underlying vision transformer model was then trained to learn parameters that reversed this forward diffusion process, conditioned on the input amino acid sequence and ionic strength (Fig. 1g). Ultimately, ensemble generation is enabled by running many inference rounds in parallel, creating many independent distance maps.

Although training requires the VAE encoder to generate latent-space distance maps, only the VAE decoder is needed for inference once the model is trained. Ultimately, the fully trained model (STARLING) combines two different models (the VAE decoder and the DDPM) that work together to enable rapid ensemble prediction. Using default settings, STARLING can generate 400 independent IDR conformations in approximately 12 s on a GPU (Nvidia A4000), 20 s on a Macbook Pro M3 CPU and approximately 6 min on an Intel CPU (Intel(R) Xeon(R) Silver 4210 R CPU @ 2.40 GHz; Extended Data Fig. 3a,b). The prediction runtime and memory are roughly independent of sequence length; 400 conformations for a 50-amino-acid IDR took approximately 10 s, whereas 400 conformations for a 350-residue IDR took approximately 14 s, and prediction times were not hampered by low-memory GPUs (Extended Data Fig. 3c–e).

We first checked that the STARLING-derived ensemble-averaged global dimensions agree with those obtained from the Mpipi-GG simulation of unseen sequences. In all cases, STARLING ensembles comprised 800 conformers. Using a held-out test set of approximately 10,000 sequences, we performed Mpipi-GG simulations at 150 mM ionic strength and calculated R_g and R_e. Our STARLING-derived R_g (RMSE = 0.85 Å, R^2 = 0.996) and R_e (RMSE = 3.48 Å, R^2 = 0.989) values are in excellent agreement with the simulations (Fig. 2a,b).

As STARLING was also trained on Mpipi-GG simulations at both 20 mM and 300 mM ionic strength, we verified that STARLING-derived ensemble-averaged global dimensions match the simulations. Using a held-out test set of approximately 3,000 unseen sequences, we performed Mpipi-GG simulations at ionic strengths of 20 mM and 300 mM. Our STARLING-derived R_g (Fig. 2c,d; RMSE = 0.98 Å and 1.12 Å, R^2 = 0.995

(1) (1) 研究无序蛋白质中的序列–构象集合关系；(2) 探究二元无序蛋白质复合物的结合态构象集合；(3) 为 IDR 特征分析、检索及设计提供基于构象的潜在表示。综上所述，我们认为 Starling 的易用性与高效性使其成为一项强大的工具，可助力实现对无序结构域（IDRs）中序列–构象集合关系的大规模探索的普及化应用。

生成式人工智能（AI）对文本到图像生成领域产生了深远影响[24,25]。在文本到图像的生成式AI中，图像是通过直接将提示词（描述所需图像的简短短语）输入到预训练的深度学习模型中而生成的；随后，该模型会生成一张与该提示词相符的图像，这一过程称为“推理”。能够进行推理的深度学习模型必须经过预先训练。对于现代文本到图像生成式AI中所使用的模型而言，训练并非简单的记忆过程，而是要学习提示词与对应图像中的特征之间的关系。因此，一旦模型完成训练，若对同一提示词进行多次调用，模型将生成许多独立的图像，这些图像与训练过程中观察到的任何单个实例均截然不同。值得注意的是，尽管这些生成的图像彼此不同，但每张图像都应与对应的提示词保持一致（图**1c**）。

将单个文本提示映射到一组不同的图像集合——其中每幅图像均与该输入提示相符——正是我们为 IDR 集合生成任务所要解决的问题。在 IDR 集合生成中，我们希望基于一个文本提示（氨基酸序列）生成一组众多互不相关且均与该提示相符的 IDR 构象集合（图**1d**）。此外，我们还希望这一生成过程运行快速（仅需数秒）且可在普通硬件（笔记本电脑或台式机）上实现。为实现这一目标，我们将变分自编码器（VAE）与离散时间去噪扩散概率模型（DDPM）相结合，构建了一个潜在扩散模型[25]。由此开发出的 starling 方法能够对 IDRs 的粗粒度构象集合进行准确且快速的预测。

Starling 模型基于约 50,000 条独特的、经理性设计或天然存在的 IDR 序列（长度为 10–384 个残基）进行了训练；这些序列均在 150 mM 离子强度条件下进行模拟（见补充表 1）。其中，约 14,000 条序列还额外在 20 mM 和 300 mM 离子强度条件下进行了模拟（图**1e**、扩展数据图 1a 及补充表 2）。理性设计的序列采用 Goose [26]工具进行设计（详见“方法”部分；扩展数据图 1c–f）。尽管天然存在的内在无序区（IDRs）所提供的训练数据侧重于最相关的序列流形，但通过纳入大量理性设计的序列，我们得以构建一个在序列空间上进行系统性梯度扫描的训练数据集，从而获得一个涵盖序列组成与模式分布极端情况的全面训练集。

在深度学习模型中，一个至关重要的决策便是确定训练数据的长度上限。我们选择将序列长度设定为最多 384 个氨基酸，这基于以下几个原因：首先，在常见模式生物中，近 95% 的天然无序蛋白质序列长度均短于 384 个氨基酸（扩展数据图 1h，i）；其次，该长度范围使得能够以合理的计算时间完成充分采样的粗粒度模拟；第三，模型的底层架构要求设定一个固定的上限（参见“方法”部分）；最后，目前尚缺乏对长度超过约 350 个残基的无序蛋白质序列进行实验表征的研究，因此难以根据实验数据来评估模型对极长无序蛋白质序列预测结果的有效性。

分子动力学模拟采用 Mpipi-GG 力场进行。Mpipi-GG [20]（原 Mpipi 力场[17]的一个变体）是一种针对无序区域开发的 “每个残基对应一个珠子”的粗粒化模型。模拟运行完成后，对长度为n的序列所得到的构象被转换为距离图——即$n \times n$矩阵；其中每个元素描述了特定构象下第 i 个残基与第 j 个残基之间的距离。这一转换将每个 IDR 构象转化为一张“图像”，使我们能够直接利用为条件图像生成所开发的技术。总体而言，我们的训练数据集包含近 1200 万张距离图。

DDPM 模型的两个核心局限性在于其巨大的内存需求以及推理过程中生成过程速度较慢。这些局限性源于在复杂、高维空间（例如尺寸为 384 × 384 像素的图像）中执行逆去噪过程。为缓解这一问题，我们开发了 aVAE，将每幅距离图压缩至较低分辨率的潜在空间（24 × 24 像素）[27]（图**1f**）。随后，去噪扩散过程可在该潜在空间中进行（潜在扩散），从而显著降低内存需求和推理时间。基于此，starling 模型分两个独立阶段进行训练。

我们首先训练了一个高精度的 VAE 模型，该模型能够将全分辨率距离图压缩至潜在空间（扩展数据图 2a）。我们的 VAE 采用 ResNet18 架构，分别学习编码器（全分辨率至潜在空间）和解码器（潜在空间至全分辨率）的参数[28]。我们通过对源自与训练集和验证集所用序列截然不同的序列所生成的距离图进行编码与解码，来评估 VAE 的精度。我们在一个包含约 16,000 次模拟、总计近 250 万幅距离图的独立测试集上对模型进行了评估。该模型的均方根重构误差（RMSE）为 1.16 Å（扩展数据图 2b）。此外，我们的模型能够准确重构键长，其均方根重构误差为 0.18 Å（扩展数据图 2c），这对于蛋白质构象建模至关重要。

在训练的第二阶段，我们开发了一个 DDPM 模型（图**1d**及扩展数据图 2d–f）。简而言之，该模型旨在学习将随机噪声映射至个体潜在空间构象距离图的参数，且该过程受相关氨基酸序列及离子强度的条件约束（详见“方法”部分）。训练数据来源于近 78,000 次模拟产生的约 1,200 万张距离图。每张距离图首先被压缩至潜在空间，随后利用固定的前向扩散过程对每张潜在空间距离图添加噪声；接着对底层视觉Transformer模型进行训练，使其学习逆向执行该前向扩散过程的参数，且该过程同样受输入的氨基酸序列及离子强度的条件约束（图**1g**）。最终，通过并行执行多次推理轮次来实现集成生成，从而生成大量独立的距离图。

尽管训练阶段需要VAE编码器生成潜在空间距离图，但模型训练完成后，仅需使用VAE解码器进行推理。最终，经过完整训练的Starling模型整合了两个不同的模型（VAE解码器与 DDPM），二者协同工作以实现快速集成预测。在默认设置下，Starling可在GPU（Nvidia A4000）上于约12秒内、在Macbook Pro M3 CPU上于20秒内、在Intel CPU（Intel® Xeon® Silver 4210 R CPU @ 2.40 GHz）上于约6分钟内生成400个独立的 IDR 构象（参见扩展数据图3a、b）。预测的运行时间与内存需求大致与序列长度无关：对于由50个氨基酸组成的 IDR，生成400个构象约需10秒；而对于由350个残基组成的 IDR，生成400个构象约需14秒；且低内存GPU并未对预测速度造成影响（参见扩展数据图3c–e）。

我们首先验证了基于椋鸟数据计算的集合平均全局维度与通过未见序列的 Mpipi-GG 模拟所得结果是否一致。在所有案例中，椋鸟数据集均包含 800 个构象异构体。利用一个包含约 10,000 条序列的留出测试集，我们在 150 mM 离子强度条件下进行了 Mpipi-GG 模拟，并计算了R_g and R_e。我们通过椋鸟数据得出的R_g（RMSE = 0.85 Å，R^2 = 0.996）and R_e（RMSE = 3.48 Å，R^2 = 0.989）数值与模拟结果高度吻合（图2a，b）。

由于 Starling 也经过了在离子强度为 20 mM 和 300 mM 条件下的 Mpipi-GG 仿真训练，我们验证了 Starling 计算得到的系综平均全局尺寸与仿真结果相符。利用一个包含约 3,000 条未见序列的独立测试集，我们分别在离子强度为 20 mM 和 300 mM 的条件下进行了 Mpipi-GG 仿真。我们通过 Starling derived R_g（图2c，d； RMSE = 0.98 Å 与 1.12 Å，R^2 = 0.995）。

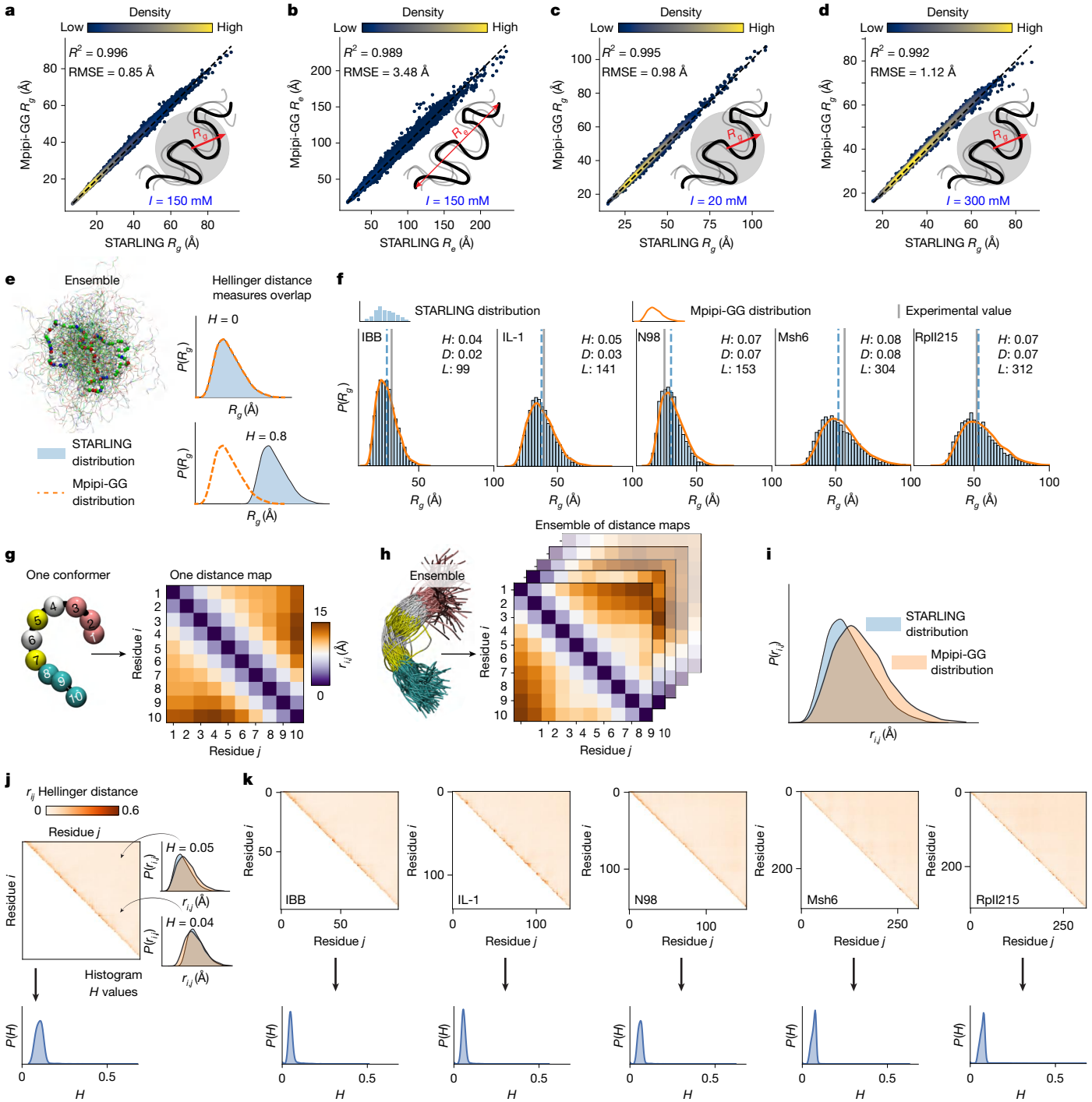

Fig. 2 | Evaluating STARLING on held-out simulation data. a, Comparison of STARLING-derived average R_g values with those from Mpipi-GG (simulations) from a set of approximately 10,000 unseen sequences. **b**, Comparison of STARLING-derived average R_e values with those from Mpipi-GG (simulations) from approximately 10,000 unseen sequences. **c**, Comparison of STARLING-derived R_g values with those from Mpipi-GG (simulations) from a set of approximately 3,000 unseen sequences at an ionic strength of 20 mM. **d**, Comparison of STARLING-derived R_g values with those from Mpipi-GG (simulations) from a set of approximately 3,000 unseen sequences at an ionic strength of 300 mM. **e**, STARLING ensembles enable the calculation of distributions of observables. Distribution similarity can be quantified by the Hellinger distance (H), where $H = 0$ is perfect overlap and $H = 1$ is no overlap. **f**, Overlap of R_g distributions obtained from STARLING (blue bars) and Mpipi-GG (orange line), with the experimental value shown for completeness. Overlap between the STARLING and Mpipi-GG distributions was quantified by the H value and the KS statistic (D) to assess the difference between the distributions and the effect size of the difference. A KS statistic of 0.02 means that the CDFs of the two distributions differ by at most 2%. Vertical dashed lines indicate the average STARLING-derived value. **g**, 3D configurations (left) can be represented as distance maps (right), where each pixel defines the distance between a pair of residues (r_{ij}). **h**, Ensembles of conformations (left) can be represented as ensembles of distance maps (right). **i**, Intra-residue distance distributions in STARLING ensembles can be compared with equivalent distributions from Mpipi-GG simulations. **j**, All possible intra-residue distances can be compared between STARLING and Mpipi-GG simulations using H to quantify overlap in distributions (top). Histograms of intra-residue H values are also shown (bottom). **k**, Pairwise inter-residue distance comparison across different exemplar ensembles reveals excellent overlap.

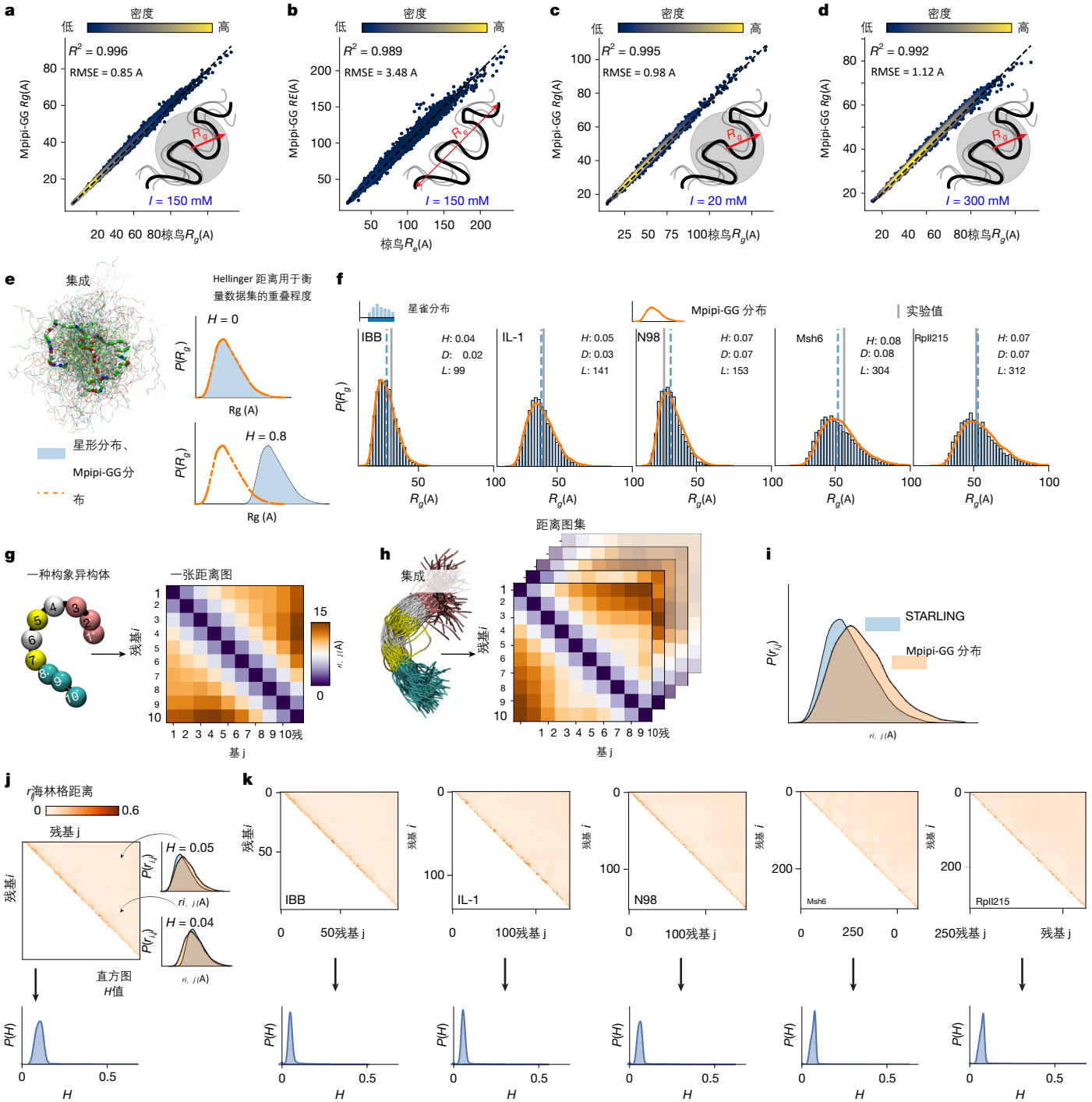

图 2 | 基于独立预留模拟数据对 Starling 模型进行评估。a，将 Starlingderived average R_g值与 Mpipi-GG（模拟）所得值在约 10,000 条未见序列集上进行比较。**b**，将 Starling 求得的average R_e值与 Mpipi-GG（模拟）所得值在约 10,000 条未见序列上进行比较。**c**，将 Starling 求得的R_g值与 Mpipi-GG（模拟）所得值在约 3,000条未见序列（离子强度为 20 mM）。**d**，将星雀模型derived R_g值与来自约 3,000 条未见序列（离子强度为 300 mM）的 Mpipi-GG（模拟）所得值进行比较。**e**，星雀模型集合可用于计算可观测物理量的分布。分布相似性可通过赫林格距离（H）进行量化，其中H= 0 表示完全重叠，H = 1 表示无重叠。**f**，由星雀模型（蓝色柱状图）与 Mpipi-GG（橙色曲线）获得的Overlap of R_g分布的重叠情况；为完整起见，同时展示了实验值。星雀模型与 Mpipi-GG 分布之间的重叠程度通过 H 值和 KS 统计量（D）进行量化，以评估两分布间的差异及该差异的效应量。KS 统计量为 0.02 表示两个分布的累积分布函数（CDF）差异至多为 2%。垂直虚线表示星雀模型推导出的平均值。**g**，三维构型（左图）可表示为距离图（右图），其中每个像素定义了一对残基之间的距离（r_{ij}）。**h**：构象集合（左图）可表示为距离映射集合（右图）。**i**：棕鸟体系中残基内距离分布可与 Mpipi-GG 模拟得到的对应分布进行比较。**j**：利用 H 值可对棕鸟体系与 Mpipi-GG 模拟中的所有可能残基内距离进行比较，以量化其分布重叠程度（上图）；同时展示了残基内 H 值的直方图（下图）。**k**：对不同示例集合进行的残基间成对距离比较显示了极佳的分布重叠性。

and 0.992 for 20 mM and 300 mM, respectively) and R_e (Extended Data Fig. 4a,b; RMSE = 4.17 Å and 4.53 Å, R^2 = 0.986 and 0.978 for 20 mM and 300 mM, respectively) are again in the excellent agreement with simulations.

Model accuracy depends on the balance between ensemble size and denoising steps. By comparing STARLING results to long-timescale Mpipi-GG simulations, we found that error in the R_g stabilizes beyond 30 denoizing steps and 400 conformations (Extended Data Fig. 4c–i). We adopted these as default parameters, although larger ensembles may be required for other order parameters.

As R_e and R_g are highly correlated with sequence length, we next assessed the accuracy of our model on a set of length-matched sequences[29]. This approach allowed us to determine whether the model had effectively captured the influence of sequence chemistry on the global dimensions of IDRs, or whether it had simply learned to relate sequence length to these dimensions. Using all 100-residue, 200-residue and 300-residue sequences with distinct sequence chemistries from the held-out test set, we found excellent agreement across a range of R_g (Extended Data Fig. 4j; RMSE = 0.78 Å, 1.18 Å and 1.20 Å, R^2 = 0.984, 0.990 and 0.995 for 100-residue, 200-residue and 300-residue lengths, respectively) and R_e (Extended Data Fig. 4k; RMSE = 3.38 Å, 4.22 Å and 5.65 Å, R^2 = 0.965, 0.984 and 0.986 for 100-residue, 200-residue, 300-residue lengths, respectively). These results gave us confidence that STARLING had learned bona fide sequence-to-ensemble rules instead of simply learning polymer scaling theory.

In addition to ensemble-averaged observables, full structural ensembles enabled us to calculate distributions of any observable of interest (Fig. 2e). We set out to assess how well STARLING-derived global dimension distributions match distributions from simulations. We used the Hellinger distance (H) as a similarity measure to compare distributions. H ranges from 0 to 1, where zero indicates that the distributions are identical, and one signifies that they are entirely disjoint (Fig. 2e). The STARLING-derived R_g distributions show excellent agreement with distributions from simulations (Fig. 2f). This overlap was quantified by a low H, confirming the high degree of similarity between the two distributions.

Finally, to quantify ensemble fidelity, we assessed all inter-residue distance distributions between ensembles generated using STARLING or Mpipi-GG simulations. Each 3D conformation can be represented as a single distance map (Fig. 2g), and the ensemble of conformations can be represented as an ensemble of distance maps (Fig. 2h). Therefore, every inter-residue distance distribution from a pair of residues in a STARLING ensemble can be compared against the corresponding distribution from an Mpipi-GG (Fig. 2i). For a 100-residue IDR, this yields 4,900 unique distribution comparisons. We could quantify the overlap between the distributions using H and visualize each pair using a heat map (Fig. 2j, top). Furthermore, we could use histograms for those H to provide an overall assessment of the ensemble agreement (Fig. 2j, bottom). Overall, we found excellent agreement across all distance distributions investigated (Fig. 2k). In summary, all evidence supports that STARLING can directly predict conformational ensembles of disordered proteins from sequence.

Having established that STARLING can recapitulate simulated ensembles, we next investigated agreement with experimental data (Fig. 3a). We used a previously curated set of 133 sequences for which high-quality small-angle X-ray scattering (SAXS) data have been assembled and computed average R_g values. We found excellent agreement between average R_g values from STARLING and SAXS-derived R_g, yielding values comparable with state-of-the-art coarse-grained simulations (RMSE = 4.53, R^2 = 0.90; Fig. 3b, Extended Data Fig. 5a–h and Supplementary Tables 3 and 4).

To further validate STARLING, we predicted ensembles for 3,417 IDRs from DisProt[30], successfully replicating established sequence-conformation trends (for example, aromatic residues drive compaction, whereas proline and glutamic acid promote expansion[3,20,31,32]; Extended Data Fig. 5i–k and Supplementary Table 5). STARLING-generated ensembles are also sensitivity to minor sequence changes; when tested on four hnRNPA1 low complexity domain (LCD) variants, the predicted dimensions of the model quantitatively matched experimental SAXS data regarding aromatic content[3] (Fig. 3c).

To go beyond simple R_g comparisons (Fig. 3b,c), we leveraged STARLING-generated ensembles to back-calculate synthetic SAXS curves using FOXS[33]. This allows for a direct comparison between our model and raw experimental SAXS data (down to 5–10 Å resolution). We demonstrated that STARLING accurately captures the subtle conformational shifts in A1-LCD variants, as reflected in their specific scattering profiles[3]. Next, we selected 12 unrelated IDRs for which high-quality scattering data were available and compared scattering profiles derived from STARLING ensembles with experimental data, finding excellent agreement across a wide range of sequence chemistries and sequence lengths (Fig. 3e and Supplementary Table 3). We further reanalysed 40 additional sets of scattering data, finding that the majority are in excellent agreement with STARLING-derived scattering profiles (Extended Data Fig. 6a and Supplementary Figs. 1 and 2). Together, our data suggest that our STARLING ensembles faithfully reproduce experimentally measured IDR ensemble behaviour across a range of length scales and sequence chemistries.

Finally, we sought to compare against an alternative experimental modality in single-molecule Förster resonance energy transfer (smFRET). Recent work has illustrated how a series of 16 length-matched sequences shows a range of end-to-end distances depending on the underlying sequence chemistry[34]. We compared predicted end-to-end distances (and, where available, SAXS-derived scattering data) for those 16 sequences, again finding good agreement between STARLING ensembles and experimental data (Fig. 3f, Extended Data Fig. 6b,c and Supplementary Table 3). Of note, our average error here across the 16 sequences (RMSE of 6.7 Å) is comparable with the differences obtained between different experimental groups in a recent smFRET benchmarking study[35]. Overall, our work suggests that STARLING can generate full coarse-grain ensembles of IDRs that accurately capture experimental data.

Having established that STARLING can reproduce extant experimental data, we applied it to investigate a set of distinct IDRs across various contexts. These vignettes illustrate the types of rapid exploration and hypothesis generation that STARLING enables.

The master transcriptional regulator Myc (439 residues) underlies cell growth, metabolism and proliferation, and is frequently dysregulated in many cancers[36]. Myc consists of a large N-terminal IDR ($Myc^{1–361}$), a C-terminal HLH DNA-binding domain ($Myc^{368–406}$) and a leucine zipper ($Myc^{413–439}$). The large size of its IDR, coupled with its chemical composition, has challenged biophysical characterization (Fig. 4a). Using STARLING, we generated conformational ensembles of $Myc^{1–361}$ to investigate sequence-encoded conformational biases in the ensemble (Fig. 4b and Extended Data Fig. 6d). In particular, we wondered whether conformational behaviour would offer insight into regions within Myc known to bind to other partners (so-called Myc boxes). Curiously, the STARLING-derived ensemble suggests that $Myc^{1–361}$ can be divided into two halves (IDR1 and IDR2). IDR1 ($Myc^{1–200}$) is more compact and engages in intramolecular interactions and some long-range interactions. IDR2 ($Myc^{201–361}$) is more expanded with locally compact regions. Finally, conformationally distinct subregions align with established Myc-box boundaries, indicating local and long-range conformational properties can align with established functional annotations. Overall, these results predict that $Myc^{1–361}$ contains strong sequence-encoded biases that may be correlated with (or even underlie) its function.

The C-terminal domain (CTD) of eukaryotic RNA polymerase II (RPB1) is a large IDR composed of tandem YSPTSPS heptad repeats, whose repeat number varies across species (52 in humans)[37] (Fig. 4c). The CTD is believed to function as a multivalent interaction platform for transcriptional regulation. Extensive previous work has shown that low-complexity domains enriched in tyrosine (such as the CTD) are

（对于 20 mM 和 300 mM 浓度，对应值分别为 0.992）and R_e（扩展数据图 4a、b； RMSE = 4.17 Å 和 4.53 Å，R^2对应于 20 mM 和 300 mM 浓度时分别为 0.986 和 0.978）再次与模拟结果高度吻合。

模型精度取决于集合规模与去噪步数之间的平衡。通过将 Starling 的计算结果与长时程 Mpipi-GG 模拟结果进行对比，we found that error in the R_g我们将这些参数设为默认值，不过对于其他序参量，可能需要采用更大的集合规模。

由于R_e and R_g与序列长度高度相关，我们随后在一组长度匹配的序列[29]上评估了模型的准确性。这一方法使我们能够判断该模型是否有效地捕捉了序列化学性质对 IDR 全局维度的影响，抑或仅仅是学会了将序列长度与这些维度相关联。通过使用来自留出测试集、具有不同序列化学性质的全部 100 个残基、200 个残基及 300 个残基序列，我们发现R_g（扩展数据图 4j；对于 100 个残基、200 个残基和 300 个残基长度， RMSE = 0.78 Å、1.18 Å 和 1.20 Å，R^2= 0.984、0.990 和 0.995）and R_e（扩展数据图 4k；对于 100 个残基、200 个残基和 300 个残基长度， RMSE = 3.38 Å、4.22 Å 和 5.65 Å，R^2= 0.965、0.984 和 0.986）在不同R_g值范围内均呈现极佳的一致性。这些结果使我们确信，Starling 模型所学习到的是真正的“序列至集合”规则，而非仅仅是学习了聚合物标度理论。

除了集合平均观测量外，完整的结构集合使我们能够计算任意感兴趣观测量的分布（图**2e**）。我们旨在评估椋鸟数据推导出的全局维度分布与模拟所得分布之间的吻合程度。我们采用赫林格距离（H）作为相似性度量指标对这些分布进行比较。*H 的取值范围*为 0 至 1：其中 0 表示两个分布完全相同，而 1 则表示它们完全不相交（图**2e**）。椋鸟数据derived R_g分布与模拟所得分布呈现出极佳的一致性（图**2f**）。这一重叠程度通过低 H 值进行量化，证实了两个分布之间存在高度相似性。

最后，为量化集合保真度，我们评估了通过 Starling 或 Mpipi-GG 模拟生成的各构象集合之间所有残基间距离分布。每个三维构象均可表示为一张单距离图（图**2g**），而整个构象集合则可表示为一组距离图集合（图**2h**）。因此，可将 Starling 构象集合中任意一对残基的残基间距离分布与 Mpipi-GG 生成的对应分布进行比较（图**2i**）。对于一个由 100 个残基组成的 IDR，由此可获得 4,900 次独特的分布比较。我们可利用 Hand 方法对每对分布进行量化分析，并通过热图对其进行可视化呈现（图**2j**，上图）。此外，还可对这些热图绘制直方图，以对整个构象集合的一致性进行整体评估（图**2j**，下图）。总体而言，我们在所研究的所有距离分布中均发现了极佳的一致性（图**2k**）。综上所述，所有证据均表明，Starling 能够直接根据蛋白质序列预测无序蛋白质的构象集合。

在确认星雀模型能够复现模拟系综后，我们接下来探究了其与实验数据的一致性（图**3a**）。我们采用了一组此前经过人工整理的 133 条序列数据集，这些序列均配有高质量的小角X射线散射（SAXS）数据，并据此计算了average R_g值。我们发现，星雀模型计算得到的between average R_g值与通过 SAXS 方法derived R_g值之间具有极佳的一致性，所得数值与当前最先进的粗粒化模拟结果相当（RMSE = 4.53，R^2= 0.90；见图**3b**、扩展数据图 5a–h 以及补充表 3 和 4）。

为进一步验证 Starling，我们针对来自 DisProt [30]的 3,417 个 IDR 预测了集合模型，成功复现了已知的序列-构象规律（例如，芳香族残基驱动结构紧凑化，而脯氨酸和谷氨酸则促进结构扩张[3,20,31,32]；扩展数据图 5i–k 以及补充表 5）。Starling 生成的集合模型对微小的序列变化也具有敏感性；当对四种 hnRNPA1 低复杂度结构域（LCD）变体进行测试时，该模型预测的维度与实验获得的关于芳香族残基含量的 SAXS 数据定量吻合[3]（图**3c**）。

为了超越简单的R_g比较（图3b，c），我们利用 Starling 生成的系综，通过 FOXS [33]对合成的 SAXS 曲线进行反演计算。这使得我们能够将我们的模型与原始实验测得的 SAXS 数据（分辨率低至 5–10 Å）进行直接比较。我们证明了 Starling 能准确捕捉 A1-LCD 变体中细微的构象变化，这一点体现在其特定的散射谱图中[3]。随后，我们选取了 12 组无亲缘关系且拥有高质量散射数据的 IDR，将基于 Starling 系综计算得到的散射谱图与实验数据进行比较，发现它们在广泛的序列化学性质和序列长度范围内均具有极佳的一致性（图3e 及补充表 3）。我们进一步对另外 40 组散射数据集进行了重新分析，发现其中大多数数据与基于 Starling 计算得到的散射谱图高度吻合（扩展数据图 6a 以及补充图 1 和 2）。综上所述，我们的数据表明，Starling 系综能够准确再现实验测得的 IDR 系综行为，且这一再现性适用于多种长度尺度和序列化学性质。

最后，我们尝试将上述方法与另一种实验手段——单分子福斯特共振能量转移（smFRET）——进行对比。近期的研究表明，由16条长度匹配的序列组成的集合所对应的端到端距离范围，取决于其底层序列的化学性质[34]。我们对这16条序列的预测端到端距离（以及在可获得的情况下，基于 SAXS 方法获得的散射数据）进行了比较；结果再次显示，星雀模型生成的系综与实验数据之间具有良好的一致性（图**3f**、扩展数据图6b、c及补充表3）。值得注意的是，我们在这16条序列上计算出的平均误差（RMSE 为6.7 Å）与近期一项smFRET基准测试研究中不同实验组之间所获得的差异相当[35]。总体而言，我们的研究结果表明，星雀模型能够生成能够准确反映实验数据的IDRs全粗粒度系综。

在确认 Starling 能够复现现有的实验数据后，我们将其应用于探究一系列在不同情境下表现各异的 IDRs。这些示例展示了 Starling 所支持的快速探索与假设生成类型。

主转录调控因子 Myc（含 439 个氨基酸残基）参与调控细胞生长、代谢及增殖过程，在多种癌症中常出现功能失调[36]。Myc 由一个较大的 N 端 IDR 结构域（Myc $^{1–361}$）、一个 C 端 HLH DNA 结合结构域（Myc $^{368–406}$）以及一个亮氨酸拉链结构域（Myc $^{413–439}$）组成。其 IDR 结构域的庞大尺寸结合其化学组成，给其生物物理表征带来了挑战（图**4a**）。利用 Starling 软件，我们构建了 Myc $^{1–361}$的构象集合，以探究该集合中由序列编码的构象偏好性（图4b及扩展数据图 6d）。特别地，我们想探究这种构象行为是否能为揭示 Myc 中已知可与其他分子结合的区域（即所谓的“Myc 盒”）提供线索。有趣的是，基于 Starling 软件构建的构象集合表明，Myc $^{1–361}$可划分为两个部分（IDR1 与 IDR 2）。IDR1（Myc $^{1–200}$）结构更为紧凑，参与分子内相互作用及部分长程相互作用；而 IDR2（Myc $^{201–361}$）结构更为伸展，并含有局部紧凑区域。最后，构象上不同的亚区与已确立的 Myc-box 边界相吻合，这表明局部及长程的构象特性可与已确立的功能注释相匹配。总体而言，这些结果预示着 Myc $^{1–361}$包含着强烈的序列编码偏好性，这些偏好性或许与其功能相关，甚至可能是其功能的基础。

真核生物RNA聚合酶II（RPB1）的C端结构域（CTD）是一个大型 IDR，由串联的ysptsps七肽重复序列构成；其重复序列的数量因物种而异（人类为52个）[37]（图**4c**）。CTD被认为可作为转录调控中的多价相互作用平台。大量前期研究已表明，富含酪氨酸的低复杂度结构域（如CTD）

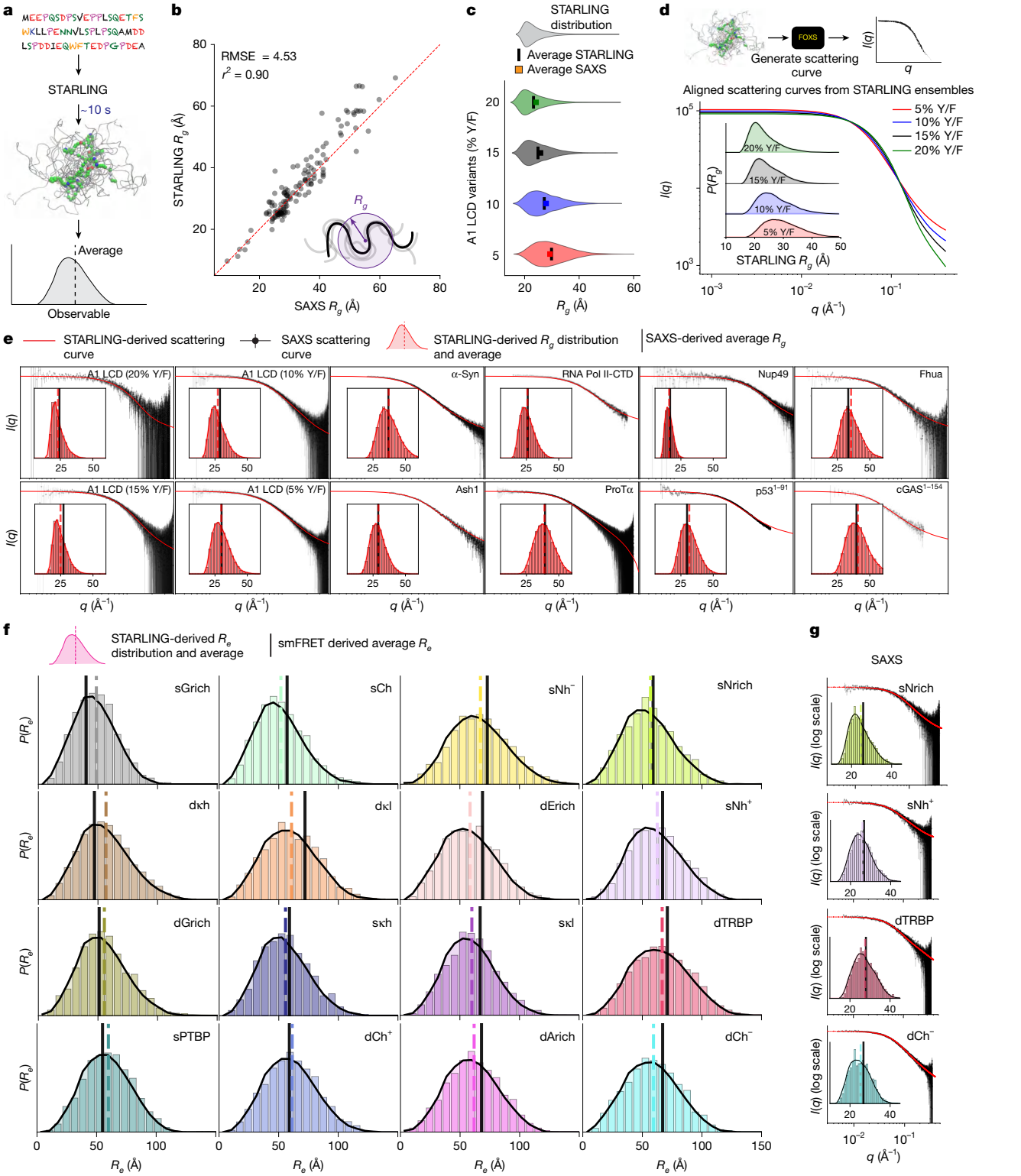

Fig. 3 | Experimental validation of STARLING. a, STARLING enables rapid generation of ensembles from sequence. **b**, STARLING shows state-of-the-art accuracy in terms of predicting the average R_g from sequence. IDR sequences (n = 133) comparing SAXS-derived and STARLING-derived average R_g values (see also Extended Data Fig. 4c–f) are shown. **c**, STARLING accurately captures the effect of small numbers of mutations, here illustrating how altering the number of aromatic residues changes the global dimensions of a low-complexity domain (LCD) taken from the RNA-binding protein hnRNPA1. Y/F, tyrosine/phenylalanine. **d**, Full scattering curves can be back-calculated from STARLING ensembles using FOXS, illustrating how changes in conformational behaviour alter both global dimensions (small angles and low q values) and the shape of the ensemble (intermediate and high q values). **e**, Comparison of 12 different SAXS profiles with scattering profiles calculated from STARLING ensembles. The insets show STARLING R_g distributions, with average R_g values from scattering data displayed as a solid black line. See also Extended Data Figs. 4j,k and 5a–h. **f**, STARLING-derived R_e distance distributions with experimental values are shown as black lines for 16 different length-matched disordered protein sequences with different sequence chemistries. See also Extended Data Fig. 5i–k. **g**, SAXS scattering curves for unlabelled proteins for a subset of those in panel **f**.

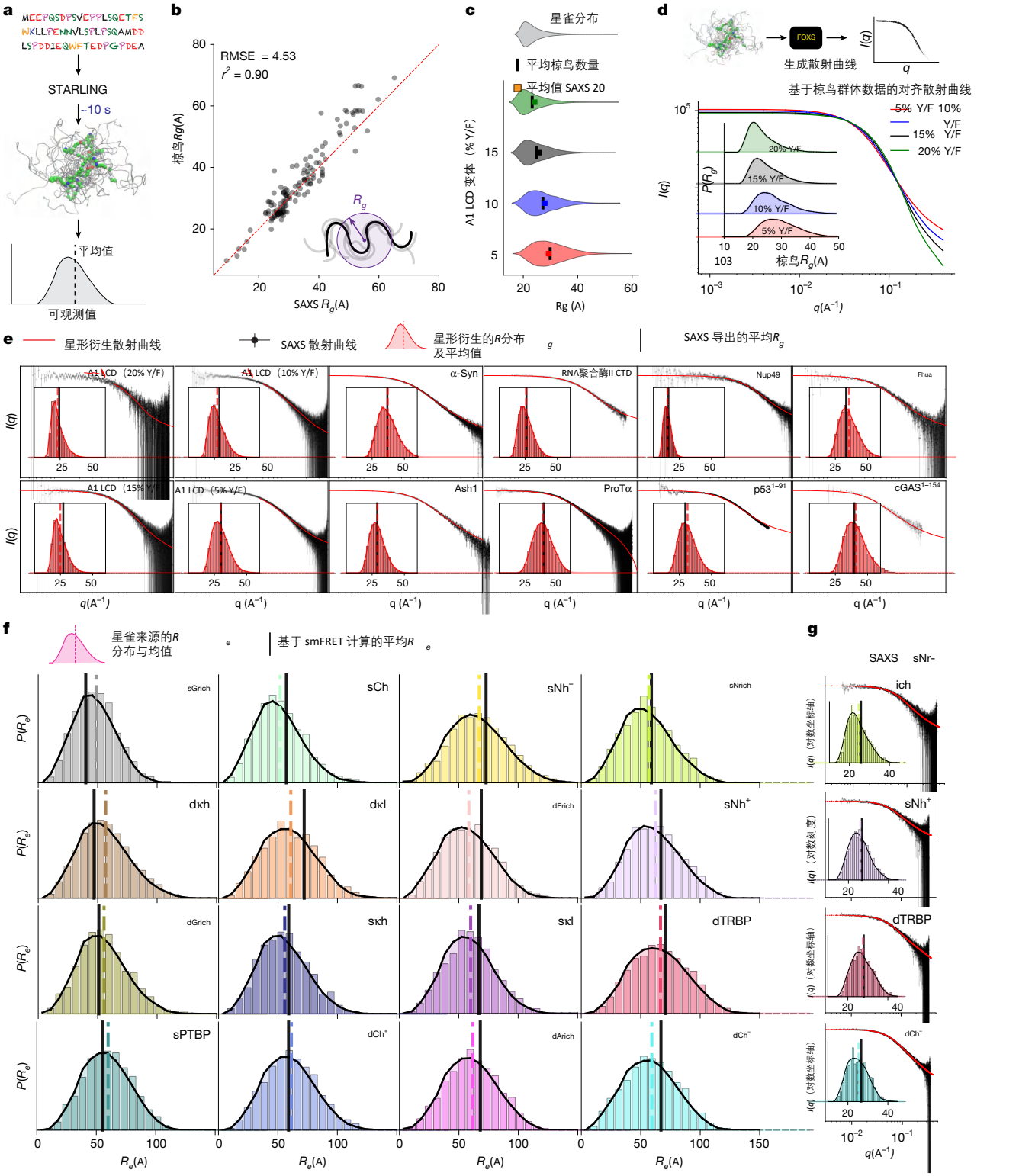

图3 | Starling 的实验验证。a：Starling 可实现基于序列快速生成集合体。**b**：Starling 在in terms of predicting the average R_g展现出最先进的准确度。图中展示了 IDR 序列（n= 133）中 SAXS 法与 Starling 法derived average R_g值对比（另见扩展数据图 4c–f）。**c**：Starling 准确捕捉了少量突变所产生的影响；本图展示了改变芳香族残基数量如何改变源自 RNA 结合蛋白 hnRNPA1 的低复杂度结构域（LCD）的全局维度。Y/F 表示酪氨酸/苯丙氨酸。**d**：通过使用 FOXS 从 Starling 集合体中反演得到完整的散射曲线，揭示了构象行为的变化如何同时影响全局维度（小角度及低 q 值）与集合体的形状（中等及高 q 值）。**e**：将 12 个不同的 SAXS 谱图与基于 Starling 集合体计算得到的散射谱图进行对比。插图show STARLING R_g分布情况，with average R_g值以黑色实线表示。另见扩展数据图 4j、k 以及 5a–h。**f**：基于椋鸟数据derived R_e距离分布，其中 16 条长度匹配且具有不同序列化学组成的无序蛋白质序列的实验值以黑色线条表示。另见扩展数据图 5i–k。**g**：针对图 f 中部分未标记蛋白质绘制的 SAXS 散射曲线。

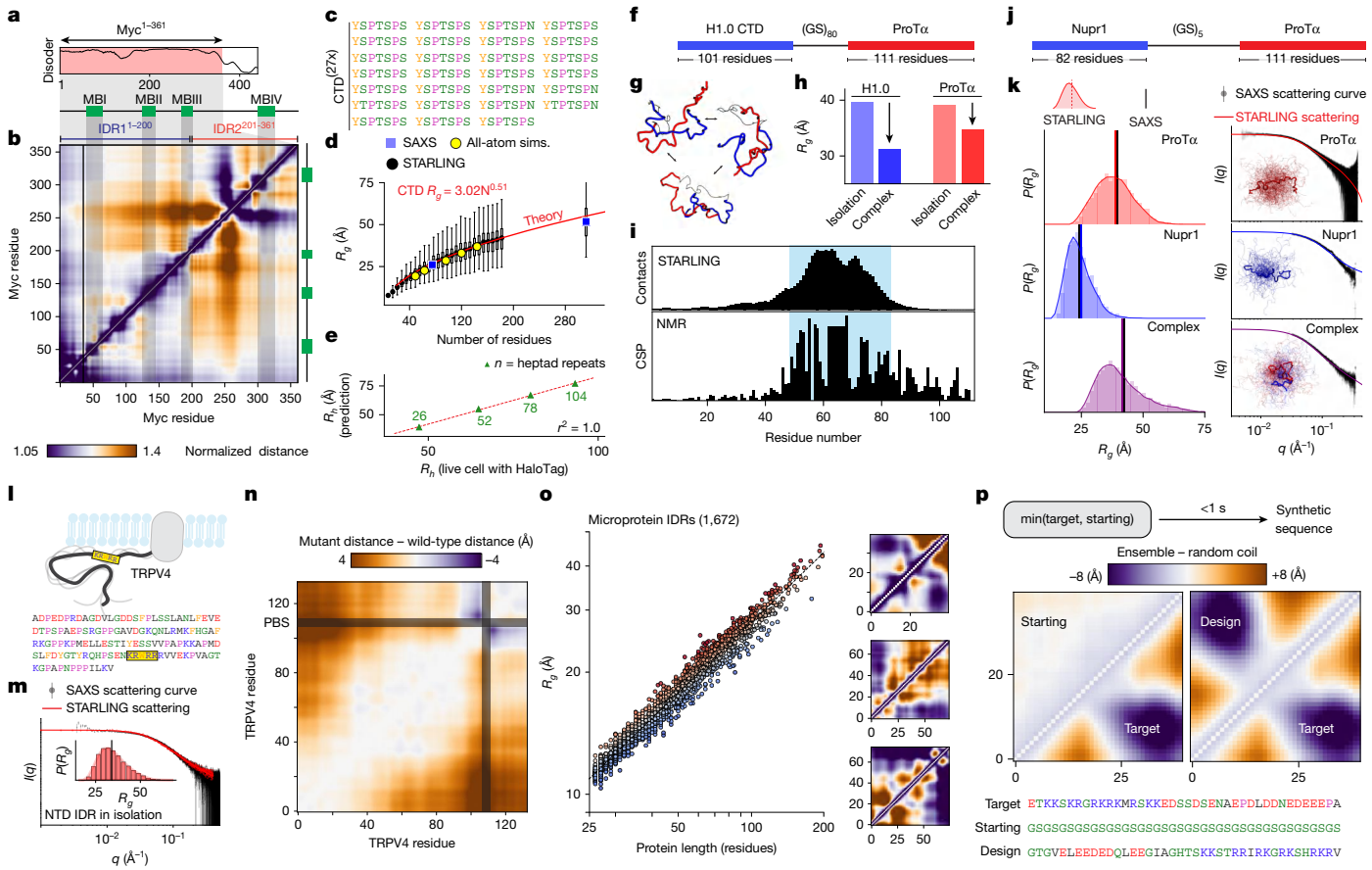

Fig. 4 | Biophysical characterization and design applications with STARLING. **a**, The transcription factor Myc is largely disordered. **b**, STARLING-derived Myc$^{1–361}$ ensembles identify distinct subregions that drive attractive and repulsive interactions. The darker colours are closer together, and the lighter colours are further apart. **c**, The RNA polymerase II CTD consists of a series of heptad repeats. **d**, STARLING ensembles show excellent agreement with SAXS and all-atom simulations (sims.), enabling polymer theory to describe CTD dimensions with a simple analytical expression. **e**, Comparison of predicted hydrodynamic radii (R_h) values based on STARLING-derived polymer scaling behaviour (Extended Data Fig. 6g) and R_h values from freely diffusing CTD-only constructs measured in nuclei. **f**, Histone H1.0 and ProTα form a high-affinity complex together. The H1.0 CTD and ProTα were tethered together by an inert $(GS)_{80}$ linker. **g**, The resulting ensemble shows robust H1.0–ProTα interaction. **h**, The H1.0 CTD and ProTα contract in the complex versus in isolation. **i**, Comparison of per-residue ProTα–H1.0 CTD contacts and NMR-derived chemical shift perturbations (CSPs) shows excellent agreement. Blue highlighting identifies core binding subregion. **j**, Nupr1 and ProTα form a complex; we used a shorter $(GS)_5$ linker to enable comparison to SAXS data. **k**, STARLING-derived ensembles show excellent agreement with Nupr1, ProTα and the Nupr1–ProTα complex. **l**, The TRPV4 channel has a large N-terminal IDR that contains a basic phosphatidylinositol 4,5-bisphosphate-binding site (PBS). NTD, N-terminal domain. **m**, STARLING-derived ensembles show good agreement with SAXS data. **n**, Difference distance map showing the difference in inter-residue distance between the mutant sequence minus the wild-type sequence. Orange colours indicate that distances are further apart in the mutant while blue means that residues are closer together in the mutant than in the wild type. The PBS engages in long-range interactions with a distal acidic region, and mutation of positively charged residues reduces these interactions, in good agreement with extant experimental work. **o**, STARLING enables systematic prediction of disordered microprotein ensembles. **p**, Using an optimized latent-space design approach, STARLING enables rapid sequence design to match a desired conformational ensemble. See also Extended Data Fig. 7d–m.

prone to intermolecular and intramolecular interactions, meaning that these domains would be expected to be relatively compact (driven by transient but numerous intramolecular contacts)[3]. Unexpectedly, previous in vitro work characterized an 81-residue fragment of the CTD with a seemingly highly extended radius of gyration, equivalent to that expected under strongly denaturing conditions (for example, 8 M urea)[38] (Extended Data Fig. 6e). This raises an apparent paradox: how can the CTD be simultaneously enriched for tyrosine, poised for intermolecular and intramolecular interactions, yet also appear to behave like a non-interacting, fully denatured polypeptide chain?

To answer this question, we generated STARLING ensembles across 26 different CTD lengths, complemented by all-atom Monte Carlo simulations across five lengths. All-atom and STARLING-derived ensembles are in excellent agreement with one another, and with extant SAXS data for CTD variants of length 81 and 312 residues (Fig. 4d). This analysis (see Methods; Extended Data Fig. 6f,g and Supplementary Fig. 3) revealed that the CTD in fact behaves with polymer statistics of an effective Gaussian chain, meaning it is poised to engage in intramolecular and intermolecular interactions[39]. The seemingly extended nature of the ensemble is driven by the stiffness imparted by the size of the tyrosines and the high proline content (Extended Data Fig. 6h,i). Furthermore, comparisons of CTD dimensions calculated from in-cell single-molecule diffusion data by Ling et al. support a model in which the STARLING-derived CTD dimensions persist in live cells[40] (Fig. 4e). These results reconcile previous observations and suggest that in cells, the CTD is primed for intermolecular interactions.

We next wondered whether we could tether two different IDRs together with a glycine-serine (GS) linker to investigate pseudo-intermolecular interactions (Fig. 4f). Tethering the C-terminal IDR from the histone H1.0 (H1.0$^{97–198}$) to the histone chaperone prothymosin-α (ProTα) enabled the formation of an electrostatically driven complex between the two proteins (Fig. 4g), as previously

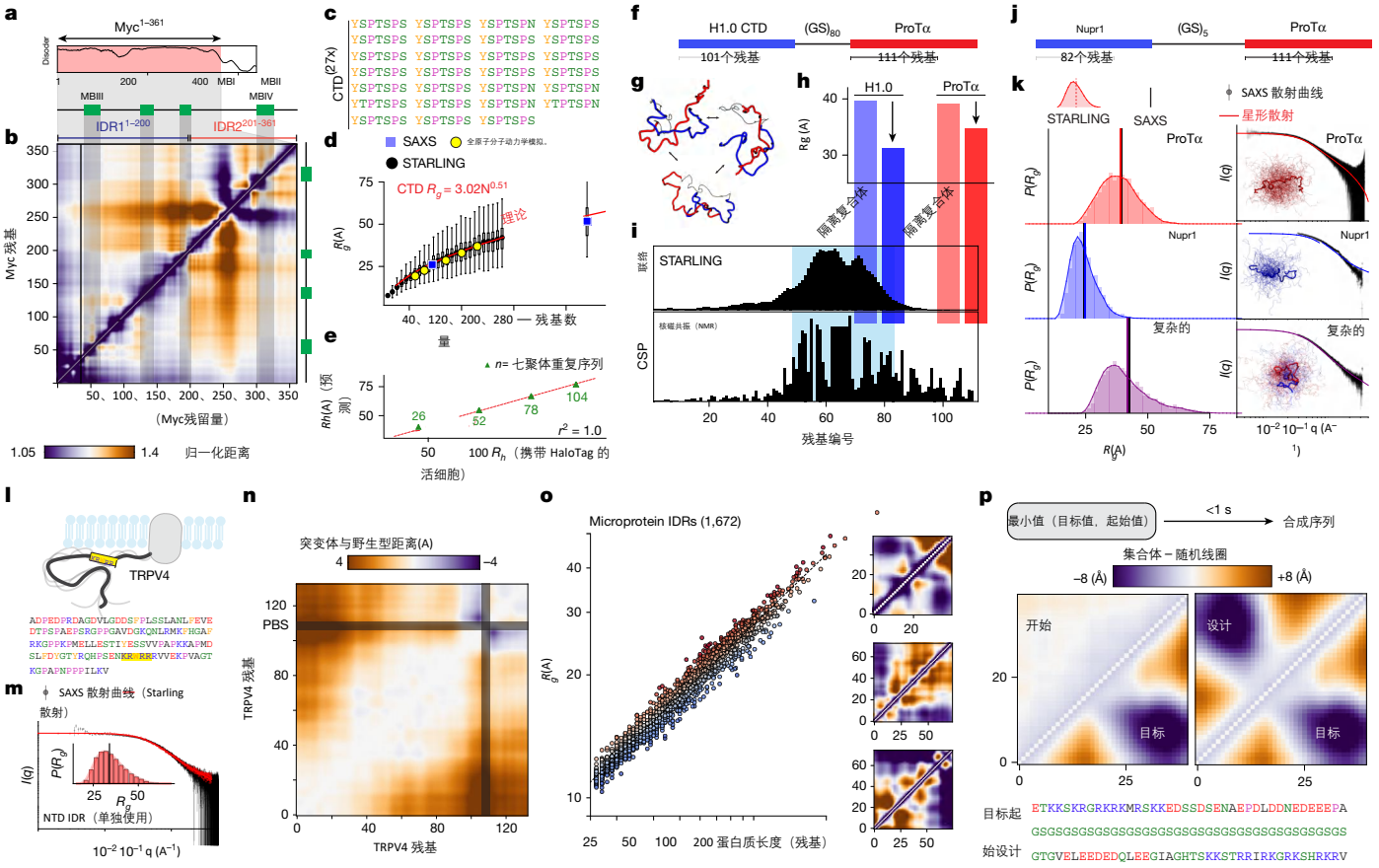

图4|基于Starling的生物物理表征与设计应用。a，转录因子Myc在很大程度上呈无序状态。**b**，基于Starling构建的Myc $^{1-361}$聚合体可识别出驱动吸引与排斥相互作用的distinct 亚区域；颜色越深表示区域越接近，颜色越浅表示区域相距较远。**c**，RNA聚合酶II CTD由一系列七肽重复单元组成。**d**，Starling聚合体与SAXS及全原子模拟结果（sims.）高度吻合，使得聚合物理论能够用简单的解析表达式描述CTD的尺寸。**e**，基于Starling衍生聚合物标度行为预测的流体动力学半径（R_h）值andR_h在细胞核内测得的自由扩散的仅含CTD构建体的Rh值进行比较。**f**，组蛋白H1.0与ProTα共同形成高亲和力复合物；H1.0 CTD与ProTα通过一个惰性（GS）$_{80}$连接体锚定在一起。**g**，所得聚合体显示出稳健的H1.0–ProTα相互作用。**h**，H1.0 CTD 与ProTα 在复合物中的收缩情况与单独存在时的收缩情况对比。**i**：对每个残基的 ProTα–H1.0 CTD 接触位点与通过 NMR 法获得的化学位移扰动（CSPs）进行比较，结果显示两者高度吻合。蓝色高亮部分标出了核心结合亚区。**j**：Nupr1 与 ProTα 形成复合物；我们采用了一个较短的（GS）$_5$连接肽，以便与 SAXS 数据进行比较。**k**：基于 Starling 构建的分子集合与 Nupr1、ProTα 以及 Nupr1–ProTα 复合物的预测结果高度吻合。**l**：TRPV4 通道的 N 端具有一个较大的 IDR 结构域，该结构域包含一个碱性磷脂酰肌醇 4,5-二磷酸结合位点（PBS）。NTD：N 端结构域。**m**：基于 Starling 构建的分子集合与 SAXS 数据吻合良好。**n**：差值距离图展示了突变序列与野生型序列之间残基间距离的差异；橙色表示突变体中残基间距更大，而蓝色则表示突变体中残基间距小于野生型。PBS 与远端酸性区域发生长程相互作用，而带正电荷残基的突变会削弱这些相互作用，这一结果与现有实验研究高度一致。**o**：Starling 可用于对无序微蛋白集合进行系统性预测。**p**：通过采用优化的潜在空间设计方法，Starling 可实现快速序列设计，以匹配所需的构象集合。另见扩展数据图 7d–m。

这些结构域易发生分子间及分子内相互作用，这意味着它们预期会呈现相对紧凑的构象（这是由瞬时但数量众多的分子内接触所驱动）[3]。然而出乎意料的是，此前的一项体外研究发现，CTD中一个由 81 个残基组成的片段具有一个看似极高的回转半径，该数值与在强变性条件下（例如 8 M 尿素）预期值相当[38]（扩展数据图 6e）。这引发了一个明显的悖论：CTD 如何既能富含酪氨酸、为分子间及分子内相互作用做好准备，又似乎表现出像一条无相互作用、完全变性的多肽链那样的行为？

为解答这一问题，我们构建了涵盖 26 种不同 CTD 长度的 Starling 系综，并辅以针对五种长度开展的全原子蒙特卡罗模拟。全原子系综与 Starling 源系综之间以及二者与现有针对长度为 81 和 312 个残基的 CTD 变体所采集的 SAXS 数据之间均呈现高度一致性（图4d）。该分析（参见“方法”部分；扩展数据图 6f、g 以及补充图 3）表明，CTD 实际上遵循着一种等效高斯链的聚合物统计学行为，这意味着它处于参与分子内及分子间相互作用的状态[39]。该系综所呈现的看似延展的特性，是由酪氨酸残基尺寸所赋予的刚性以及较高的脯氨酸含量所驱动的（扩展数据图 6h、i）。此外，通过对比 Linge 等人基于细胞内单分子扩散数据计算得出的 CTD 维度，支持了这样一个模型：即 Starling 源系的 CTD 维度在活细胞中依然保持不变[40]（图4e）。这些结果与此前的观测结果相吻合，并提示在细胞内，CTD 已处于参与分子间相互作用的状态。

接下来，我们探究是否可通过甘氨酸-丝氨酸（GS）连接子将两种不同的IDR连接起来，以研究伪分子间相互作用（图4f）。将组蛋白H1.0的C端 IDR 区段（H1.0 $^{97-198}$）与组蛋白伴侣Prothymosin-α（ProTα）进行连接，成功诱导了这两种蛋白质之间形成静电驱动的复合物（图4g），这与先前报道的结果一致[41]。在该复合物中，两种蛋白质均发生

reported[41]. Both proteins contract in the complex (Fig. 4h), consistent with experimental data and theory. Moreover, the specific residues in ProTα identified by nuclear magnetic resonance (NMR)-derived chemical shift perturbations match those residues from the STARLING complex ensemble that drive intermolecular interactions (Fig. 4i). We also investigated the complex between ProTα and the fully disordered 82-residue protein Nupr1 (ref. 42) (Fig. 4j). Here again, we saw agreement with SAXS data for the individual proteins and, using a shorter GS linker, for the disordered bimolecular complex (Fig. 4k). These results indicate that STARLING-derived binary complex ensembles can capture elements of IDR–IDR interaction ensembles.

Although our analyses thus far have primarily focused on ensemble differences between very different sequences, we wondered whether STARLING could interpret relatively small amino acid changes. The N-terminal IDR from the transient receptor potential vanilloid 4 (TRPV4) ion channel has a long-range regulatory interaction between a basic phosphatidylinositol 4,5-bisphosphate-binding site and an acidic cluster[43] (Fig. 4l). STARLING-derived ensembles of this region show good agreement with SAXS data (Fig. 4m). Moreover, comparing the differences in intramolecular distances between the wild type and a KRWRR-to-AAWAA mutant that reduces long-range interactions in vitro yields a commensurate reduction in long-range interactions in STARLING ensembles (Fig. 4n). These illustrate how STARLING can be used to understand biophysical mechanisms of functional regulation in proteins, even with relatively subtle changes in sequence.

We were also interested in investigating the application of STARLING to larger-scale ensemble prediction. Recent work has characterized a large set of microproteins — non-canonical open reading frames typically under 100 amino acids in size — using a combination of mass spectrometry and Ribo-seq-based approaches[44]. Although these microproteins are highly disordered, they possess distinct amino acid sequence biases from 'canonical' open reading frame-derived IDRs (Extended Data Fig. 7a–c and Supplementary Table 6). From 1,785 microproteins in these data, we identified 1,672 IDRs and predicted ensembles for all of them. Although most short IDRs are relatively expanded, many had extensive intramolecular contacts driven primarily between aromatic and arginine residues (Fig. 4o). These results suggest that even short-disordered proteins can possess sequence-encoded conformational biases.

Beyond accurate, rapid ensemble prediction, STARLING addresses the inverse-design problem: generating sequences whose conformational ensembles match a chosen target. To do this, we exploited the ensemble-aware sequence representations of the model by implementing a latent-alignment objective that maximizes cosine similarity between target and candidate learned sequence representations using the sequence optimization framework in GOOSE (see Methods)[26]. This allowed us to quickly design in this ensemble-aware latent space, rather than generating ensembles in 3D, enabling the rapid optimization of bespoke disordered sequences with prescribed ensemble properties.

As a proof of concept, we used STARLING in conjunction with our IDR design framework, GOOSE to design 100 sequences targeting a short IDR from the protein CTCF ($CTCF^{590–629}$; UniProt ID: P49711)[26]. Starting from length-matched GS sequences, generation averaged 0.7 s per sequence with high structural accuracy (mean absolute error of 0.94 Å; average Hamming distance of 35; Extended Data Fig. 7d). Furthermore, we found that ensemble-matching designs can be generated within tens of seconds, even for long sequences (Extended Data Fig. 7e–m and Supplementary Fig. 4). These results, alongside performance tests on longer sequences, demonstrate that STARLING enables rapid, systematic design of IDRs with specific conformational properties.

Predictions up until this point have focused on ensemble generation at 150 mM ionic strength. However, the STARLING sequence encoder was explicitly conditioned on ionic strength (see Methods). Consequently, STARLING can also generate ensembles at ionic strengths between 20 mM and 300 mM (Fig. 5a).

To assess how well STARLING captures ionic-strength dependence — as defined by the underlying Mpipi-GG model — we analysed 30 strong polyampholytic sequences consisting solely of lysine and glutamate with distinct charge patterning ('Das–Pappu sequences')[45]. Figure 5b shows the end-to-end distance distributions from STARLING and Mpipi-GG at the three ionic strengths used during training, showing excellent agreement as measured by both the H and the KS statistic (note that these are not fitted to one another). This trend holds across the Das–Pappu sequence variants (Extended Data Fig. 8a,b).

We next asked whether our ionic-strength model, conditioned with three anchor points (ionic strength of 20 mM, 150 mM and 300 mM), would generalize to new ionic strengths not seen in training. We evaluated 16 ionic strengths spanning 20–300 mM (13 never seen in training) and analysed the mean end-to-end distance of each ensemble. STARLING interpolates smoothly between the three training ionic strengths and reproduces Mpipi-GG means with high concordance (Pearson $r = 0.98$; Spearman $\rho = 0.97$; Fig. 5c). Beyond just global dimensions, normalized distance maps compared between ensembles generated via STARLING and Mpipi-GG simulations revealed that STARLING accurately learns the ionic-strength-dependent changes in the underlying conformational distributions of inter-residue distances (Fig. 5d and Extended Data Fig. 8c). Furthermore, STARLING can qualitatively reproduce ionic-strength dependence of the ProTα–$H1.0^{CTD}$ interaction[41] (Extended Data Fig. 8d–f). Together, these results suggest STARLING can be used to investigate solution-dependent IDR interactions using a limited number of anchor points and interpolating between them.

Although STARLING aligns with Mpipi-GG and experimental data, it is not a substitute for rigorous biophysical measurements. Instead, experiments can complement STARLING by providing restraints to refine and integrate into IDR ensembles via two routes. The first route uses guidance during ensemble generation to push conformers towards target observables (see Methods; Extended Data Fig. 9a–c and Supplementary Fig. 5), an approach well-suited for generating ensembles far from the unbiased distribution (for example, cyclic peptides), although guided ensembles likely deviate from the canonical ensemble. The second route uses Bayesian maximum entropy (BME) reweighting — reimplemented natively within STARLING — to reweight ensembles to match experimental observables[15]. BME treats the STARLING ensemble as the prior and incorporates data (for example, SAXS and smFRET) through a likelihood-based restraint with uncertainties, yielding ensembles that better match experiment (Fig. 5e). As a proof of concept, we used BME to reweight four of the ensembles for sequences measured by Holla et al.[34], treating the experimental end-to-end distance as a restraint in the BME objective. The resulting ensembles not only improve agreement with end-to-end distance but also agreement with the SAXS-derived R_g (Fig. 5f). These results suggest that STARLING provides a powerful and easy-to-use platform for integrative modelling of experimental data of IDRs.

Finally, beyond the strict generation of ensembles, a novel application enabled by the unique architecture of STARLING is the ability to perform fast, large-scale disordered-protein biophysical ensemble-similarity search directly from sequence (Fig. 5g). By encoding sequences into a conformationally aware latent space using the sequence encoder (Extended Data Fig. 2d), search can be performed within this embedded space, enabling the identification of conformationally similar IDRs.

As a proof of principle, we created a database of all predicted IDRs in UniRef50 with fewer than 380 residues (approximately 35 million sequences). We then created STARLING ensemble-aware embeddings for each of these sequences, implementing a methodology for rapid, large-scale approximate nearest-neighbour search (see Methods). We tested this approach by querying the protein SERF (UniProt ID: O75920) against this database[46] (Fig. 5g). Next, we extracted the candidate with the most similar and least similar vector embeddings and built ensemble-averaged distance maps for each construct. Normalizing

收缩（图4h），这一现象与实验数据及理论预测相符。此外，通过核磁共振（NMR）化学位移扰动法鉴定出的ProTα中特定残基，与星雀复合物集合中驱动分子间相互作用的残基相匹配（图4i）。我们还研究了ProTα与完全无序的含82个残基的蛋白质Nupr1（参考文献42）形成的复合物（图4j）。在此案例中，我们再次观察到该复合物的SAXS数据与单个蛋白质的SAXS数据相符；而在使用较短GS连接子时，该无序双分子复合物的SAXS数据也与之吻合（图4k）。这些结果表明，椋鸟来源的二元复合物集合能够捕捉 IDR– IDR 相互作用集合中的若干要素。

尽管迄今为止我们的分析主要聚焦于差异巨大的序列之间的集合差异，但我们想知道星雀是否能够解析相对微小的氨基酸变化。瞬时受体电位香草酸受体4（TRPV4）离子通道的N端 IDR 区域存在一个碱性磷脂酰肌醇4,5-二磷酸结合位点与一个酸性氨基酸簇[43]之间的长程调控相互作用（图4l）。由星雀构建的该区域的集合数据与 SAXS 实验数据高度吻合（图4m）。此外，通过比较野生型与在体外实验中削弱长程相互作用的aKRWRR→ AAWAA 突变体之间的分子内距离差异，可以发现星雀构建的集合中长程相互作用也相应减少（图4n）。这些结果表明，即使在序列存在相对细微变化的情况下，亦可利用星雀方法来解析蛋白质功能调控的生物物理机制。

我们还致力于探究 Starling 方法在更大规模集合预报中的应用。近期的一项研究利用质谱分析与基于 Ribo-seq 的方法相结合的技术，对大量微蛋白——即通常长度小于 100 个氨基酸的非典型开放阅读框——进行了表征[44]。尽管这些微蛋白具有高度无序性，但其氨基酸序列偏好性与源自“典型”开放阅读框的 IDR 不同（参见扩展数据图 7a–c 及补充表 6）。在这些数据中的 1,785 个微蛋白中，我们鉴定出了 1,672 个 IDR，并为所有这些微蛋白预测了相应的集合。尽管大多数短 IDR 相对而言较为伸展，但其中许多 IDR 具有广泛的分子内相互作用，这些相互作用主要发生在芳香族残基与精氨酸残基之间（图4o）。这些结果表明，即使对于短链无序蛋白而言，也存在由序列编码的构象偏好性。

除了实现精确、快速的集合预测外，Starling 还解决了逆设计问题：即生成其构象集合与给定目标相匹配的序列。为实现这一目标，我们利用了该模型中具有集合感知特性的序列表示方法，通过实现一个潜向对齐目标来最大化目标序列与候选学习到的序列表示之间的余弦相似度；该目标基于 Goose 中的序列优化框架实现（参见“方法”部分）[26]。这使得我们能够在该具有集合感知特性的潜空间中快速进行设计，而无需在三维空间中生成构象集合，从而能够对具有指定集合性质的定制化无序序列进行快速优化。

作为概念验证，我们利用 Starling 与自研的 IDR 设计框架 Goose 进行结合，设计了 100 条序列，旨在针对蛋白质 CTCF（$^{590–629}$；UniProt ID：P49711）[26]的短 IDR 进行设计。从长度匹配的 GS 序列出发，序列生成平均耗时 0.7 秒，且具有较高的结构精度（平均绝对误差为 0.94 Å；平均汉明距离为 35；见扩展数据图 7d）。此外，我们发现即使对于较长的序列，集成匹配设计也可在数十秒内完成生成（见扩展数据图 7e–m 以及补充图 4）。这些结果结合对较长序列的性能测试，表明 Starling 能够实现具有特定构象特性的内在无序区域（IDRs）的快速、系统化设计。

此前的所有预测均聚焦于 150 mM 离子强度条件下的集合生成。然而，Starling 序列编码器是明确基于离子强度进行条件化的（参见“方法”部分）。因此，Starling 亦可生成在 20 mM 至 300 mM 离子强度范围内的集合（图5a）。

为评估 Starling 模型对离子强度依赖性的拟合程度（该依赖性由基础的 Mpipi-GG 模型定义），我们分析了 30 条仅由赖氨酸与谷氨酸构成且具有不同电荷分布模式的强两性离子序列（“Das–Pappu 序列”）[45]。图5b展示了在训练过程中所采用的三种离子强度下，Starling 模型与 Mpipi-GG 模型计算得到的端到端距离分布；通过 Hand-KS 统计量可观察到两者之间存在极佳的一致性（需注意，这两组数据并未进行相互拟合）。这一趋势在各类 Das–Pappu 序列变体中均成立（扩展数据图 8a，b）。

接下来，我们探究了采用三个锚定点（离子强度分别为 20 mM、150 mM 和 300 mM）进行条件化的离子强度模型是否能泛化到训练集中未出现过的新离子强度条件下。我们评估了涵盖 20–300 mM 范围内的 16 个离子强度值（其中 13 个在训练集中从未出现），并分析了每个模拟集合的平均端到端距离。Starling 模型在三个训练用离子强度之间实现了平滑插值，并能以较高的一致性重现 Mpipi-GG 的均值（Pearson r= 0.98；Spearman ρ= 0.97；图5c）。除全局维度外，通过比较基于 Starling 与 Mpipi-GG 模拟生成的模拟集合所得到的归一化距离图表明，Starling 能准确学习到残基间距离的底层构象分布中随离子强度变化而产生的变化（图5d 及延伸数据图 8c）。此外，Starling 能定性地重现 ProTα–H1.0 ^{CTD}相互作用[41]的离子强度依赖性（延伸数据图 8d–f）。综上所述，这些结果表明，可利用星形图来研究依赖于溶液环境的 IDR 相互作用——仅需使用有限数量的锚定点并在其间进行插值计算即可。

尽管 Starling 与 Mpipi-GG 及实验数据相符，但它并不能替代严谨的生物物理测量方法；相反，实验可通过两种途径为 Starling 提供约束，从而对其进行优化并整合到 IDR 系综中。第一种途径在系综生成过程中引入引导机制，将构象异构体推向目标可观测值（参见“方法”部分；扩展数据图 9a–c 以及补充图 5）——这种方法非常适合生成远离无偏分布的系综（例如环肽），尽管引导后的系综很可能偏离经典系综。第二种途径采用贝叶斯最大熵（BME）重加权法——该方法已原生集成于 Starling 中——对系综进行重加权，使其与实验观测值相匹配[15]。BME 将 Starling 系综视为先验分布，并通过一种包含不确定性的基于似然函数的约束条件引入实验数据（例如 SAXS 和 smFRET），从而得到与实验结果更吻合的系综（图5e）。作为概念验证，我们对 Holla 等人测得的四个序列系综采用了 BME 重加权法。[34]，将实验测得的端到端距离作为 BME 目标函数中的约束条件。由此得到的模型集合不仅提高了与端到端距离的吻合度，也提高了与derived R_g的吻合度（图5f）。这些结果表明，Starling 提供了一个功能强大且易于使用的平台，可用于对 IDRs 的实验数据进行整合建模。

最后，除了严格的集合生成之外，得益于 Starling 独特的架构，该工具还支持一种新型应用——即直接基于序列进行快速、大规模的无序蛋白质生物物理集合相似性搜索（图5g）。通过使用序列编码器将序列编码到一个具有构象感知能力的潜在空间中（扩展数据图 2d），可在该嵌入空间内进行搜索，从而识别出构象相似的内在无序区域（IDRs）。作为原理验证，我们构建了一个包含所有在 UniRef50 中预测且残基数少于 380 个的 IDR（约 3500 万条序列）的数据库；随后，针对这些序列分别构建了 Starling 基于集合感知的嵌入表示，并实现了一种用于快速、大规模近似最近邻搜索的方法（详见“方法”部分）。我们通过将蛋白质 SERF（UniProt ID:O75920）在该数据库中进行查询[46]（图5g）来验证此方法的有效性。接着，我们提取了具有最相似和最不相似向量嵌入表示的候选序列，并为每个构建体构建了集合平均距离图。归一化

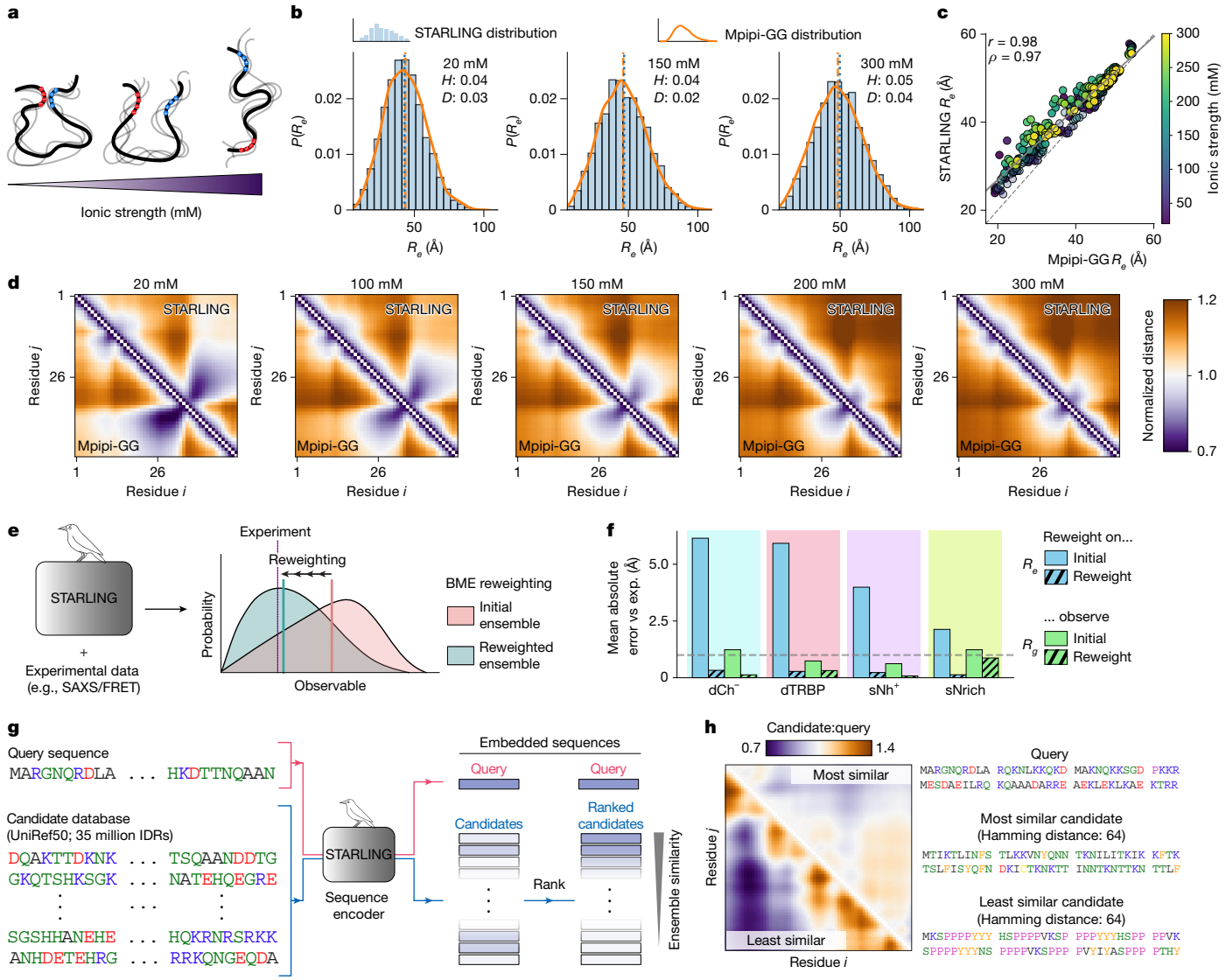

Fig. 5 | STARLING enables solution-dependent ensemble generation, Bayesian reweighting and ensemble-aware embedding for search. **a**, Schematic representation of sequence-dependent changes in conformational behaviour in response to changes in ionic strength for a blocky polyampholytic sequence. **b**, STARLING-derived ionic-strength-dependent R_e distributions for a highly charged, ionic-strength-responsive sequence. Mpipi-GG simulations shown as a kernel density estimate plot. In each subpanel, we show H and the KS statistic to quantify the difference between distributions and the effect size of the difference. A KS statistic of 0.03 means the CDFs of the two distributions differ by at most 3%. **c**, R_e correlation between the STARLING and Mpipi-GG simulations performed at various ionic strengths for the highly charged Das–Pappu sequences. Despite being trained only at three different ionic strengths, STARLING can interpolate across ionic strengths unseen during training. **d**, AFRC-normalized average inter-residue distance maps for STARLING (upper) compared with Mpipi-GG simulations (lower) across five different ionic strengths. **e**, Schematic representation of incorporating experimental data into STARLING for integrative biophysical modelling. **f**, Reweighting based on single-molecule Förster resonance energy transfer (smFRET) data for R_e improves agreement with SAXS data (R_g), highlighting how ensemble reweighting through one modality can improve agreement with others. Exp., experiment. **g**, Schematic representation of the ensemble-aware latent embedding of STARLING as used for sequence search. **h**, Heatmap of the ratio (candidate:query) of average inter-residue distance maps for the fully disordered query protein SERF *Saccharomyces cerevisiae* (UniProt ID: Q3E7B7). The upper triangle shows the top-ranked candidate; the lower triangle shows the lowest-ranked candidate, where candidates were ranked by cosine similarity of their embedding maps to the query SERF after scanning a database of IDRs from UniRef50. The colours indicate deviation from SERF: white ≈ 1.0 (similar), purple < 1.0 (more compact) and orange > 1.0 (more expanded). Serendipitously, the most and least similar candidates are equidistant from SERF in terms of sequence changes.

each of these candidates by the query protein SERF reveals that the most similar length-matched candidate has a highly similar averaged pairwise distance map (Fig. 5h). By contrast, the least similar length-matched candidate has stark differences with subregions that are both overly compact and overly expanded. Together, the STARLING sequence encoder enables IDR sequences to be represented in a high-dimensional conformationally aware space, with a broad range of potential applications, from design (Fig. 4p) to search (Fig. 5g,h) to the development of new protein representations that combine IDR-specific embeddings with general protein language embeddings.

Overall, these vignettes demonstrate the versatile range of applications that STARLING enables. Of note, the speed and ease with which STARLING generates ensembles mean that these investigations can be accomplished in hours or even minutes.

Discussion and conclusion

STARLING is a generative model that can produce coarse-grained IDR ensembles directly from protein sequences in seconds. Despite several caveats and limitations, one of the major features of STARLING

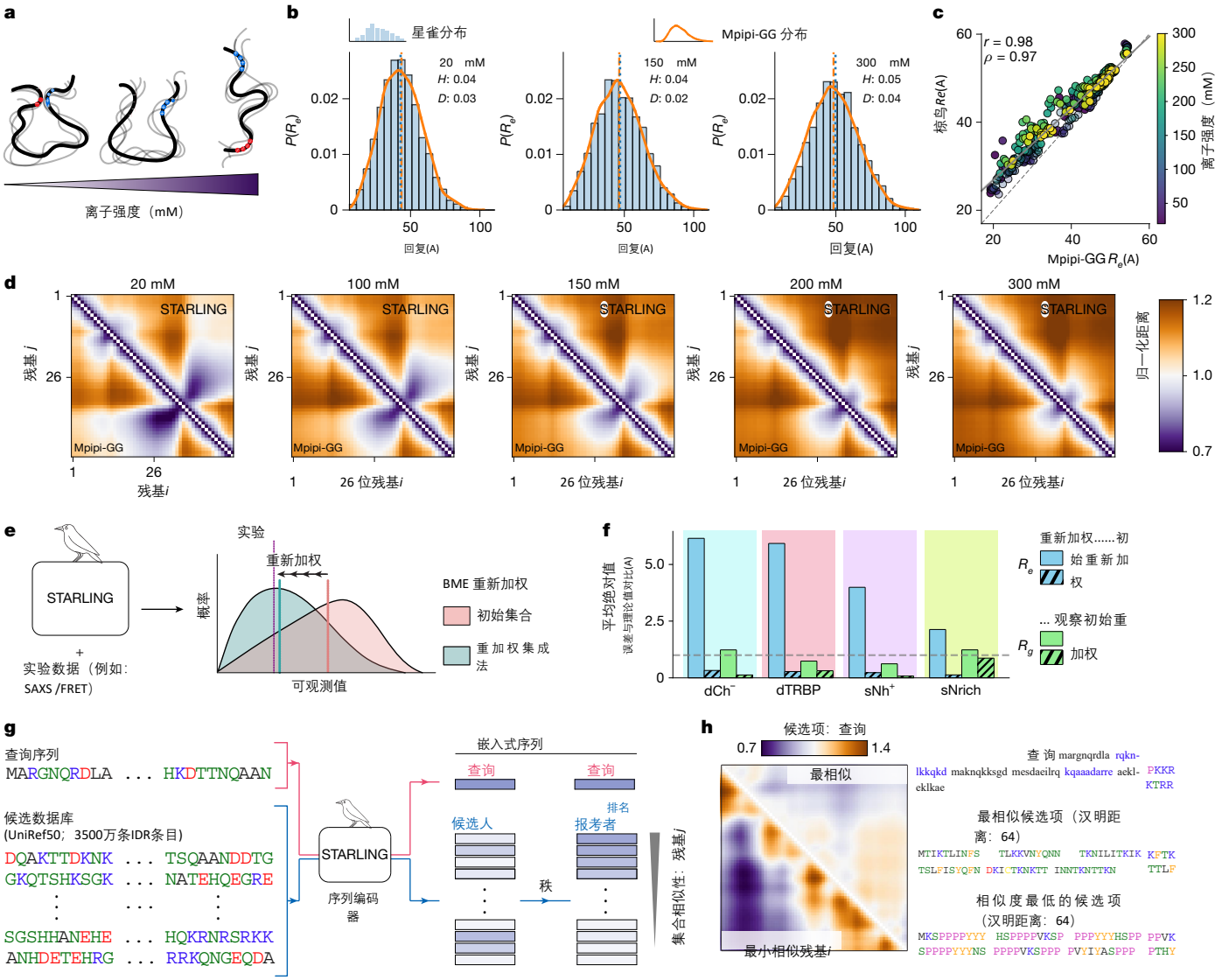

图 5 | Starling 支持基于解决方案的集成模型生成、贝叶斯重加权以及面向搜索的集成感知嵌入技术。

a，针对块状两性离子序列，展示其构象行为随离子强度变化而产生的序列依赖性变化的示意图。**b**，基于 Starling 模型构建的、具有高电荷且对离子强度响应的序列的dependent R_e分布情况；Mpipi-GG 模拟结果以核密度估计图形式呈现。在每个子图中，我们展示了 Hand 与 KS 统计量，用于量化两个分布之间的差异及其差异的效应量。KS 统计量为 0.03 表示两个分布的累积分布函数（CDF）之间的差异最多为 3%。**c**，针对高电荷 Das–Pappu 序列，在不同离子强度下进行的 Starling 与 Mpipi-GG 模拟之间的R_e相关性。尽管 Starling 模型仅在三种不同离子强度下进行过训练，但它能够对训练过程中未涉及的离子强度进行插值预测。**d**：星雀蛋白的 AFRC 归一化平均残基间距离图（上图）与 Mpipi-GG 模拟结果（下图）在五种不同离子强度条件下的对比。**e**，将实验数据整合至 Starling 用于整合性生物物理建模的示意图。**f**，基于单分子福斯特共振能量转移（smFRET）data for R_e使其与 SAXS 数据（R_g）的吻合度提高，这凸显了通过单一模态进行系综重新加权如何提升与其他模态数据的吻合度。Exp.：实验。**g**，用于序列搜索的 Starling 系综感知潜在嵌入表示法示意图。**h**，针对完全无序查询蛋白 SERF*酵母菌*（UniProt ID：Q3E7B7）的平均残基间距离图比值（候选蛋白：查询蛋白）热力图。上三角区域显示排名最高的候选蛋白；下三角区域显示排名最低的候选蛋白——这些候选蛋白是通过扫描 UniRef50 中的 IDRs 数据库后，根据其嵌入图与查询蛋白 SERF 之间的余弦相似度进行排序的。颜色表示与 SERF 的偏差：白色 ≈ 1.0（相似），紫色 < 1.0（更紧凑），橙色 > 1.0（更扩展）。巧合的是，相似度最高与最低的候选序列在序列变化方面与 SERF 的距离相等。

对这些候选序列分别与查询蛋白 SERF 进行比对后发现，长度匹配度最高的候选序列具有高度相似的平均成对距离图（图5h）；相比之下，长度匹配度最低的候选序列在部分亚区域上存在显著差异——这些亚区域既呈现过度紧凑状态，又呈现过度扩张状态。综上所述，Starling 序列编码器使得 IDR 序列能够被表示在高维的、具有构象感知能力的空间中，从而具有广泛的应用前景，涵盖从设计（图4p）到搜索（图5g，h）乃至开发新型蛋白质表征方法（将 IDR 特异性嵌入表示与通用蛋白质语言嵌入表示相结合）等多个领域。

总体而言，这些示例充分展示了 Starling 所支持的多样化应用场景。值得注意的是，Starling 生成集成模型的速度与便捷性意味着这些研究可在数小时甚至数分钟内完成。

讨论与结论

Starling 是一款生成式模型，能够仅需数秒即可直接从蛋白质序列生成粗粒度的 IDR 等效集。尽管该模型存在若干注意事项与局限性，但其核心优势之一在于它显著降低了获取依赖于序列的 IDR 构象预测的门

is that it drastically reduces the barrier to obtaining conformational predictions of sequence-dependent IDR behaviour. Although IDRs are often ignored or visualized as AlphaFold 'orange spaghetti', STARLING makes it straightforwards for anyone to obtain realistic coarse-grained ensembles. Rather than 'the end' of an investigation into IDR sequence-ensemble behaviour, we view STARLING as a beginning, making it straightforwards to develop hypotheses as to how the sequence of an IDR may determine its conformational ensemble and/or how it may influence interactions with other IDRs.

A key feature of STARLING is its ability to generate conformational ensembles rather than just predict ensemble-average values. Although ensemble average properties (for example, average R_g or average R_e) can be informative[20,21,47], they do not enable a 'structural' investigation into the determinants of a given behaviour. With a conformational ensemble, the specific sets of residues responsible for an observed behaviour can be directly identified. Moreover, substantial effort has been placed into developing methods to back-calculate experimental observables from simulation trajectories to interpret experimental data[15]. STARLING ensembles are amenable to these analyses, allowing comparison with a range of experimental results. However, we note that all comparisons between STARLING predictions and experimental data rely on computational tools to predict observables from protein structures which can introduce their own approximations. As such, agreement with experiment reflects both STARLING and the accuracy of these tools and their assumptions.

Although STARLING generates ensembles in seconds, it has key limitations (see also Methods). Because STARLING was trained with simulations from a one-bead-per-residue coarse-grained force fields that do not include secondary structure, it cannot predict secondary structure elements. In addition, it cannot incorporate flanking folded domains. However, we view the inability to incorporate folded domains as a potential 'feature'; discrepancies between predictions of isolated IDRs and experimental data (where flanking folded domains are present) may reveal biophysically meaningful interactions.

STARLING is also poorly poised to capture many environmental effects or post-translational modifications. IDRs can act as environmental sensors, with their conformational behaviour and interactions being tuned by their local chemical environment (salt, temperature and osmolytes, and so on)[48]. Furthermore, charge regulation can 'rewrite' the charge state of a protein, up-shifting or down-shifting intrinsic pK_a values[49]. Because STARLING was trained using simulations performed under 'standard conditions' (infinite dilution, 298 K), it does not enable explicit incorporation of many of these physicochemical modulators. Moreover, post-translational modifications and non-canonical amino acids are currently not accounted for. Looking ahead, generalization to environmental sensitivity or non-canonical residues remains challenging, in part due to limited benchmarking data. Although our ionic strength conditional results are encouraging (Supplementary Fig. 6) — at least conceptually, see Methods for Debye–Hückel limitations — broader environmental response modelling will require expanded training sets and dedicated experimental benchmarks to establish reliability (see Methods).

Unlike conventional molecular dynamics, STARLING generates conformational ensembles without time-resolved trajectories. Although molecular dynamics provides explicit dynamics, it often struggles to accrue independent samples for long chains that reconfigure slowly, typically requiring advanced kinetic frameworks for robust rate estimates. By contrast, STARLING produces decorrelated samples through independent stochastic denoising, maintaining high speed regardless of sequence length. Consequently, STARLING avoids many of the challenges in generating ensembles for longer sequences seen in molecular dynamics. Furthermore, we suggest that STARLING-generated conformers could be used to seed large numbers of shorter molecular dynamics simulations, enabling more reliable kinetic analyses.

Several recent sequence-to-ensemble approaches have emerged, beginning with Taneja and Lasker's two-stage DDPM, which established the feasibility of learning these mappings for heterogeneous sequences[50]. More recently, idpSAM utilized a latent-space DDPM trained on all-atom simulations, although its predictive power is generally limited to sequences under 60 residues[51] (Extended Data Fig. 9d). Other methods, such as IDPFold and IDPForge, integrate protein language models with experimental or molecular dynamics data[52,53] (Extended Data Fig. 9e–r and Supplementary Fig. 7). Unlike the coarse-grained approach of STARLING, these methods (and others) provide all-atom molecular detail and, in some cases, the ability to include folded domains, albeit at the expense of runtime performance, hardware requirements, and potentially accuracy with respect to global dimensions and long-range IDR-mediated interactions[52–56]. Other emerging strategies include combining deep learning with Bayesian reweighting[10,54,57] or predicting Hamiltonians rather than ensembles[58]. Overall, however, these tools often fall short of the balance of accuracy, performance and accessibility of STARLING. Furthermore, the ability of STARLING to reweight against experimental data (Extended Data Fig. 10) and encode and search ensembles offers unique research opportunities when integrated with bioinformatics pipelines or protein language models.

STARLING enables systematic, library-scale design of IDRs to test the 'ensemble–function relationship': the hypothesis that IDR function depends directly on conformational behaviour[1,2,26]. Historically, designing sequences with targeted ensembles was challenging due to the astronomical sequence space and the low-throughput nature of experimental or computational ensemble characterization. Although recent exciting ensemble-based design methods have emerged, they are not practical for thousands of sequences[59,60]. By providing a direct optimization handle within a biophysically meaningful latent space, STARLING (integrated with GOOSE) transforms this underconstrained search into a tractable design problem, allowing for the creation of large-scale libraries to test sequence–ensemble–function hypotheses[26].

Finally, the ensemble-aware embedding of STARLING enables sequence search and design based on conformational behaviour rather than primary structure. This provides a biophysically grounded alternative to BLAST, which often fails with poorly conserved or low-complexity IDRs. By facilitating the homology-independent discovery of 'biophysical look-alikes', this approach enables hypothesis transfer to uncharacterized regions and identifies natural templates for inverse design, even when traditional alignments and composition heuristics fail.

In conclusion, we hope STARLING serves the disordered protein community as a highly accessible tool for biophysical investigation. Unlike many deep learning tools, STARLING is designed for ease of use; it can be installed via pip and run with a single command. By enabling 'on-the-fly' predictions on commodity hardware, STARLING fills a unique gap in IDR characterization. Although it does not replace existing technologies, it provides a practical way to explore conformational behaviour easily, quickly, accurately and at scale.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-026-10141-2.

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槛。尽管内在无序区域（IDRs）常被忽视或被形象地描述为 AlphaFold 的“橙色意大利面状结构”，但 Starling 使得任何研究者都能轻松获取真实的粗粒度等效集。我们并不将 Starling 视为对 IDR 序列-等效集行为研究的“终点”，而是将其视为一个起点——这使得研究人员能够 straightforward 地提出相关假说：即 IDR 的序列如何决定其构象等效集，以及/或该序列如何影响其与其它 IDR 之间的相互作用。

Starling 的一个关键特性在于其能够生成构象系综，而非仅仅预测系综平均值。尽管系综平均性质（例如average R_g或average R_e）可提供有价值的信息[20,21,47]，但它们无法对特定行为的决定因素进行“结构”层面的探究。通过构象系综，可直接识别出导致观测到的特定行为的特定残基集合。此外，学界已投入大量精力开发相关方法，通过模拟轨迹反向计算实验可观测量以解释实验数据[15]。Starling 的系综适用于此类分析，从而可与一系列实验结果进行比较。然而，我们注意到，所有 Starling 预测结果与实验数据之间的比较均依赖于利用蛋白质结构预测可观测量的计算工具，而这些工具本身可能存在一定的近似性。因此，与实验结果的一致性既反映了 Starling 的性能，也反映了这些工具及其假设的准确性。

尽管 Starling 能在数秒内生成结构集合，但其存在关键局限性（另见“方法”部分）。由于 Starling 是基于每个残基对应一个珠子的粗粒化力场模拟数据进行训练的（该力场不包含二级结构信息），因此它无法预测二级结构元件；此外，它也无法整合侧翼折叠结构域。然而，我们将这种无法整合折叠结构域的能力视为一种潜在的“特征”；孤立的IDR预测结果与实验数据（存在侧翼折叠结构域时）之间的差异，或许能揭示具有生物物理意义的相互作用。

Starling 模型在捕捉多种环境效应或翻译后修饰方面也存在局限性。IDRscan 作为环境传感器，其构象行为与相互作用受其局部化学环境（如盐浓度、温度、渗透调节剂等）的调控[48]。此外，电荷调控可“重写”蛋白质的电荷状态，从而上调或下调其固有的pK_a值[49]。由于 Starling 是基于在“标准条件”（无限稀释、298 K）下进行的模拟实验训练的，因此无法显式地纳入上述诸多物理化学调节因子；此外，目前尚未考虑翻译后修饰及非经典氨基酸的影响。展望未来，将模型泛化至环境敏感性或非经典残基识别仍具挑战性，这部分归因于基准测试数据有限。尽管我们针对离子强度条件下的实验结果令人鼓舞（补充图）。6) —至少在概念上，参见“德拜-休克尔极限”相关方法 — 更广泛的环境响应建模需采用扩展的训练集及专门的实验基准测试以验证其可靠性（参见“方法”部分）。

与传统的分子动力学方法不同，Starling 可生成构象系综，而无需依赖时间分辨轨迹。尽管分子动力学能够提供显式动力学信息，但对于构象变化缓慢的长链分子体系而言，该方法往往难以采集到独立的采样数据——通常需要借助先进的动力学框架才能获得可靠的速率估计值。相比之下，Starling 通过独立的随机去噪技术生成去相关采样数据，无论序列长度如何均能保持高速计算效率。因此，Starling 避免了分子动力学方法在生成长序列系综时所面临的诸多挑战。此外，我们认为，通过 Starling 生成的构象异构体可作为大量短序列分子动力学模拟的初始构型，从而实现更可靠的动力学分析。

近年来，多种序列到集成的建模方法相继出现；其中，Taneja与Lasker提出的两阶段 DDPM 方法首次证实了为异质性序列学习此类映射关系的可行性[50]。近期，idpSAM采用了基于全原子模拟数据训练的潜在空间 DDPM 方法，但其预测能力通常仅适用于长度小于60个残基的序列[51]（扩展数据图9d）。其他方法（如IDPFold和IDPForge）则将蛋白质语言模型与实验数据或分子动力学数据相结合[52,53]（扩展数据图9e–r及补充图7）。与Starling采用的粗粒度方法不同，这些方法（及其他方法）能够提供全原子级的分子细节，在某些情况下还可纳入折叠结构域，但此举往往需要牺牲运行时性能、硬件资源需求，以及在全局维度和长程 IDR 介导的相互作用方面的预测精度[52–56]。其他新兴策略包括将深度学习与贝叶斯重加权技术相结合[10,54,57]，或直接预测哈密顿量而非系综[58]。然而总体而言，这些工具在准确度、性能和易用性方面往往难以媲美 Starling。此外，Starling 根据实验数据进行重加权（扩展数据图 10）以及对系综进行编码与搜索的能力，在与生物信息学流程或蛋白质语言模型相结合时，可提供独特的研究机遇。

Starling支持系统化、库规模的IDR设计，用于验证“集合–函数关系”：即IDR功能直接取决于构象行为的假说[1,2,26]。从历史上看，由于序列空间规模巨大以及实验或计算层面的集合表征方法通量较低，采用靶向集合策略进行序列设计颇具挑战性。尽管近年来涌现出一些令人振奋的基于集合的设计方法，但这些方法对于数千条序列而言并不实用[59,60]。通过在具有生物物理意义的潜在空间中提供直接的优化手段，Starling（与Goose集成）将这种受限搜索问题转化为一个可处理的设计问题，从而能够构建大规模序列库以验证序列–集合–函数假说[26]。

最终，基于椋鸟蛋白的集成感知嵌入方法使得序列搜索与设计能够基于构象行为而非一级结构进行。这为 Blast 方法提供了一种基于生物物理学原理的替代方案——后者在处理保守性较低或复杂度较低的 IDR 区域时往往难以奏效。通过促进对“生物物理相似结构”进行不依赖于同源性的发现，该方法实现了将假设迁移至尚未表征的区域，并为逆向设计识别出天然模板，即使在传统序列比对与组成启发式方法失效的情况下亦然。

总之，我们希望 Starling 能作为一款操作简便的工具，服务于蛋白质结构解析领域中的生物物理研究工作。与许多深度学习工具不同，Starling 专为易用性而设计——用户可通过 `pip` 安装该工具，仅需一条命令即可启动运行。通过支持在常规硬件上进行“实时”预测，Starling 填补了 IDR 表征领域中的一个独特空白。尽管它无法替代现有的各类技术，但它提供了一种实用的方法，使研究人员能够轻松、快速、准确且大规模地探究蛋白质的构象行为。

在线内容

所有方法、补充参考文献、《Nature Portfolio》报道摘要、原始数据、扩展数据、补充信息、致谢、同行评审信息；作者贡献与利益冲突详情；以及数据与代码可用性声明，均可在https://doi.org/10.1038/s41586-026-10141-2获取。

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Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The code and data associated with the figures in this article are available on GitHub (https://github.com/holehouse-lab/supportingdata/tree/master/2026/starling_2026). The Zenodo repository for UniRef50 IDR sequence embeddings is available[62] (https://doi.org/10.5281/zenodo.17342150). Any data missing are available from the corresponding author.

Code availability

STARLING is fully open source and available on GitHub (https://github.com/idptools/starling/). The documentation for STARLING is available (https://idptools-starling.readthedocs.io). STARLING is also provided as a Colab notebook (https://github.com/idptools/idpcolab/blob/main/STARLING/STARLING_demo.ipynb).

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Additional information

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报告摘要

有关研究设计的更多信息，请参阅本文所链接的《Nature Portfolio Reporting Summary》。

数据可用性

本文中各图表所对应的代码与数据均可在 GitHub 上获取（https://github.com/holehouse-lab/supportingdata/tree/master/2026/starling_2026）。UniRef50 IDR 序列嵌入式数据的 Zenodo 仓库可访问[62]（https://doi.org/10.5281/zenodo.17342150）。任何缺失的数据均可向通讯作者索取。

代码可用性

starling 是一款完全开源的工具，可在 GitHub 上获取（https://github.com/idptools/starling/）。starling 的文档可查阅（https://idptools-starling.readthedocs.io）。starling 还提供为 Colab 笔记本形式的版本（https://github.com/idptools/idpcolab/blob/main/starling/STARLING_demo.ipynb）。

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附加信息

补充信息：在线版本包含补充材料，可访问https://doi.org/10.1038/s41586-026-10141-2获取。

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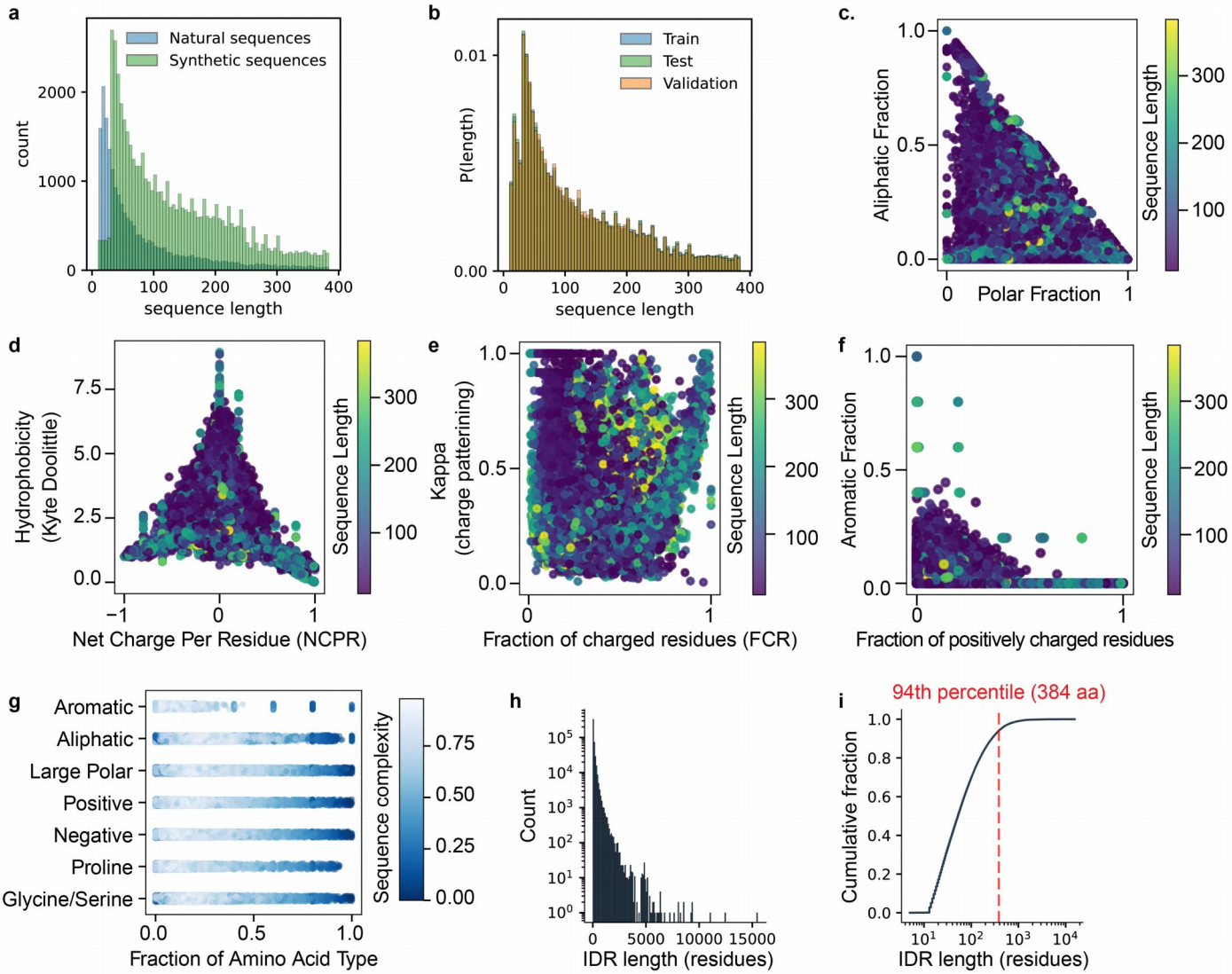
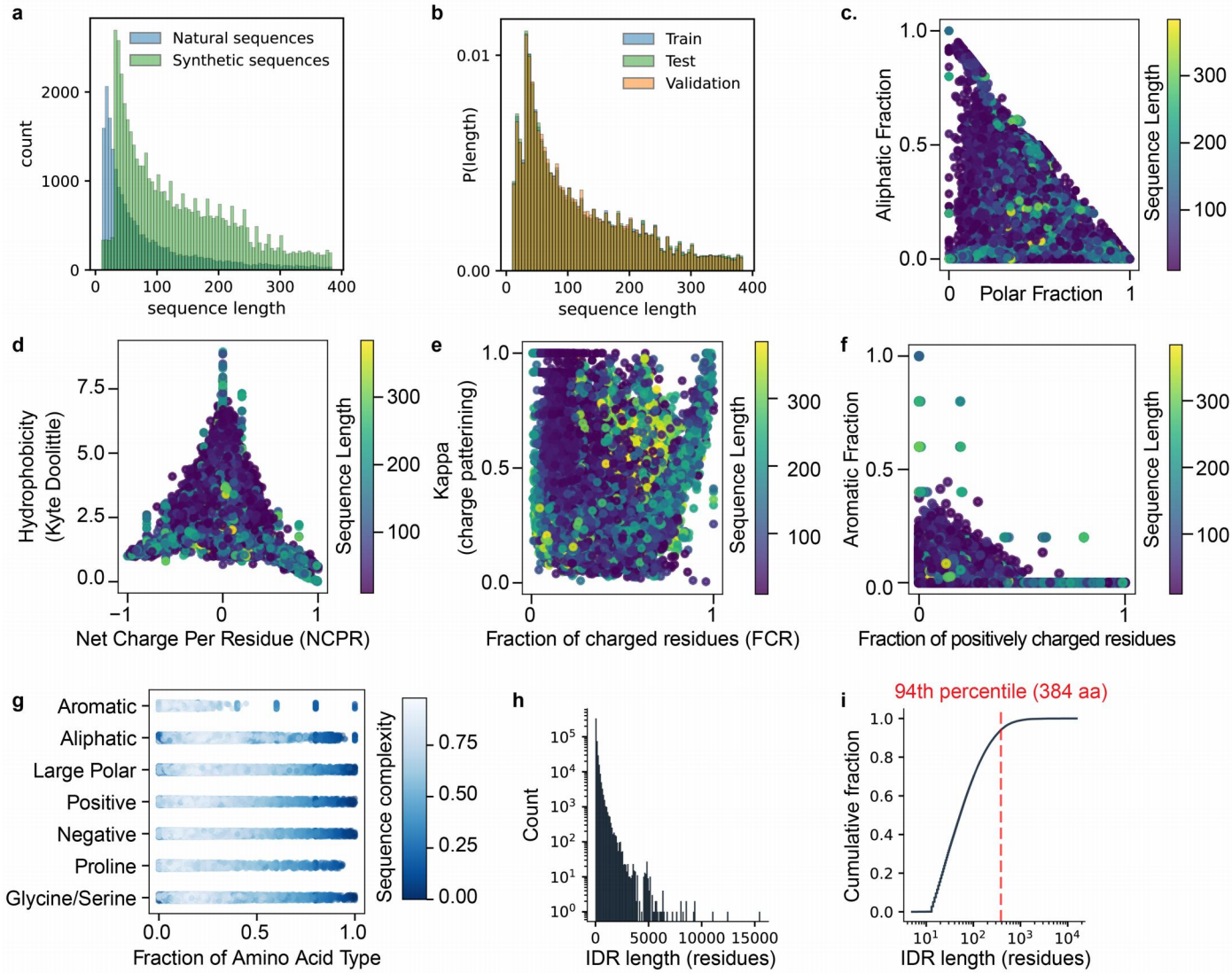
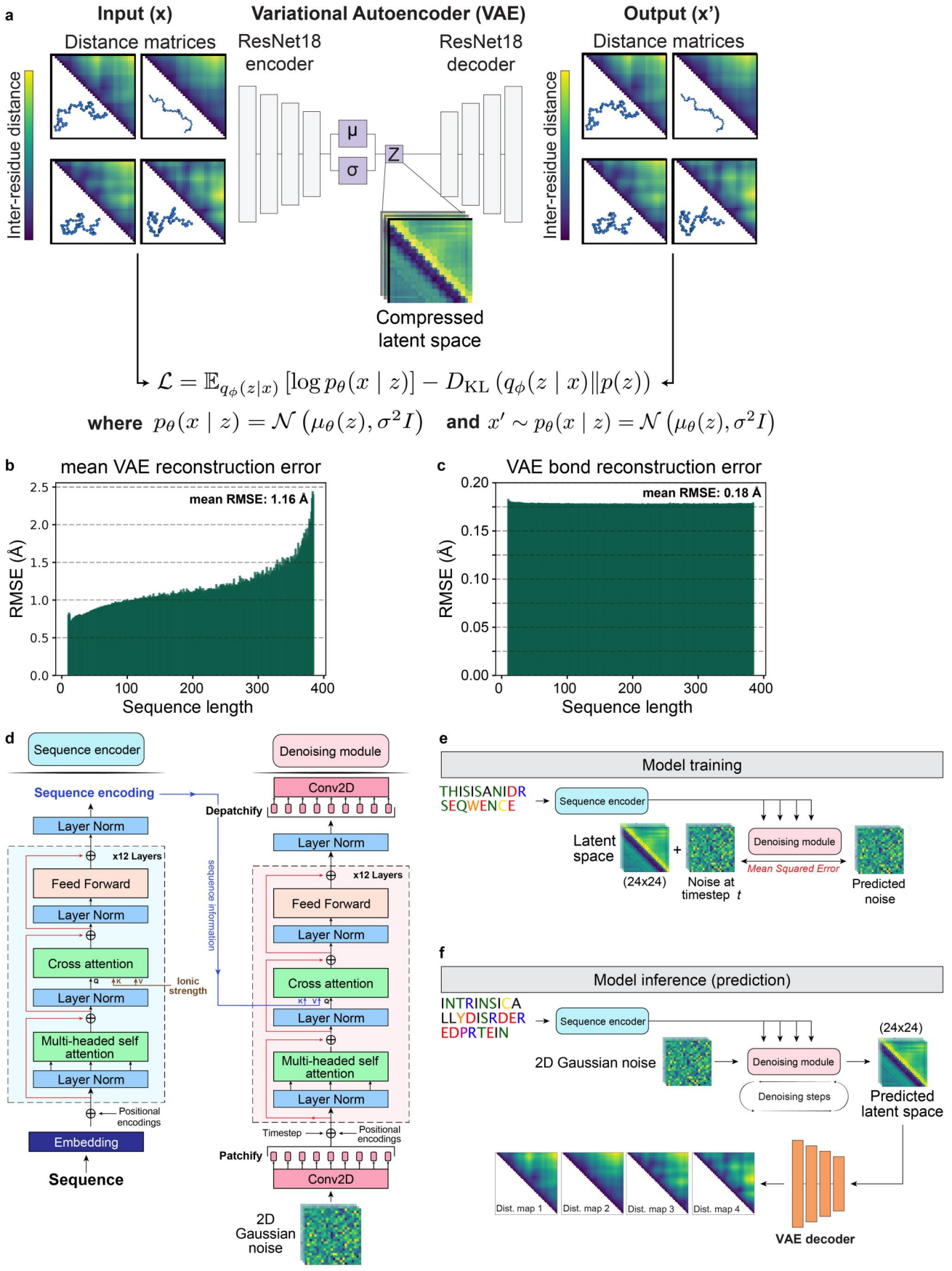


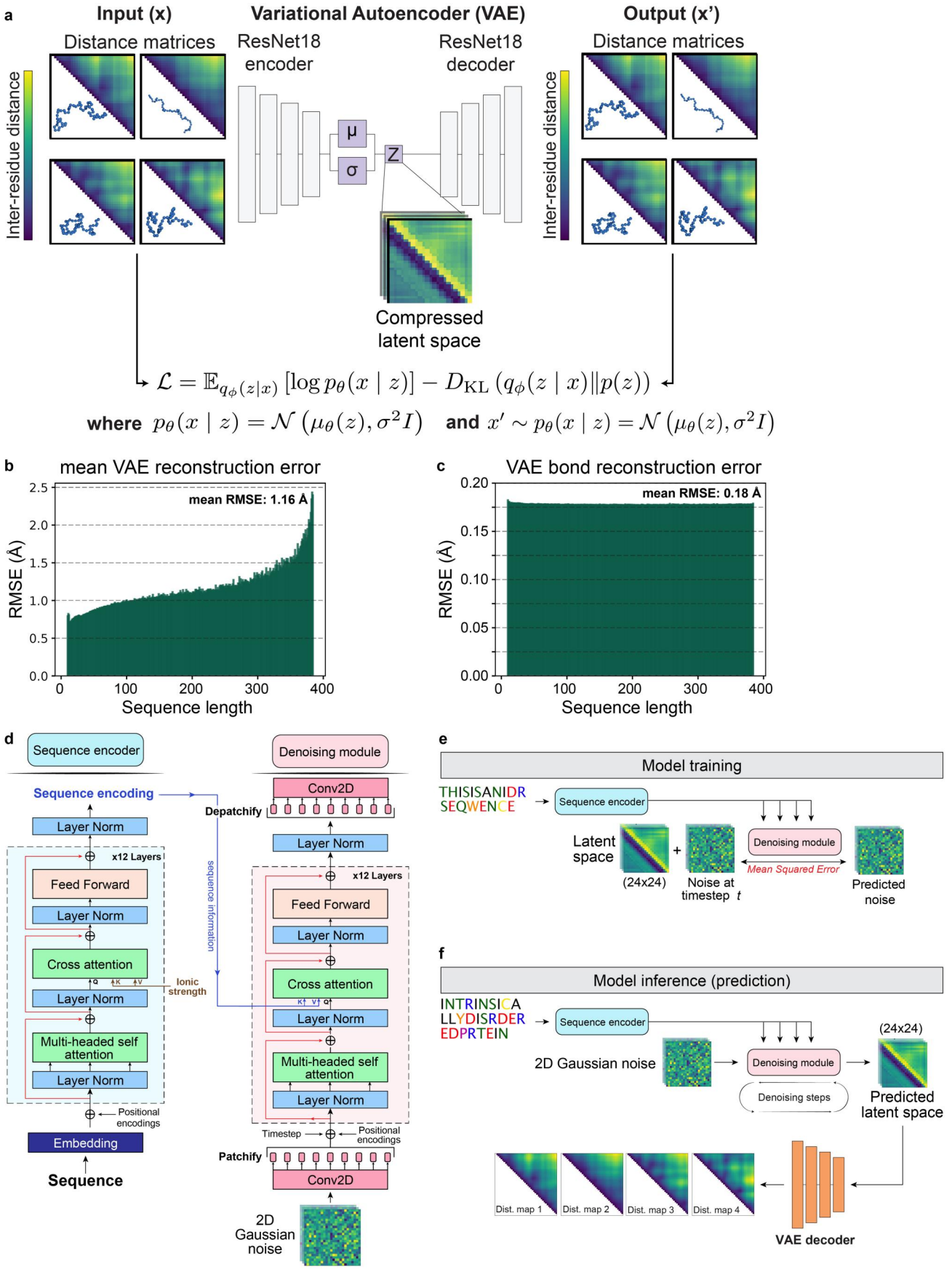
Extended Data Fig. 1 | Sequences used in STARLING training, test, and validation. a, Our dataset of 70,563 consists of both natural sequences (N = 20,349) and synthetic sequences designed by varying sequence features known to affect IDR ensembles (N = 50,214). **b**, The sequences were clustered and split into train (N = 49,423), validation (N = 10,703), and test (N = 10,437) sets while maintaining the distribution of sequence lengths across the splits consistent (right). **c**, Residues used for STARLING training, test, and validation, represented in terms of aliphatic vs. polar content. **d**, Residues used for STARLING training, test, and validation, represented in terms of net charge vs. hydrophobicity. **e**, Residues used for STARLING training, test, and validation, represented in terms of kappa (charge patterning) vs. fraction of charged residues. **f**, Residues used for STARLING training, test, and validation, represented in terms of aromatic vs. positively charged residues. **g**, Overall sequence complexity as a function of amino acid content. We note that the data in panels C-G mirror our prior quantification of sequences[20], albeit with an expanded dataset. **h**, Histogram of the length distribution of naturally occurring IDRs across 12 model organisms. **i**, A cumulative density function of the length distribution is presented in **a**. 94% of the IDRs across common model organisms fall between 0 and 384 residues (the length cutoff of STARLING).



扩展数据图 1 | 鸟类训练、测试及验证所用序列。a、我们的数据集包含 70,563 条序列，其中既包含自然序列 (N = 20,349) 以及通过改变已知影响 IDR 簇的序列特征而设计的合成序列（N = 50,214）。**b**：这些序列经过聚类后分为训练集（N = 49,423）、验证集（N = 10,703）和测试集（N = 10,437），同时确保各子集间序列长度的分布保持一致（右图）。**c**：用于 Starling 训练、测试和验证的残基，按脂肪族与极性残基含量表示。**d**：用于 Starling 训练、测试和验证的残基，按净电荷与疏水性表示。**e**：用于 Starling 训练、测试和验证的残基，按 κ 值（电荷分布模式）与带电残基占比表示。**f**：用于 Starling 训练、测试和验证的残基，按芳香族残基与带正电残基占比表示。**g**：整体序列复杂度随氨基酸含量变化的函数关系。我们注意到，图 C–G 中的数据与我们此前对序列的定量分析结果[20]相一致，尽管所用数据集规模更大。**h**，12种模式生物中天然存在的IDR序列长度分布的直方图。**i**，图a展示了该长度分布的累积分布函数；在常见模式生物中，94%的IDR序列长度介于0至384个残基之间（即“星雀”模型设定的长度截断值）。



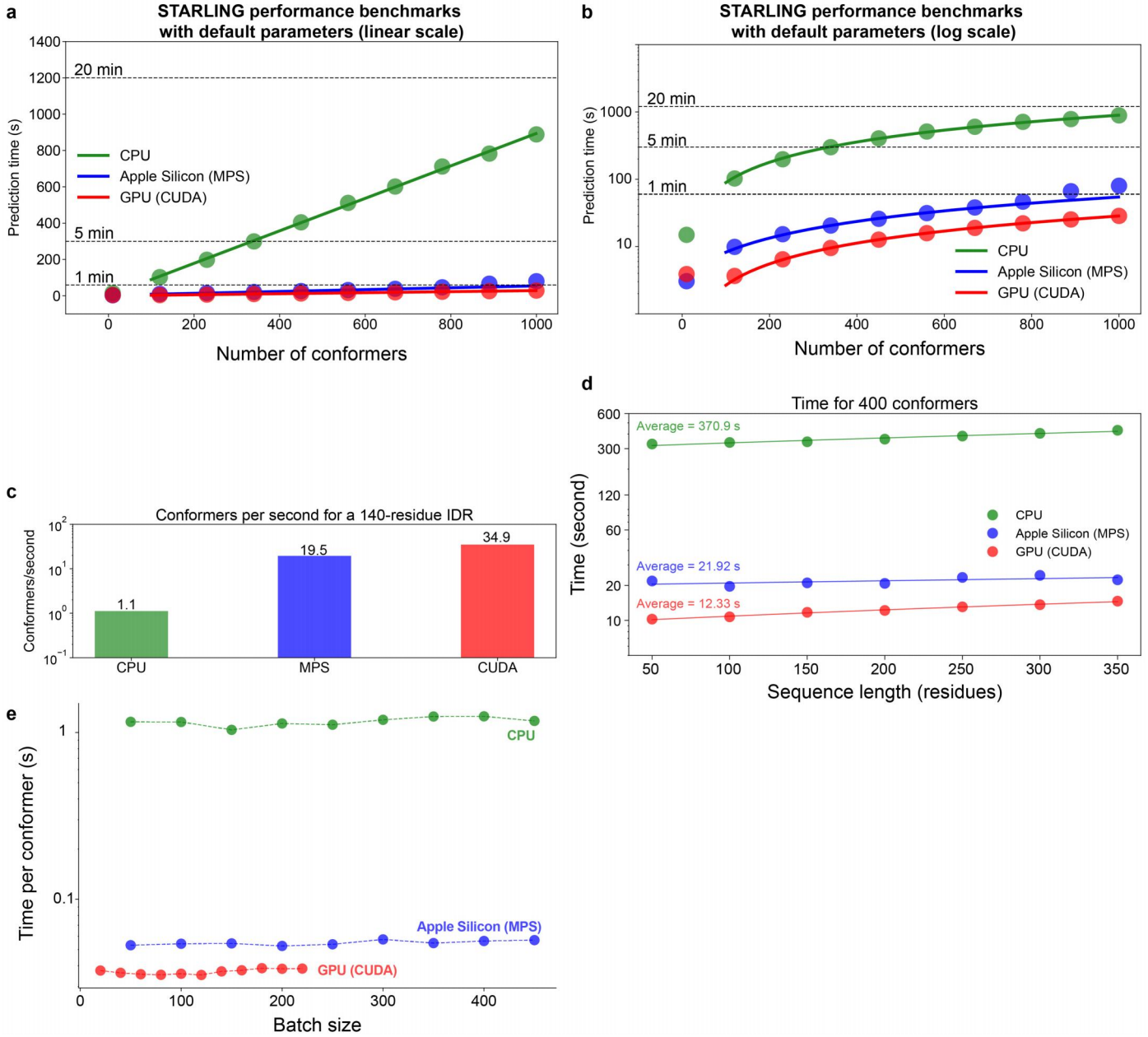
Extended Data Fig. 2 | See next page for caption.



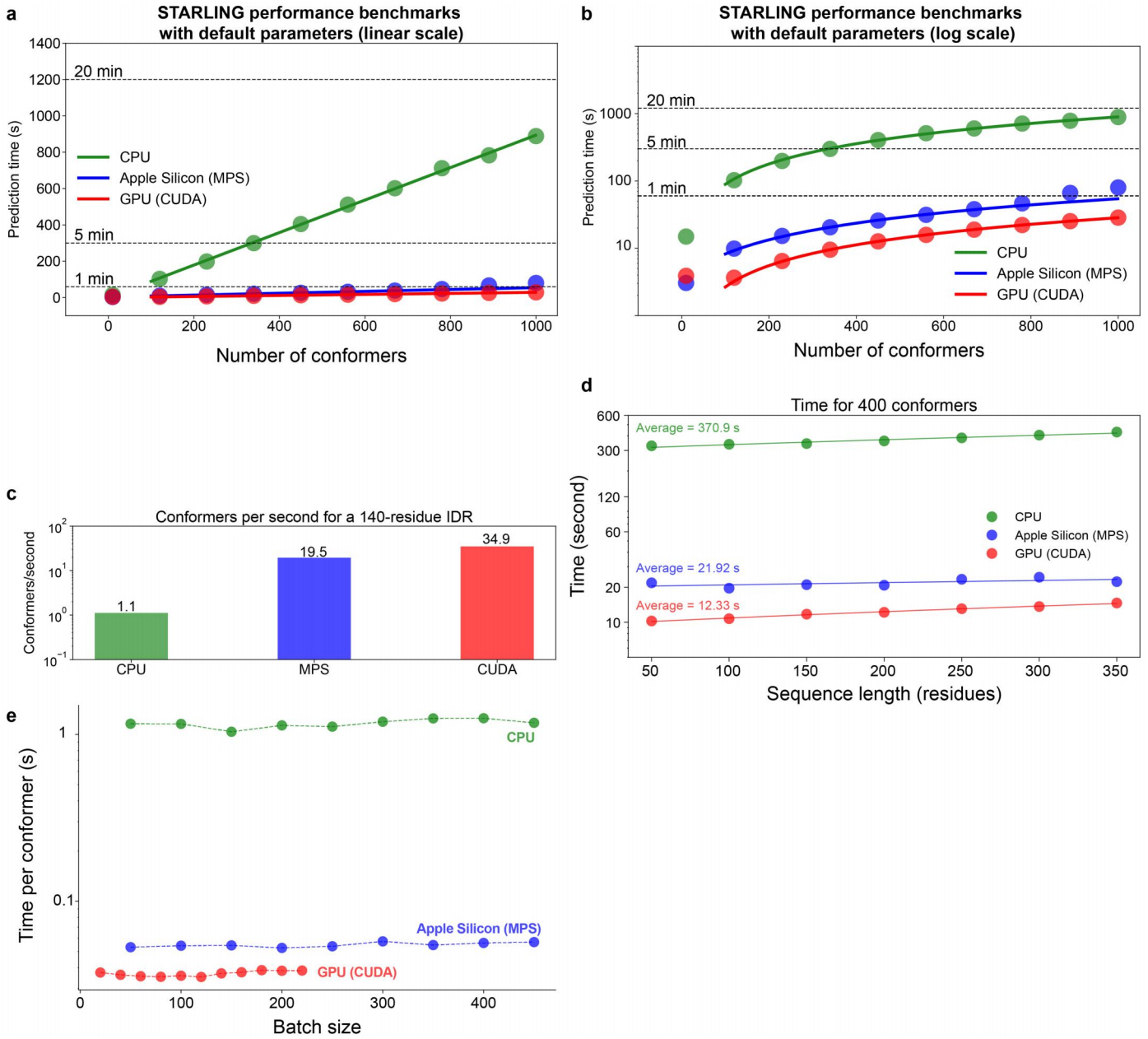
扩展数据图 2 | 图注请参阅下一页。

Extended Data Fig. 2 | VAE overview, performance, and overall detailed model architecture. a, Schematic representation of the variational autoencoder (VAE). The input to our variational autoencoder is a collection of 2D inter-residue distance maps that undergo feature extraction and spatial compression through a ResNet18 encoder, which learns the μ and σ parameters of a normal distribution that parameterizes the latent features. The VAE is trained using an evidence-based lower bound (ELBO) loss that comprises a reconstruction term and a Kullback–Leibler (KL) divergence term. **b**, The variational autoencoder (VAE) is evaluated on a held-out test set of distance maps (N = 2,422,839). The VAE encoder spatially compresses these distance maps, which the VAE decoder then reconstructs from their compressed representations. The root-mean-square error (RMSE) is computed between the input and output distance maps. The trained VAE shows a low mean RMSE across the upper triangle of each distance map (1.16 Å). **c**, Accurate bond reconstruction is essential for generating physically realistic conformations, and our model achieves a very low mean RMSE (0.18 Å). **d**, Extended schematic of the STARLING discrete-time denoising diffusion probabilistic model (DDPM). The model begins with 2D Gaussian noise (1 channel, 24×24 latent space) and is projected to 64 channels via a convolutional upsampling layer. Spatial features are then patchified into 3×3 regions (8 total) and passed through a 12-layer vision transformer (ViT). The ViT output is stitched back (depatchified) and reduced back to 1 channel by a convolutional downsampling layer. To generate sequence-specific distance maps, the ViT is conditioned on sequence features extracted by a transformer-based encoder. The input sequence is ordinally encoded, embedded into high-dimensional vectors, followed by the addition of sinusoidal positional embeddings, and processed by a 12-layer transformer. Ionic strength information is additionally incorporated through cross-attention, yielding environment-specific sequence features that are fed into the denoising module's cross-attention layers. **e**, Model overview as set up during training. **f**, Model overview as used for inference.

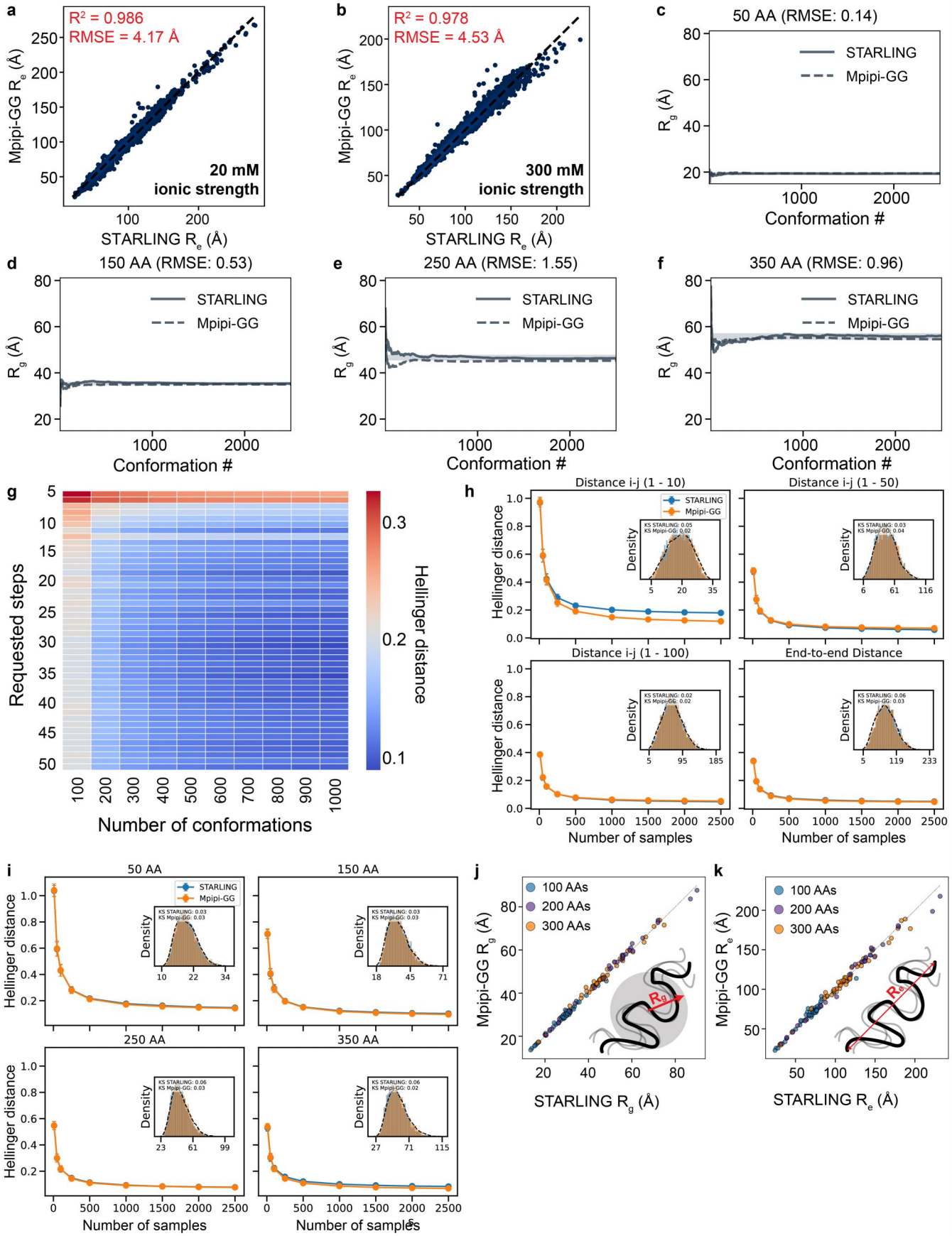
扩展数据图 2 | VAE 的总体概览、性能及整体详细模型架构。a，变分自编码器（VAE）的示意图。我们所使用的变分自编码器的输入是一组二维残基间距离图；这些距离图经过 ResNet18 编码器进行特征提取和空间压缩，该编码器学习描述潜在特征的正态分布的 μ 和 σ 参数。该 VAE 采用基于证据的下界（ELBO）损失函数进行训练，该损失函数由重构项和一个 Kullback–Leibler（KL）散度项组成。**b**，变分自编码器（VAE）在独立测试集（包含 2,422,839 张距离图）上进行评估。 VAE 编码器对这些距离图进行空间压缩，随后 VAE 解码器根据其压缩表示对这些距离图进行重构。计算输入与输出距离图之间的均方根误差（RMSE）。经过训练的 VAE 在每张距离图的上三角区域均显示出较低的平均 RMSE 值（1.16 Å）。**c**：精确的键重建对于生成符合物理规律的构象至关重要，而我们的模型实现了极低的平均 RMSE 值（0.18 Å）。**d**：星雀离散时间去噪扩散概率模型（DDPM）的扩展示意图。该模型始于二维高斯噪声（1个通道，24×24潜在空间），随后通过一个卷积上采样层投影至64个通道；空间特征随后被分割为3×3的区域（共8个区域），并输入至一个12层的视觉Transformer（ViT）中；ViT的输出经回拼接（去分割）后，通过一个卷积下采样层还原为1个通道；为生成序列特异性距离图，ViT基于通过基于Transformer的编码器提取的序列特征进行条件化处理；输入序列经过序数编码、嵌入为高维向量，随后添加正弦位置嵌入，并由一个12层的Transformer进行处理；离子强度信息通过交叉注意力机制引入，从而生成针对特定环境的序列特征，这些特征随后输入到去噪模块的交叉注意力层中。**e**，训练期间配置的模型概览；**f**，用于推理的模型概览。



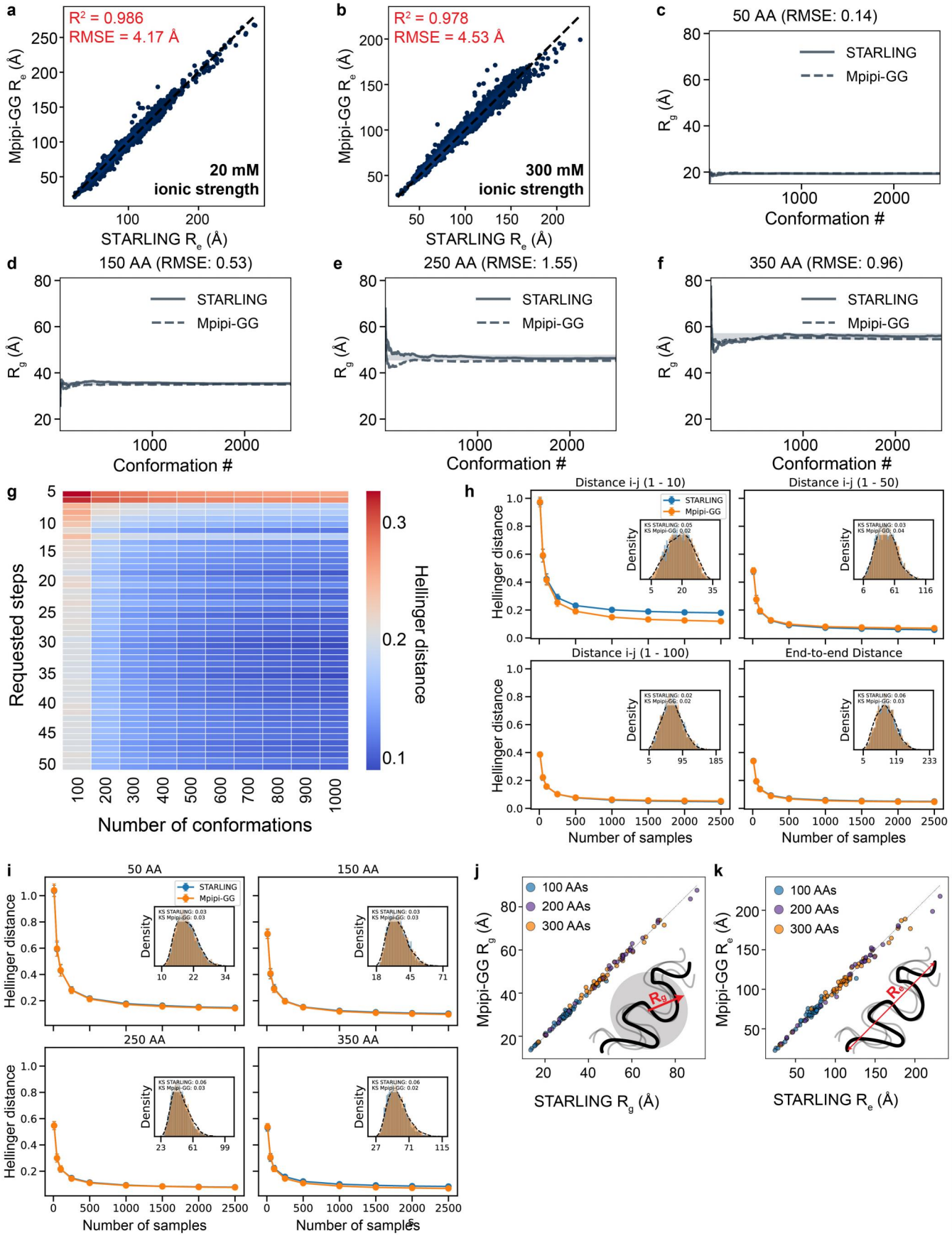
Extended Data Fig. 3 | Starling performance comparisons. a,**b**, STARLING benchmarking on different types of hardware in linear (**a**) and log (**b**) space. Runtime depends linearly on the number of conformers (although sequence length has almost no impact on runtime). Tested here using a 140-residue sequence. On GPUs and Apple Silicon (facilitated by CUDA and MPS, respectively), STARLING is highly performant, offering ensembles sufficient for detailed biophysical investigation in 10-15 s (GPU) or 30-60 s (MPS). The specific hardware tested here includes the Nvidia A4000 (GPU), MacBook Pro M3 Max CPU (MPS), and Intel(R) Xeon(R) Silver 4210 R CPU @ 2.40 GHz (CPU). We note that even on a five-year bottom-of-the-line GPU (GTX-1660 from 2019), ensemble generation takes ~50 s for 400 conformers. **c**, Number of conformers generated per second on different hardware types. The specific hardware tested here includes the Nvidia A4000 (GPU), MacBook Pro M3 Max CPU (MPS), and Intel(R) Xeon(R) Silver 4210 R CPU @ 2.40 GHz (CPU). **d**, STARLING ensemble predictions exhibit almost no runtime or memory scaling with respect to sequence length. On a GPU, generating 400 confomers takes ~10 s for a 50-residue IDR and ~14 s for a 350-residue IDR. **e**, STARLING enables users to select a batch size for concurrent ensemble prediction. A larger batch size requires more GPU-accessible memory. However, we have implemented STARLING in a manner that increasing the batch size does not materially affect the overall per-conformer prediction time. This means even GPUs with modest memory resources can work with STARLING-based predictions by selecting a reduced batch size.



扩展数据图 3 | Starling 性能对比。a、b：Starling 在不同硬件平台上的基准测试结果，分别在线性（**a**）和对数（**b**）空间下进行。运行时间与构象体数量呈线性关系（尽管序列长度对运行时间几乎没有影响）。本实验使用包含 140 个残基的序列进行测试。在 GPU 和 Apple Silicon 平台上（分别由 CUDA 和 MPS 支持），Starling 具有极高的性能，其生成的构象体集合足以在 10–15 秒（GPU）或 30–60 秒（MPS）内完成详细的生物物理研究。本次测试的具体硬件包括 Nvidia A4000（GPU）、MacBook Pro M3 Max CPU（MPS）以及 Intel® Xeon® Silver 4210 R CPU @ 2.40 GHz（CPU）。需注意的是，即使在一台使用年限达五年的低端 GPU（2019 年款 GTX -1660）上，生成 400 个构象体也需约 50 秒。**c：**不同硬件类型下每秒生成的构象体数量。本次测试的具体硬件包括 Nvidia A4000（GPU）、MacBook Pro M3 Max CPU（MPS）以及 Intel® Xeon® Silver 4210 R CPU @ 2.40 GHz（CPU）。**d：**Starling 集成预测算法在运行时或内存占用方面，几乎不受序列长度的影响。在 GPU 上，生成 400 个构象需约 10 秒（针对 50 个残基的 IDR）或约 14 秒（针对 350 个残基的 IDR）。**e：**Starling 允许用户为并行集成预测设置批处理大小。较大的批处理大小需要更多可由 GPU 访问的内存；然而，我们对 Starling 的实现方式确保了增大批处理大小不会对每个构象的总体预测时间产生实质性影响。这意味着，即使内存资源有限的 GPU 也可通过选择较小的批处理大小来使用基于 Starling 的预测方法。



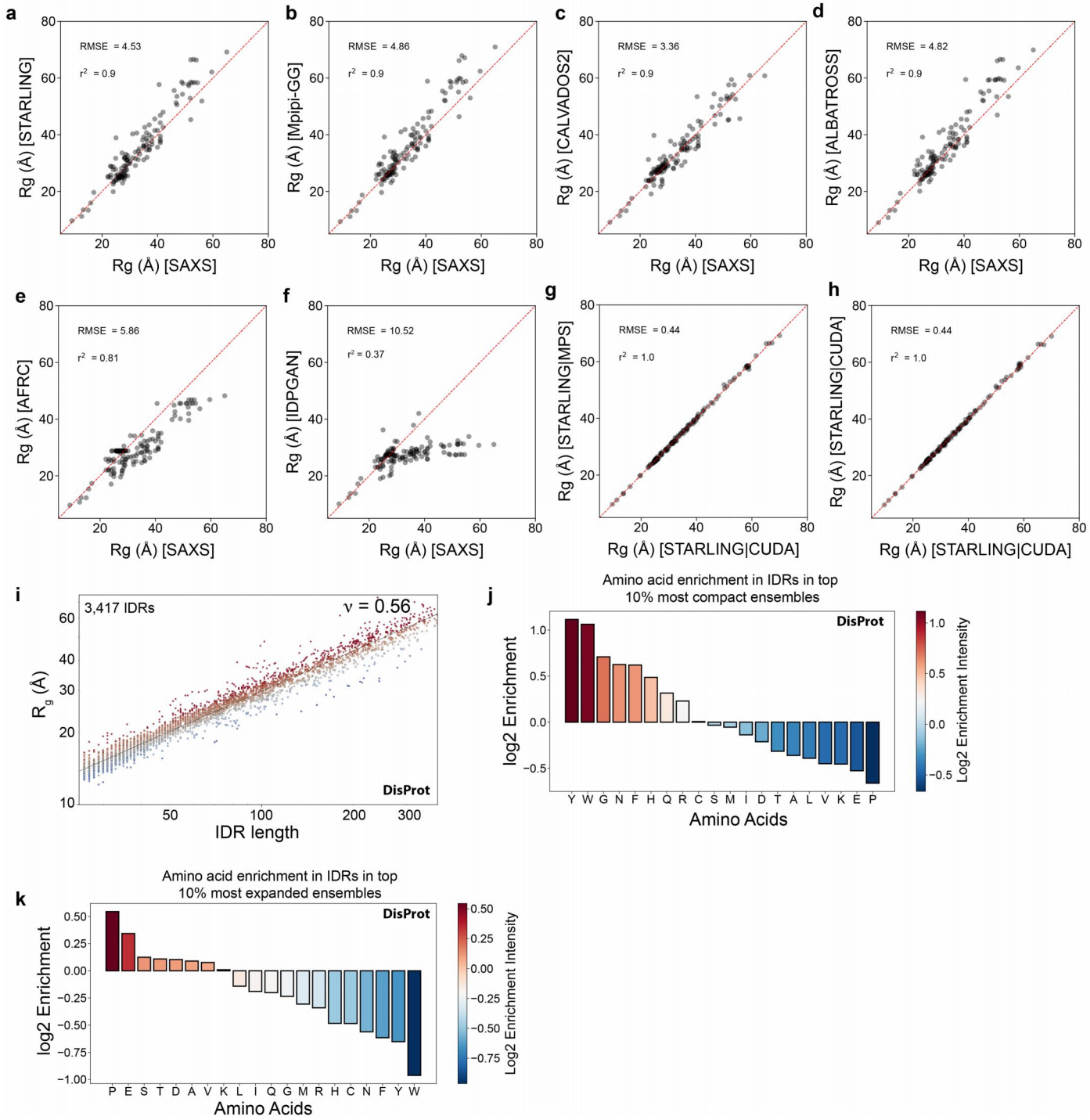
Extended Data Fig. 4 | See next page for caption.



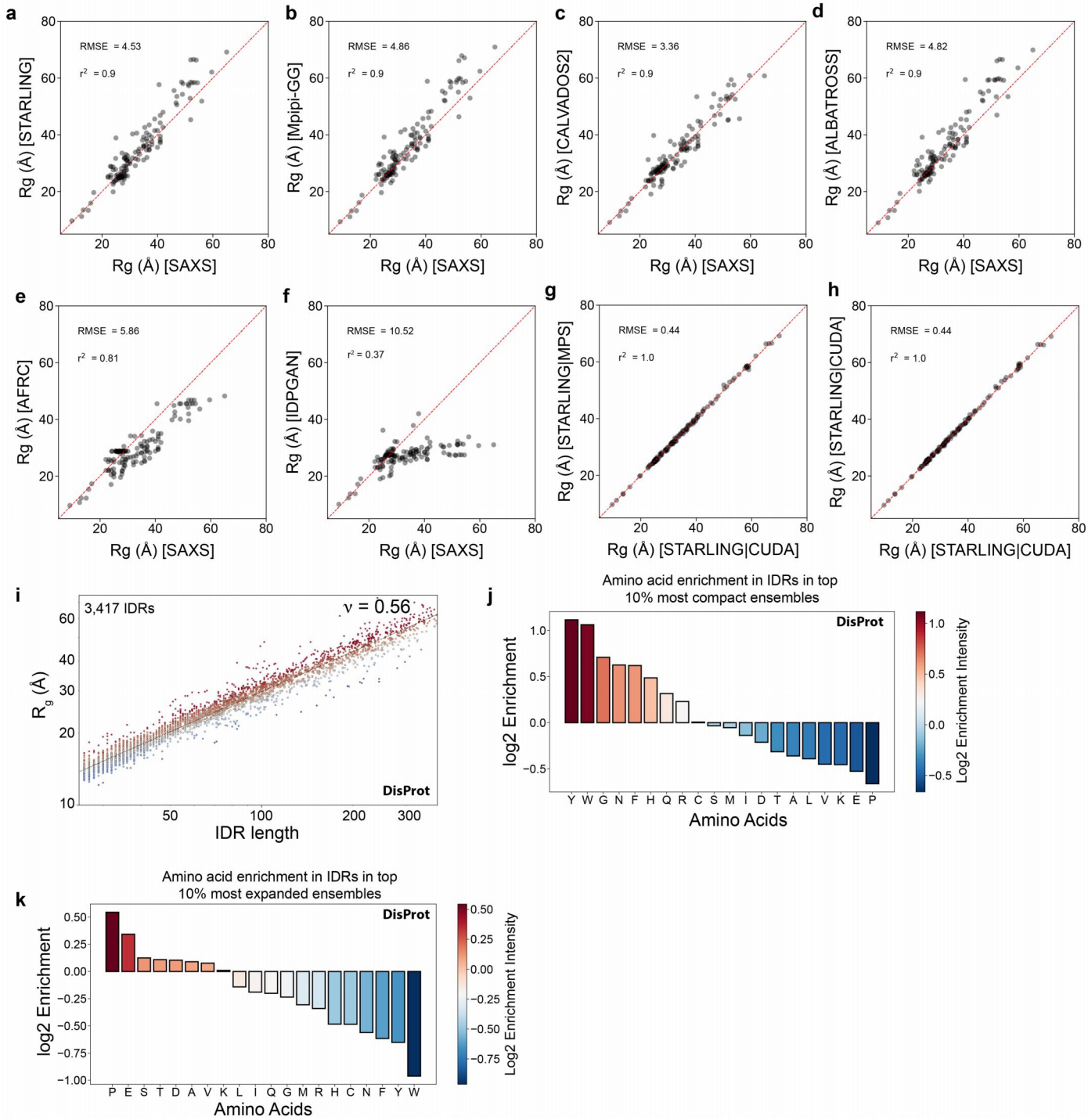
扩展数据图 4 | 图注请参阅下一页。

Extended Data Fig. 4 | Accuracy and convergence of STARLING compared to Mpipi-GG simulations. a, Comparison of STARLING-derived average end-to-end distance (R_e values) with those from Mpipi-GG (simulations) from a set of ~3,000 unseen sequences at an ionic strength of 20 mM. **b**, Comparison of STARLING-derived R_e values with those from Mpipi-GG (simulations) from a set of ~3,000 unseen sequences at an ionic strength of 300 mM. **c**–**f**, Convergence of the radius of gyration (R_g) as a function of ensemble size for IDRs of varying lengths (50, 150, 250, and 350 residues). Dashed lines represent Mpipi-GG convergence, solid lines correspond to STARLING convergence, and the shaded area indicates the ensemble-averaged R_g from long Mpipi-GG simulations. STARLING-derived R_g values converge within the first 800 conformers, exhibiting minimal error compared to Mpipi-GG ensembles. Based on this analysis, we selected 800 STARLING conformers for R_g prediction comparisons in all benchmarking used here. **g**, Convergence plot depicting the relationship between Hellinger distance (*H*), the number of conformations sampled from STARLING, and the number of denoising steps for various an IDR of length 350. The Hellinger distance quantifies the similarity between two distributions, where a value of zero indicates identical distributions and a value of one signifies no overlap. The Hellinger distance decreases as the number of conformations and denoising steps increase, with optimal STARLING performance observed at 1000 conformers and 30 denoising steps. **h**, Convergence of short-range and long-range distance distributions as a function of ensemble size. A 10 μs Mpipi-GG simulation of a 250-residue disordered protein was split into two 5 μs segments, with one serving as the reference and the other for comparison. A 2,500-conformer STARLING ensemble was also compared to the reference segment. Panels show convergence of STARLING and Mpipi-GG distance distributions relative to the reference for short-range residue pair 1–10, medium-range residue pair 1–50, long-range residue pair 1–100, and the end-to-end distance (residue pair 1–250). Error bars represent bootstrap-estimated standard deviations. Insets display the full distance distributions, along with Kolmogorov–Smirnov statistics for STARLING and Mpipi-GG (legend: blue, STARLING; orange, Mpipi-GG; black dashed line, reference). **i**, Convergence of radius-of-gyration from short to long sequence lengths. A 10 μs Mpipi-GG simulation of a disordered protein was split into two 5 μs segments, with one serving as the reference and the other for comparison. A 2,500-conformer STARLING ensemble was also compared to the reference segment. Panels show convergence of STARLING and Mpipi-GG radii-of-gyration distributions relative to the reference for a 50-residue-long disordered protein, a 150-residue-long disordered protein, a 250-residue-long disordered protein, and a 350-residue-long disordered protein. Error bars represent bootstrap-estimated standard deviations. Insets display the complete radii-of-gyration distributions, along with Kolmogorov–Smirnov statistics for STARLING and Mpipi-GG (legend: blue, STARLING; orange, Mpipi-GG; black dashed line, reference). **j**, Comparison of STARLING-derived average radii of gyration (R_g values) for length-matched sequences with a broad range of chemistries for those from Mpipi-GG (simulations) from a set of 137 unseen sequences. This is done to verify that STARLING has learned real sequence chemistry, not just that longer sequences are bigger. **k**, Comparison of STARLING-derived average radii of gyration (R_e values) for length-matched sequences with a broad range of chemistries for those from Mpipi-GG (simulations) from a set of 137 unseen sequences. This is done to verify that STARLING has learned real sequence chemistry, not just that longer sequences are bigger.

扩展数据图4 | Starling 方法与 Mpipi-GG 仿真方法的精度与收敛性对比。a，在离子强度为 20 mM 时，将 Starling 方法计算得到的平均端到端距离（$\mathbf{R_e}$值）与 Mpipi-GG（仿真）所得结果进行对比，对比对象为约 3,000 条未见序列；**b**，在离子强度为 300 mM 时，将 Starling 方法计算得到的$\mathbf{R_e}$值与 Mpipi-GG（仿真）所得结果进行对比，对比对象为约 3,000 条未见序列；**c–f**，对于不同长度（50、150、250 和 350 个残基）的 IDR，回转半径（$\mathbf{R_g}$）随系综规模变化的收敛性曲线。虚线表示 Mpipi-GG 的收敛曲线，实线表示 Starling 的收敛曲线，阴影区域表示基于长时程 Mpipi-GG 仿真的系综平均$\mathbf{R_g}$值。Starling 方法计算得到的$\mathbf{R_g}$值在前 800 个构象中即趋于收敛，与 Mpipi-GG 系综相比误差极小。基于此分析，我们选取了 800 个 Starling 构象用于本文所有基准测试中的$\mathbf{R_g}$预测对比。**g**，收敛性曲线图，展示了在不同长度为 350 的 IDR 下，Hellinger 距离（*H*）、从 Starling 中采样的构象数量与去噪步数之间的关系。Hellinger 距离用于量化两个分布之间的相似性：值为 0 表示两个分布完全相同，值为 1 则表示两个分布互不重叠。随着构象数量和去噪步数的增加，Hellinger 距离逐渐减小；在 1000 个构象和 30 个去噪步数时观察到 Starling 的最佳性能。**h**，短程与长程距离分布随系综规模变化的收敛性曲线。对一个由 250 个残基组成的无序蛋白质进行的 10 μs MpipiGG 模拟被分为两个 5 μs 的片段，其中一个作为参考片段，另一个用于对比。还将一个包含 2,500 个构象的 Starling 系综与参考片段进行对比。各子图展示了 Starling 与 Mpipi-GG 的距离分布相对于参考分布的收敛情况，对比对象分别为：短程残基对 1–10、中程残基对 1–50、长程残基对 1–100 以及端到端距离（残基对 1–250）。误差线 r **i**表示通过Bootstrap法估算的标准差；插图展示了完整的距离分布曲线以及星雀（Starling）与Mpipi-GG的Kolmogorov–Smirnov统计量（图例：蓝色——星雀；橙色——Mpipi-GG；黑色虚线——参考组）。图中展示了从短序列长度到长序列长度下回转半径的收敛情况：对一个时长为10 μs的无序蛋白质Mpipi-GG模拟结果进行分割，分为两个时长为5 μs的片段，其中一个作为参考组，另一个用于对比分析；同时将包含2,500个构象的星雀系综与参考组进行比较。各子图展示了星雀与Mpipi-GG的回转半径分布相对于参考组的收敛情况，涵盖时长分别为50个残基、150个残基、250个残基及350个残基的无序蛋白质。误差线表示通过Bootstrap法估算的标准差；插图展示了完整的回转半径分布曲线以及星雀与Mpipi-GG的Kolmogorov–Smirnov统计量（图例：蓝色——星雀；橙色——Mpipi-GG；黑色虚线——参考组）。**j**：对来自 137 组未见序列的、长度匹配且化学性质范围广泛的序列，与来自 Mpipi-GG（模拟数据）的序列进行星雀模型推导的平均转动半径（$\mathbf{R_g}$值）比较。此举旨在验证星雀模型是否真正学习到了真实的序列化学特性，而非仅仅因为较长序列的半径更大。**k**：对来自 137 组未见序列的、长度匹配且化学性质范围广泛的序列，与来自 Mpipi-GG（模拟数据）的序列进行星雀模型推导的平均转动半径（$\mathbf{R_e}$值）比较。此举旨在验证星雀模型是否真正学习到了真实的序列化学特性，而非仅仅因为较长序列的半径更大。



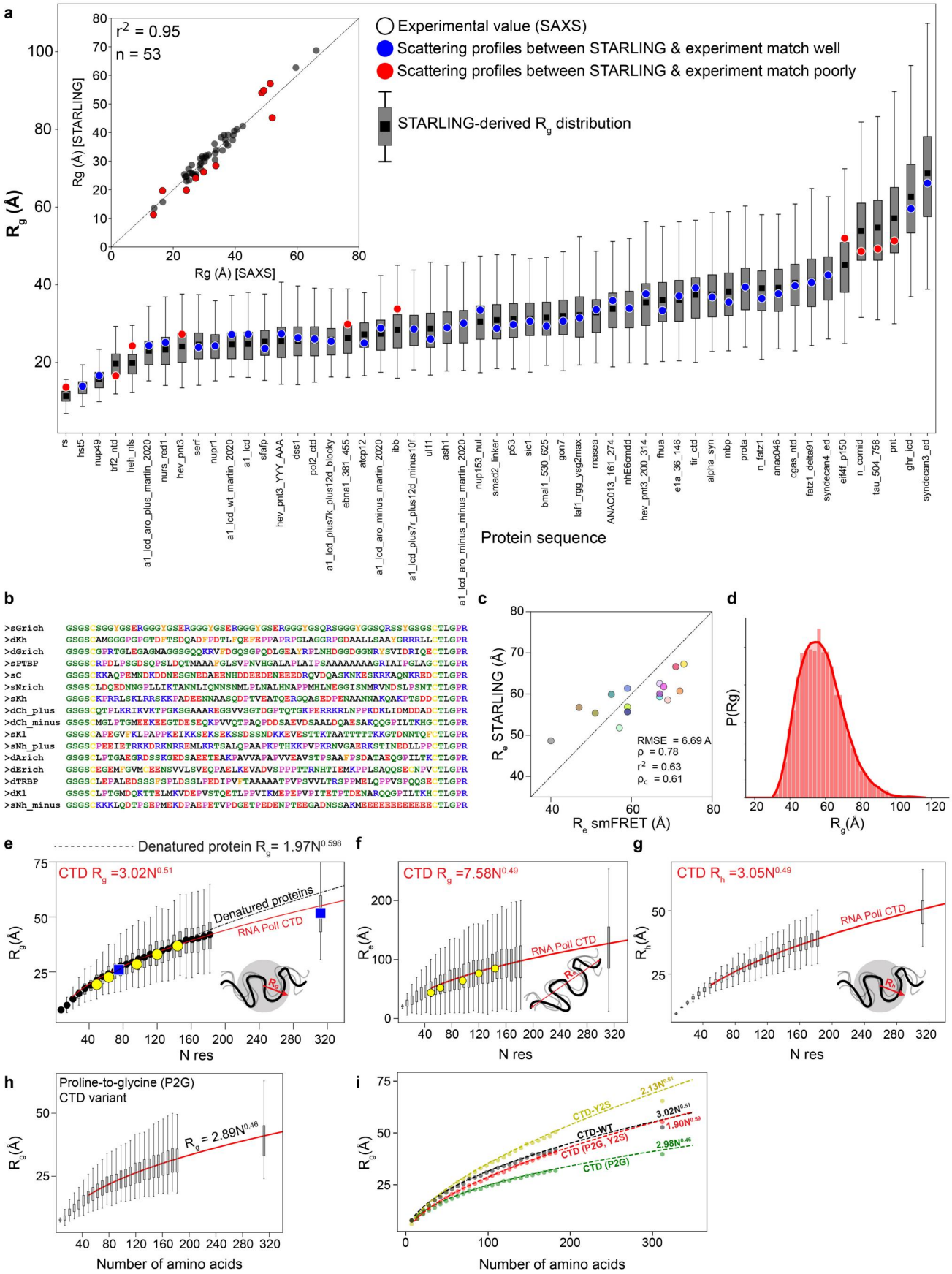
Extended Data Fig. 5 | See next page for caption.



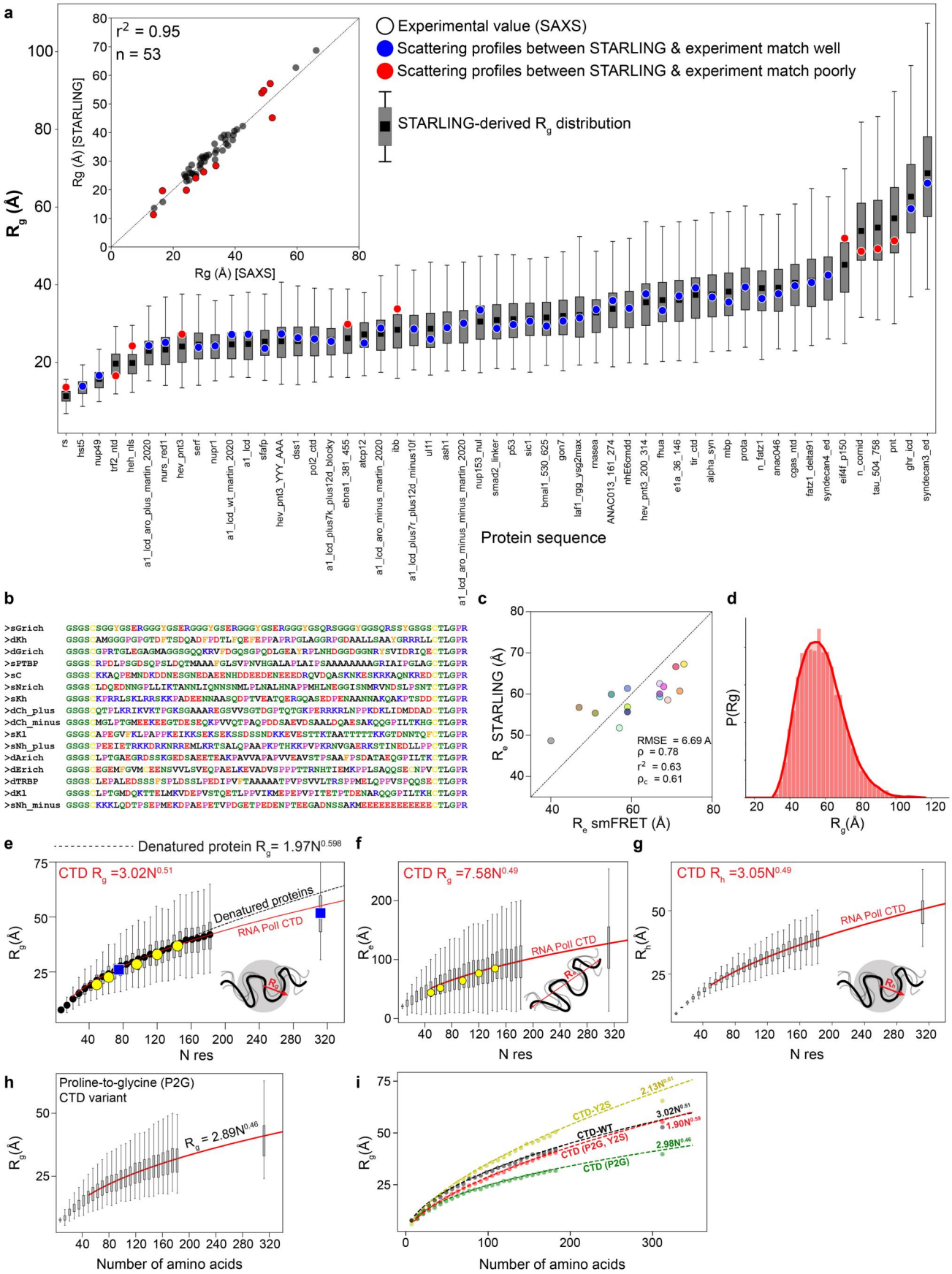
扩展数据图5| 图注请参阅下一页。

Extended Data Fig. 5 | Benchmarking STARLING vs. other methods, and assessment of DisProt ensembles. Comparison of experimental R_g values obtained from 133 different experimental datasets with various state-of-the-art tools for ensemble prediction (**a**) STARLING (this work), (**b**) Mpipi-GG, coarse-grained molecular dynamics simulations, (**c**) CALVADOS2, coarse-grained molecular dynamics simulations, (**d**) ALBATROSS, a deep-learning-based predictor that can only predict average ensemble values, (**e**) The Analytical Random Flory Coil (AFRC), a limiting null model that considers IDPs as Gaussian-like chains with no interactions between residues. We include this because it represents the true "lower bound" for what a model should achieve in terms of accuracy and RMSE. **f**, idpGAN, a Generative Adversarial Network for predicting ensembles from the sequence. **g**, We also compared R_g predictions on GPU vs. MPS implementations for completeness. This revealed 1:1 agreement that matched the RMSE and correlation obtained when the same dataset was analysed twice on the same hardware (**h**). **i**, Analysis of sequences from DisProt – a database of experimentally verified disordered regions – based on STARLING ensembles (sequences <380 residues). All ensembles contain 400 conformers, and the full DisProt was ~15 h on a desktop computer with an NVIDIA A4000 GPU (i.e., 3-4 proteins per minute). Fitting ensemble dimensions vs. length reveals an apparent scaling exponent of 0.56, indicating these IDRs are relatively expanded. However, we note an unavoidable acquisition bias in experimentally characterized IDRs, meaning that highly soluble (and hence more expanded, with fewer intra- and intermolecular interactions) proteins are preferred for biophysical characterization because they do not aggregate/crash out of solution, and many techniques used to characterize IDPs (SAXS, NMR, etc.) require high protein concentrations. We also note that we underestimate the contributions of aliphatic hydrophobes in driving intramolecular interactions, as well as the effects of local helicity and the long-range consequences of intramolecular helix interactions. Finally, given that IDRs are heteropolymers, average scaling behavior, by definition, masks both attractive and repulsive intramolecular interactions. As such, we strongly caution against interpreting a value of 0.56 as indicating that all IDRs are highly expanded. **j**, Analysis of sequences in the most compact 10% of IDRs identified aromatic residues as being over-represented here, in line with prior work. **k**, An analogous analysis of sequences in the top 10% of expanded IDRs also reveals that negatively charged residues and proline are enriched in sequences that drive expansion, again in agreement with prior work.

扩展数据图 5| 鸟鸣鸟与其他方法的基准测试，以及 DisProt 簇的评估。实验 R 值的比较 g 数据来源于133个不同的实验数据集，采用多种前沿工具进行集成预测（**a**）：Starling（本研究）；（**b**）Mpipi-GG，粗粒度分子动力学模拟；（**c**）CALVADOS2，粗粒度分子动力学模拟；（**d**）Albatross——一个仅能预测平均集成值的基于深度学习的预测器；（**e**）Analytical Random Flory Coil（AFRC）——一个极限零模型，将IDPs视为无残基间相互作用的类高斯链。我们纳入该模型是因为它代表了模型在准确度和RMSE方面应达到的真实“下限”；**f**，idpGAN——一种用于根据序列预测集成值的生成对抗网络；**g**，为确保完整性，我们还比较了基于GPU与基于MPS实现的R_g预测结果。结果显示两者完全一致，与在相同硬件上对同一数据集进行两次分析时获得的 RMSE 值和相关性相匹配（**h**）；**i**，对DisProt中的序列进行分析。– 这是一个基于星形集合（序列长度 <380 个残基）构建的、经实验验证的无序区域数据库。所有集合均包含 400 个构象异构体；在配备 NVIDIA A4000 GPU 的台式计算机上，完整构建的 DisProt 数据库的处理速度约为每分钟 3–4 个蛋白质。通过分析集合维度与序列长度之间的关系，发现其表观标度指数为 0.56，表明这些无序结构域（IDRs）相对较为伸展。然而，我们注意到在实验表征的 IDR 数据集中存在一种不可避免的采集偏差：即那些溶解度较高（因而结构更为伸展、分子内及分子间相互作用较少）的蛋白质更易被用于生物物理特性表征，因为这类蛋白质不会发生聚集或从溶液中析出；而许多用于表征无序蛋白质（IDPs）的技术（如 SAXS 、NMR 等）均要求较高的蛋白质浓度。此外，我们还发现，对于驱动分子内相互作用的脂肪族疏水作用、局部螺旋度以及分子内螺旋相互作用所引发的长程效应，其贡献度被低估了。最后，鉴于 IDR 是异聚物，其平均标度行为在定义上掩盖了分子内既存在吸引力也存在排斥力的相互作用。因此，我们强烈建议不要将数值 0.56 解释为意味着所有 IDR 均处于高度扩张状态。**j**对占比最低的 10% IDR 中的序列进行分析后发现，芳香族残基在这些序列中呈现过表达现象，这与此前的研究结果一致。**k**对占比最高的 10% 扩张 IDR 中的序列进行类似分析后亦发现，带负电荷的残基与脯氨酸在驱动序列扩张的序列中富集，这一发现同样与既往研究结果相符。



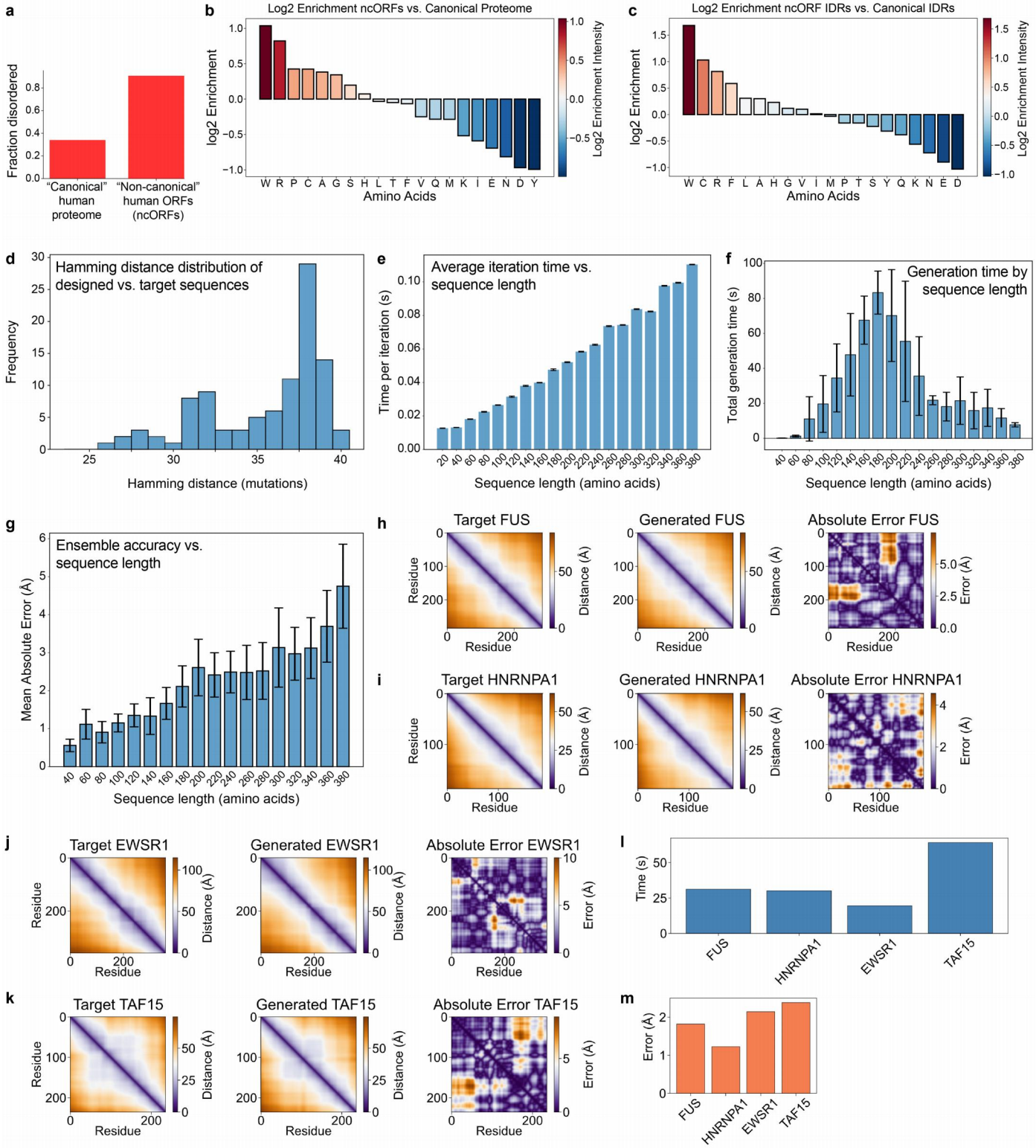
Extended Data Fig. 6 | See next page for caption.



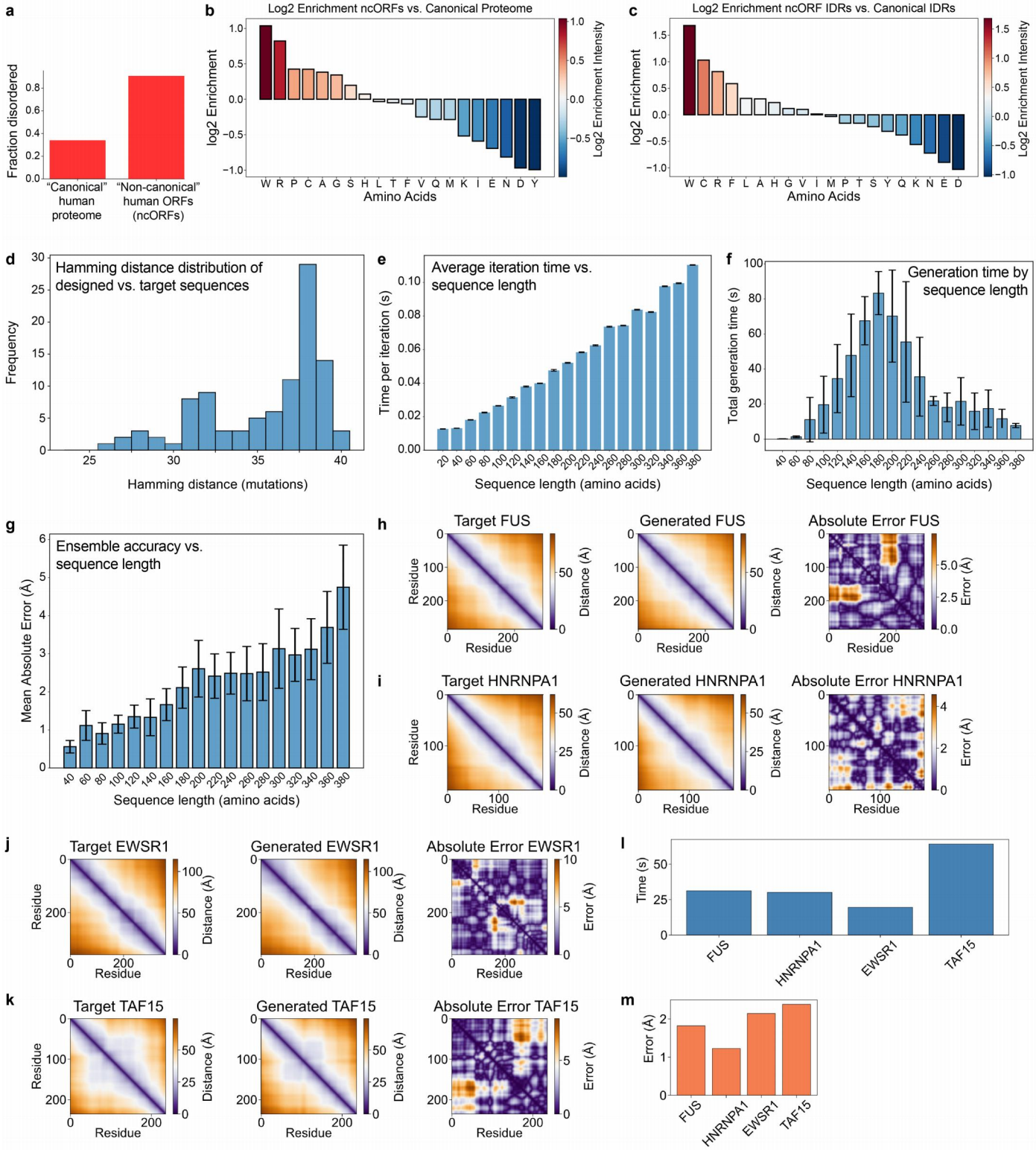
扩展数据图 6 | 图注请参阅下一页。

Extended Data Fig. 6 | Comparison of STARLING-derived ensembles with experimental data. a, Comparison of STARLING-derived R_g values vs. SAXS for high-quality consistently reanalysed SAXS data. Comparison of SAXS-derived radii of gyration vs. STARLING-derived radii of gyration for the same 53 sequences as described in Supplementary Figs. 1 and 2. Sequences in which experimental data are represented as a red dot are those where scattering profiles were in sub-optimal agreement with STARLING-derived scattering data—overall, focusing on this high-quality dataset (each of which we re-analysed to obtain R_g values using the Molecular Form Factor (MFF) of ref. 63), we find even stronger agreement with STARLING-derived predictions (inset). **b**, Amino acid sequences for those IDRs investigated by smFRET in ref. 34. **c**, Correlation between STARLING-derived end-to-end distances and smFRET-derived end-to-end distances. The agreement between STARLING and the experiment is assessed in various ways, including root-mean-square error (RMSE), Spearman correlation coefficient (ρ), squared Pearson correlation coefficient (r^2), and concordance correlation coefficient (ρ_c). We note that STARLING appears to underestimate end-to-end distances for more expanded chains here. However, we do not take dyes into explicit consideration, and it is possible that for the expanded negatively charged sequence, a small amount of (negatively-charged) dye:protein repulsions could account for the slight underestimation here. **d**, R_g distribution for $cMyc^{1\text{-}361}$ obtained from STARLING ensembles. We predict an average Rg of 58 Å. We note this value may be an overestimate given $cMyc^{1\text{-}361}$ is predicted to contain several regions of transient helicity, which may reduce the R_g, such that we suggest a value closer to 53-54 Å may be a more accurate reflection of the expected dimensions for monomeric $cMyc^{1\text{-}361}$. **e**, The same figure is shown for Fig. 4d, with an additional polymer scaling line (black dashed) derived from fully denatured proteins measured by Kohn et al.[64], also superimposed. Fitting the radius of gyration *vs*. number of amino acids to a polymer scaling law of the form $R_g = B_0N^v$ (where R_g is the radius of gyration, B_0 is a prefactor term that captures a combination of chain stiffness and monomer excluded volume, and v is the apparent Flory scaling exponent) revealed well-fit parameters of $B_0 = 3.02$ Å and $v = 0.51$. Black circles are STARLING average values, with grey box-plots showing the distribution of values from the ensemble. Yellow circles are results from all-atom Monte Carlo simulations. Blue squares are experimental (SAXS) data. **f**, Equivalent analysis to that shown in **a**, but with the end-to-end distance (R_e) fit instead of the radius of gyration. Grey box-plots show the distribution of values from the STARLING ensemble. Yellow circles are results from all-atom Monte Carlo simulations. **g**, Equivalent analysis to that shown in panel A, but the hydrodynamic radius (R_h) is fit instead of the radius of gyration. Grey box-plots show the distribution of values from the STARLING ensemble. Yellow circles are results from all-atom Monte Carlo simulations. **h**, R_g values for CTD variants in which all proline residues have been converted to glycine (P2G). Grey box-plots show the distribution of values from the STARLING ensemble. **i**, Comparison of scaling behavior for wild-type (black), P2G (green), Y2S (yellow), and P2G, Y2S (red) variants of the CTD. The large prefactor (3.02) observed in the wild-type sequence arises from a combination of the bulky tyrosine residue and the stiff proline residue; the double Y2S, G2P variant exhibits a prefactor comparable to unfolded-state scaling behaviour (see **e**).

扩展数据图 6 | 鸟鸣鸟源生成的集合与实验数据的对比。a：针对高质量、经一致性重新分析的 SAXS 数据，对鸟鸣鸟源生成的R_g值与 SAXS 值进行对比；针对与补充图 1 和 2 中所述相同的 53 条序列，对比 SAXS 源生成的回转半径与鸟鸣鸟源生成的回转半径。其中实验数据以红点表示的序列，是指其散射谱与鸟鸣鸟源生成的散射数据之间存在次优一致性的情况——总体而言，针对这一高质量数据集（每条序列均经分析以利用参考文献63中的分子形状因子 (MFF) 获取R_g值），我们发现其与鸟鸣鸟源生成的预测结果具有更强的一致性（插图）。**b**：参考文献34中通过 smFRET 研究的 IDRs 对应的氨基酸序列。**c**：鸟鸣鸟源生成的端到端距离与 smFRET 源生成的端到端距离之间的相关性。星形模型与实验结果之间的吻合度通过多种方法进行评估，包括均方根误差（RMSE）、Spearman相关系数(ρ)、Pearson相关系数平方（r^2）以及一致性相关系数（ρ_c）。我们注意到，在本研究中，星形模型对较扩展的链段所对应的端到端距离存在低估现象。然而，我们并未将染料因素 explicitly 考虑在内；对于扩展的带负电荷序列，少量（带负电荷的）染料与蛋白质之间的排斥作用可能是导致此次轻微低估的原因。**d**与R_g的分布情况适用于从星形模型系综中获得的cMyc $^{1\text{-}361}$，我们预测其平均Rg值为58 Å。我们注意到，鉴于预测认为cMyc $^{1\text{-}361}$包含多个瞬时螺旋区域（这些区域可能会降低R_g），该数值可能偏高；因此，我们认为更接近53–54 Å的数值或许能更准确地反映单体cMyc $^{1\text{-}361}$的预期尺寸。**e**，该图与图中所示内容一致。**4d**，另附加一条基于 Kohn 等人[64]测得的完全变性蛋白质所得的聚合物标度线（黑色虚线），该标度线亦叠加显示。通过将回转半径与氨基酸数量拟合至形式为$R_g = B_0 N^{\nu}$的聚合物标度律（其中R_g为回转半径，B_0为捕获链刚度与单体排除体积之综合效应的前置因子项，ν 为表观 Flory 标度指数），得到了良好的fit parameters of B_0= 3.02 Å 且 ν = 0.51。黑色圆圈表示星形平均值，灰色箱线图显示了该系综中数值的分布情况；黄色圆圈为全原子蒙特卡罗模拟结果；蓝色方块为实验（SAXS）数据。**f**：与图 a 中所示分析方法相同，但采用端到端距离（R_e）进行拟合而非回转半径；灰色箱线图显示了星形系综中数值的分布情况；黄色圆圈为全原子蒙特卡罗模拟结果。**g**，与图 A 所示的分析方法相同，但采用水动力半径 (R_h) 进行拟合，而非回转半径。灰色箱线图展示了椋鸟系综中各数值的分布情况；黄色圆圈表示全原子蒙特卡罗模拟所得结果。**h**与R_g值对应于所有脯氨酸残基均被替换为甘氨酸（P2G）的CTD变体；灰色箱线图展示了来自Starling系综的数值分布。**i**：野生型（黑色）、P2G（绿色）、Y2S（黄色）以及P2G与Y2S双变体（红色）的CTD的标度行为对比。在野生型序列中观察到的较大前置因子（3.02）源于庞大酪氨酸残基与刚性脯氨酸残基的共同作用；而双Y2S、G2P变体所呈现的前置因子与未折叠态的标度行为相当（参见e）。



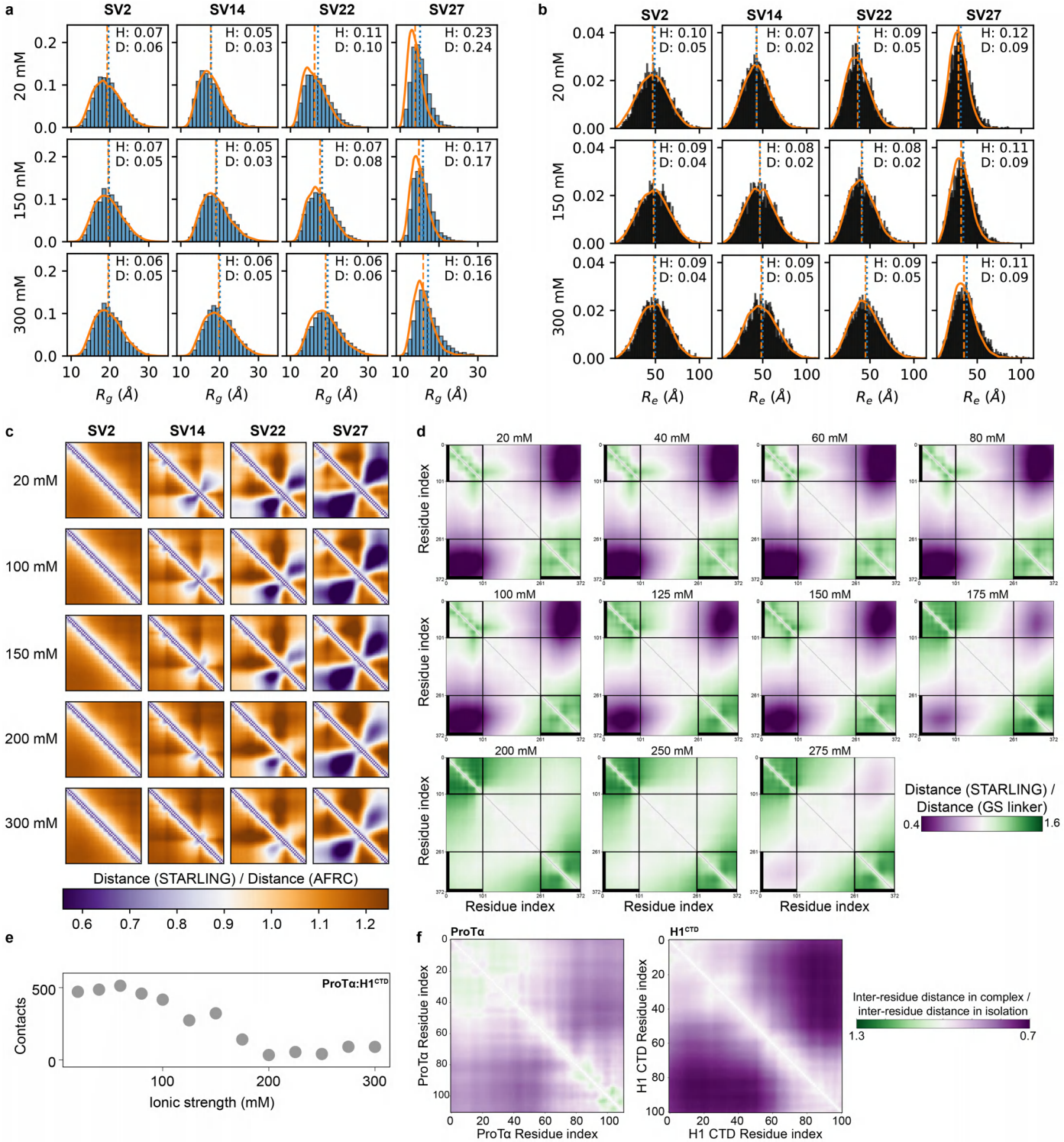
Extended Data Fig. 7 | See next page for caption.



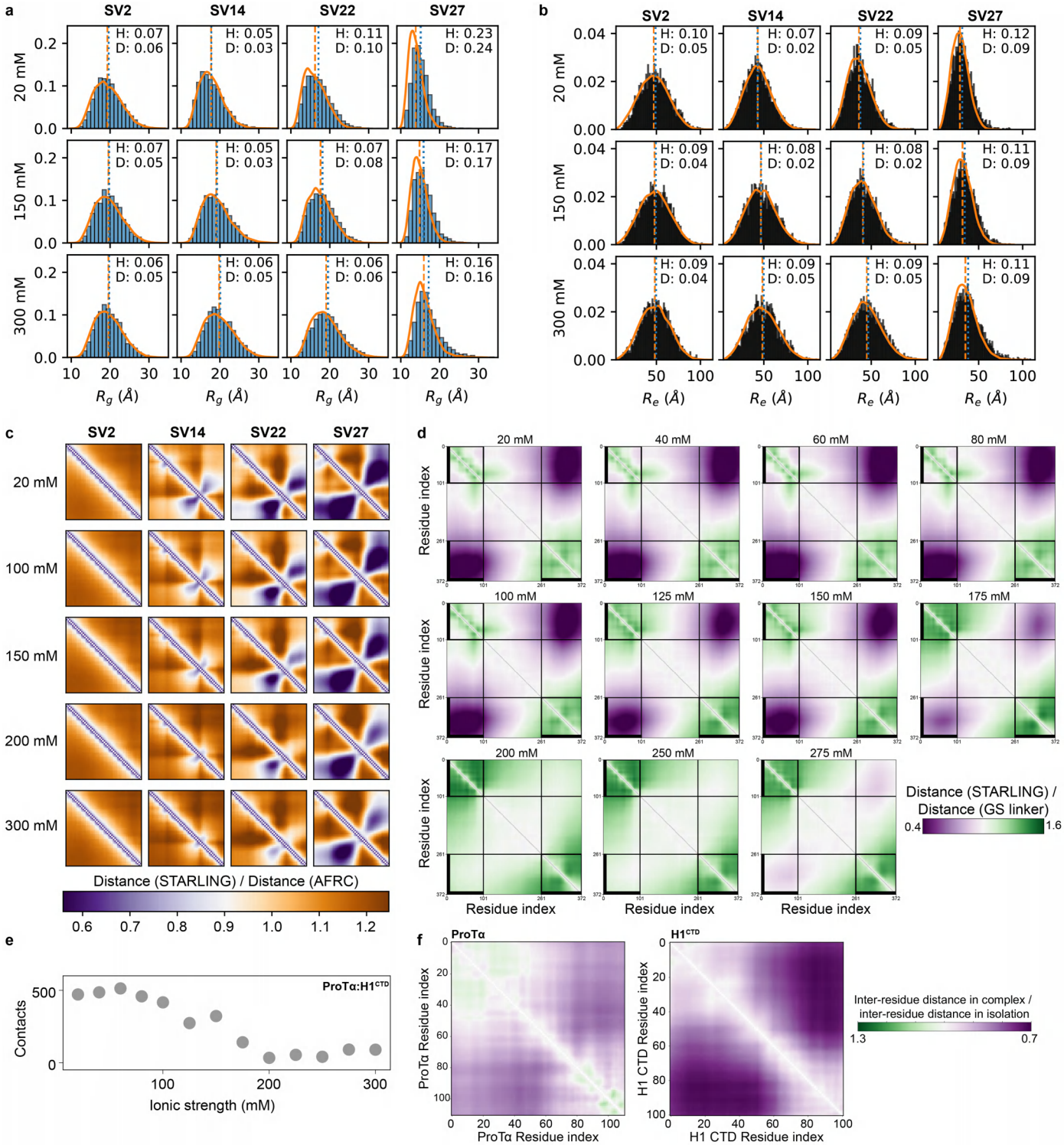
扩展数据图 7| 图注请参阅下一页。

Extended Data Fig. 7 | Microprotein analysis and assessment of latent-space design approach. a, Microproteins are predicted to be highly disordered. The comparison here is the fraction of amino acids in the canonical human proteome that are found in disordered regions vs. the fraction of amino acids in the Deutsch et al.[44] set of microproteins (the 'microproteome') that are found in disordered regions. The disorder was predicted using metapredict V3. **b**, Statistical enrichment for each of the 20 amino acids, comparing frequency observed across all canonical proteins vs. frequency observed in microproteins. We excluded initiator methionine from both datasets, given its disproportionate influence on shorter compared to longer sequences. As observed previously for metazoan microproteins, we find enrichment for tryptophan and arginine and depletion of acidic residues. **c**, To establish whether the enrichment observed here reflects something specific to microproteins or the enhanced fraction of disorder, we also compared enrichment in microprotein-derived IDRs vs. canonical-proteome-derived IDRs. This revealed an even stronger signature: microprotein IDRs are *highly* enriched for tryptophan, cysteine, arginine, and phenylalanine. In contrast, they are highly depleted for lysine, asparagine, glutamic acid, and aspartic acid. **d**, Assessing designed variants of $CTCF^{590\text{-}629}$ (a 40-residue IDR from CTCF, UniProt ID P49711) as a toy sequence for rational design. Hamming distance of 100 $CTCF^{590\text{-}629}$ designs compared to $CTCF^{590\text{-}629}$. A Hamming distance of 40 means every residue is different from the wild-type sequence. **e**, Average iteration time of the sequence optimizer based on sequence length using 20 replicates per length. **f**, Generation time based on sequence length using 20 replicates per length. Design targets are subregions of eIF4B ($eIF4B^{158\text{-}611}$, Uniprot ID P23588), excised in 20-amino-acid increments, ranging from 40 to 380 amino acids. **g**, Ensemble accuracy versus sequence length using the same cosine similarity cutoff value of 0.97. Ensemble accuracy was assessed as the mean absolute error (MAE) between the generated and target sequences, expressed in Angstroms. The sequences were the same as those used for panel **f**. We note that while accuracy worsens as sequence length increases, this can be addressed by tightening the similarity cutoff. **h**–**k**, Examples of IDRs designed to match low-complexity sequences, in all cases done to optimize cosine similarity to a value of 0.97. Target IDRs are as follows: **h**, $FUS^{1\text{-}285}$ (UniProtID P35637); **i**, $hnRNPA^{184\text{-}372}$ (UniProtID P09651); **j**, $EWSR1^{1\text{-}359}$ (UniProtID Q01844); **k**, $TAF15^{1\text{-}236}$ (UniProtID Q92804). **l**, Per-sequence generation times for the four sequences. **m**, Per-sequence mean absolute error (MAE) comparing the distance map from the designed sequence vs. the target sequence, as measured in Angstroms.

延伸数据图 7| 微蛋白分析与潜在空间设计方法的评估。a：预测微蛋白具有高度无序性。本研究中的对比对象为：在典型人类蛋白质组中发现于无序区域的氨基酸占比，与在 Deutsch 等人[44]所构建的微蛋白集合（即“微蛋白质组”）中发现于无序区域的氨基酸占比。该无序性是使用 metapredict V3 软件进行预测的。**b**：对 20 种氨基酸分别进行统计学富集分析，比较其在所有典型蛋白质中的观测频率与在微蛋白质中的观测频率。鉴于起始甲硫氨酸对较短序列的影响程度远高于较长序列，我们将其从两个数据集中均剔除。正如此前在后生动物微蛋白质中观察到的那样，我们发现色氨酸和精氨酸的富集现象，而酸性残基则呈现缺失状态。**c**：为确定此处观察到的富集现象是微蛋白质特有的还是源于无序结构比例的增加，我们还比较了微蛋白质来源的 IDR 与典型蛋白质组来源的 IDR 的富集程度。结果揭示了更为显著的特征：微蛋白质 IDR 中色氨酸、半胱氨酸、精氨酸和苯丙氨酸高度富集；相反，其赖氨酸、天冬酰胺、谷氨酸和天冬氨酸则高度缺失。**d**：评估 CTCF $^{590\text{-}629}$（源自 CTCF 的一个由 40 个残基组成的 IDR，UniProt ID P49711）作为理性设计的示例序列。比较 100 个 CTCF $^{590\text{-}629}$ 设计序列与 CTCF $^{590\text{-}629}$ 的汉明距离。汉明距离为 40 意味着每个残基均与野生型序列不同。**e**，基于序列长度的序列优化器的平均迭代时间（每种序列长度进行 20 次重复实验）。**f**，基于序列长度的生成时间（每种序列长度进行 20 次重复实验）。设计目标为 eIF4B 的亚区域（eIF4B $^{158\text{-}611}$，Uniprot ID P23588），以 20 个氨基酸为增量进行截断，长度范围从 40 至 380 个氨基酸。**g**，集成准确率与序列长度的关系图（采用相同的余弦相似度截断值 0.97）。集成准确率通过生成序列与目标序列之间的平均绝对误差（MAE）来评估，结果以埃为单位表示。所用序列与图 f 中所用序列相同。我们注意到，尽管随着序列长度的增加，准确率会下降，但这一问题可通过提高相似度截断值来解决。**h–k**，旨在匹配低复杂度序列的内在无序区（IDRs）设计示例；所有案例均旨在将余弦相似度优化至 0.97。目标IDR区域如下：**h**，FUS $^{1\text{-}285}$（UniProtID P35637）；**i**，hnRNPA $^{184\text{-}372}$（UniProtID P09651）；**j**，EWSR1 $^{1\text{-}359}$（UniProtID Q01844）；**k**，TAF15 $^{1\text{-}236}$（UniProtID Q92804）；**l**，四个序列的逐序列生成时间；**m**，比较设计序列与目标序列所得距离图时的逐序列平均绝对误差（MAE），单位为埃。



Extended Data Fig. 8 | See next page for caption.



扩展数据图 8 | 图注请参阅下一页。

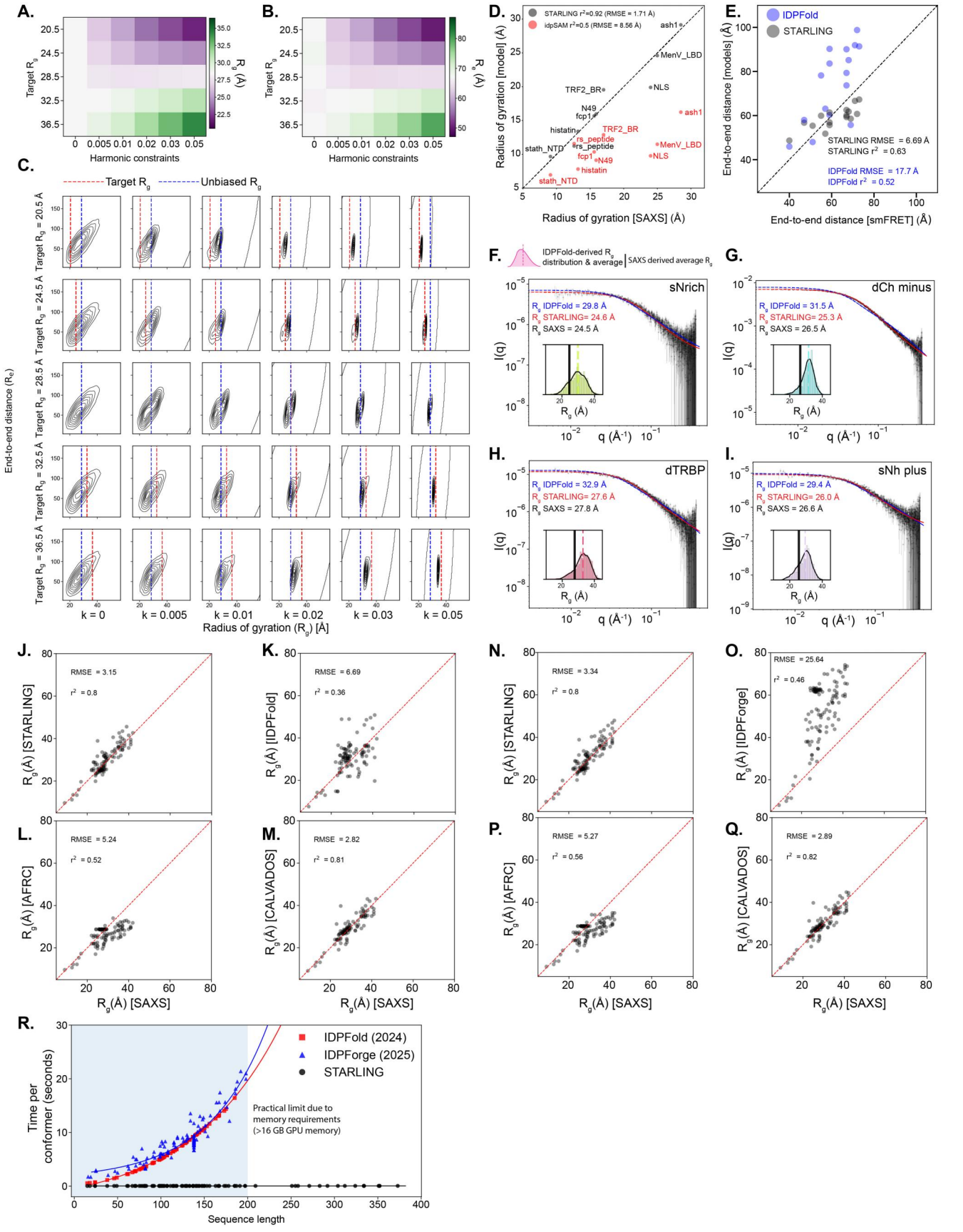
Extended Data Fig. 8 | Assessment of STARLING ionic strength effects. **a**, Overlap between Mpipi-GG and STARLING R_g distributions for a subset of the Das-Pappu sequences. R_g histograms for four of the Das-Pappu sequences with distinct sequence patterning as a function of ionic strength. Blue bars represent STARLING results; the orange line represents Mpipi-GG results. Histogram overlap is quantified by the Hellinger distance (*H*) or the KS statistic (*D*). **b**, Overlap between Mpipi-GG and STARLING R_e distributions for a subset of the Das-Pappu sequences. R_e histograms for four of the Das-Pappu sequences with distinct sequence patterning as a function of ionic strength. Bars represent STARLING results; the orange line represents Mpipi-GG results. Histogram overlap is quantified by the Hellinger distance (*H*) or the KS statistic (*D*). **c**, Normalized distance maps for Das-Pappu sequences. Normalized distance maps for four of the Das-Pappu sequences with distinct sequence patterning as a function of ionic strength. The top-right triangle is STARLING, and the bottom-left triangle is Mpipi-GG. **d**, Normalized distance maps for ProTα-$(GS)_{80}$-$H1^{CTD}$ constructs as a function of ionic strength. Distances here are normalized by a length-matched GS linker sequence (e.g., $(GS)_{186}$) to provide a self-consistent "null model" through which we expected the central $(GS)_{80}$ linker to reach a normalized distance value of 1, implying the linker is behaving as if it were not tethered. While not a prerequisite for these types of pseudo-intermolecular interaction ensembles, this provides us with confidence that we are observing behavior asymptotically approaching the limit in which the central linker has no meaningful impact on the resulting biomolecular complex. **e**, Pseudo-intermolecular contacts between ProTα and $H1^{CTD}$ quantified as a function of ionic strength. Prior work (Borgia et al. 2018) has established that the ProTα:H1 interaction is highly salt dependent, consistent with our conclusions here. **f**, Quantification of intramolecular distances in the complex divided by the same distances for the IDR in isolation. This analysis reveals specific subregions that are more locally compact than others in the complex, and that $H1^{CTD}$ undergoes a greater degree of compaction than ProTα.

扩展数据图 8| 鸟鸣鸟离子强度效应评估。

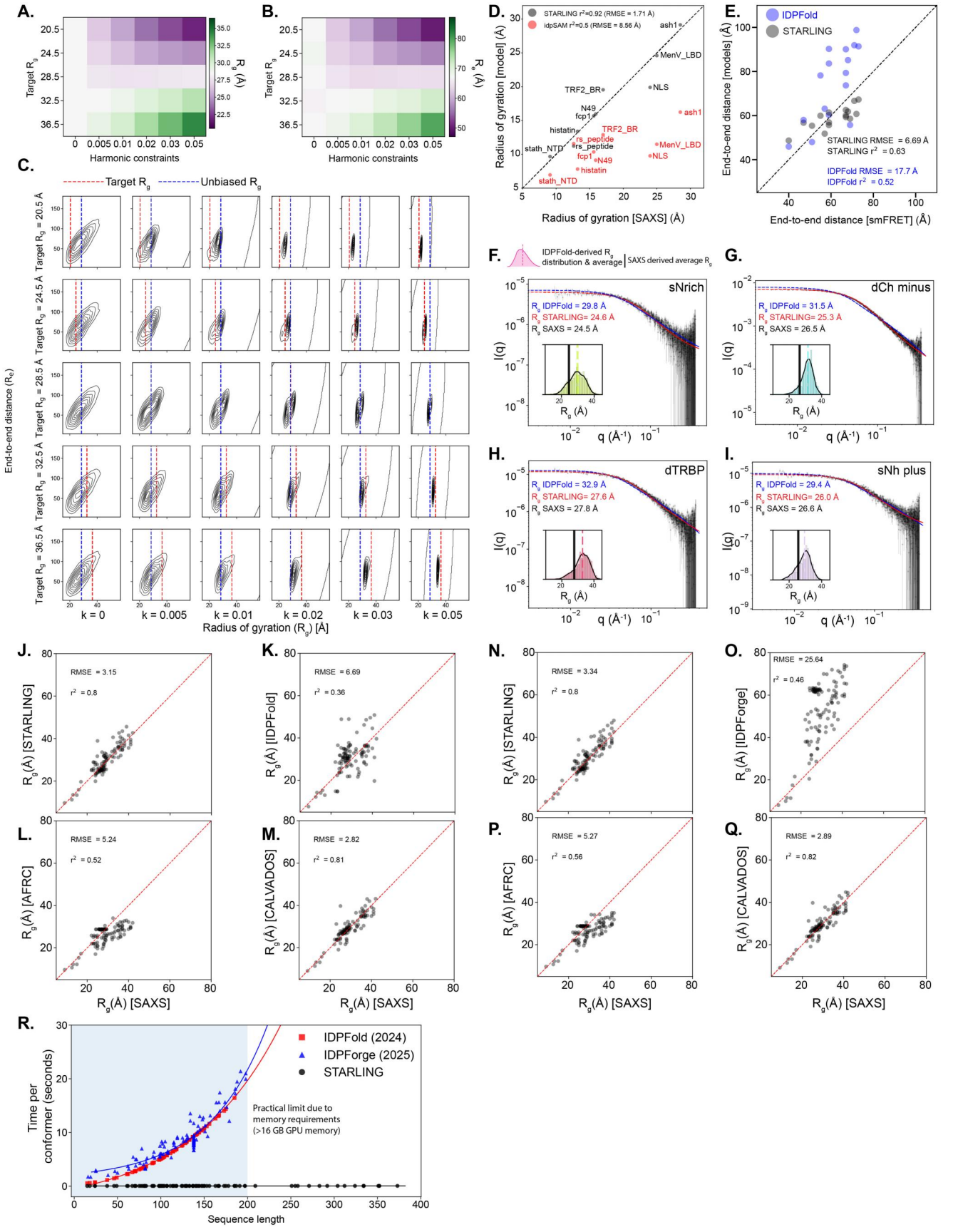
a，针对部分 Das-Pappu 序列，Mpipi-GG 与 Starling R_g分布之间的重叠情况。R_g直方图展示四个具有不同序列模式的 Das-Pappu 序列随离子强度变化的分布情况。蓝色柱状图代表 Starling 的结果；橙色线条代表 Mpipi-GG 的结果。直方图重叠程度通过 Hellinger 距离（*H*）或 KS 统计量（*D*）进行量化。**b**，针对部分 Das-Pappu 序列，Mpipi-GG 与 Starling R_e分布之间的重叠情况。R_e直方图展示四个具有不同序列模式的 Das-Pappu 序列随离子强度变化的分布情况。柱状图代表 Starling 的结果；橙色线条代表 Mpipi-GG 的结果。直方图重叠程度通过 Hellinger 距离（*H*）或 KS 统计量（*D*）进行量化。

c，Das-Pappu 序列的归一化距离图。展示了四种具有不同序列模式的 Das-Pappu 序列的归一化距离图，这些序列是离子强度的函数。右上角三角形代表 Starling 序列，左下角三角形代表 Mpipi-GG 序列。**d**，ProTα-$(GS)_{80}$-H1 ^{CTD}的归一化距离图。构建ASA作为离子强度的函数。此处距离通过一个长度匹配的GS连接子序列（例如，$(GS)_{186}$）进行归一化，以构建一个自洽的“无效模型”——通过该模型，我们预期中心（GS）$_{80}$连接子将达到归一化距离值为1，这意味着该连接子的行为如同未被束缚一般。尽管这并非此类伪分子间相互作用集合的先决条件，但这一结果使我们确信，我们观察到的行为正渐近地趋近于这样一个极限状态：在此状态下，中心连接子对最终形成的生物分子复合物不产生任何有意义的影响。**e**：ProTα与H1 ^{CTD}之间的伪分子间接触，量化为ASA作为离子强度的函数。先前的研究（Borgia 等，2018）已证实ProTα与H1之间的相互作用具有高度盐依赖性，这与我们在此得出的结论一致。

f：对复合物中分子内距离进行定量分析，并将其除以在 IDRin 分离过程中测得的相同距离。该分析揭示了在复合物中某些特定亚区域比其他区域具有更强的局部紧凑性，且 H1 ^{CTD}所经历的压缩程度高于 ProTα。



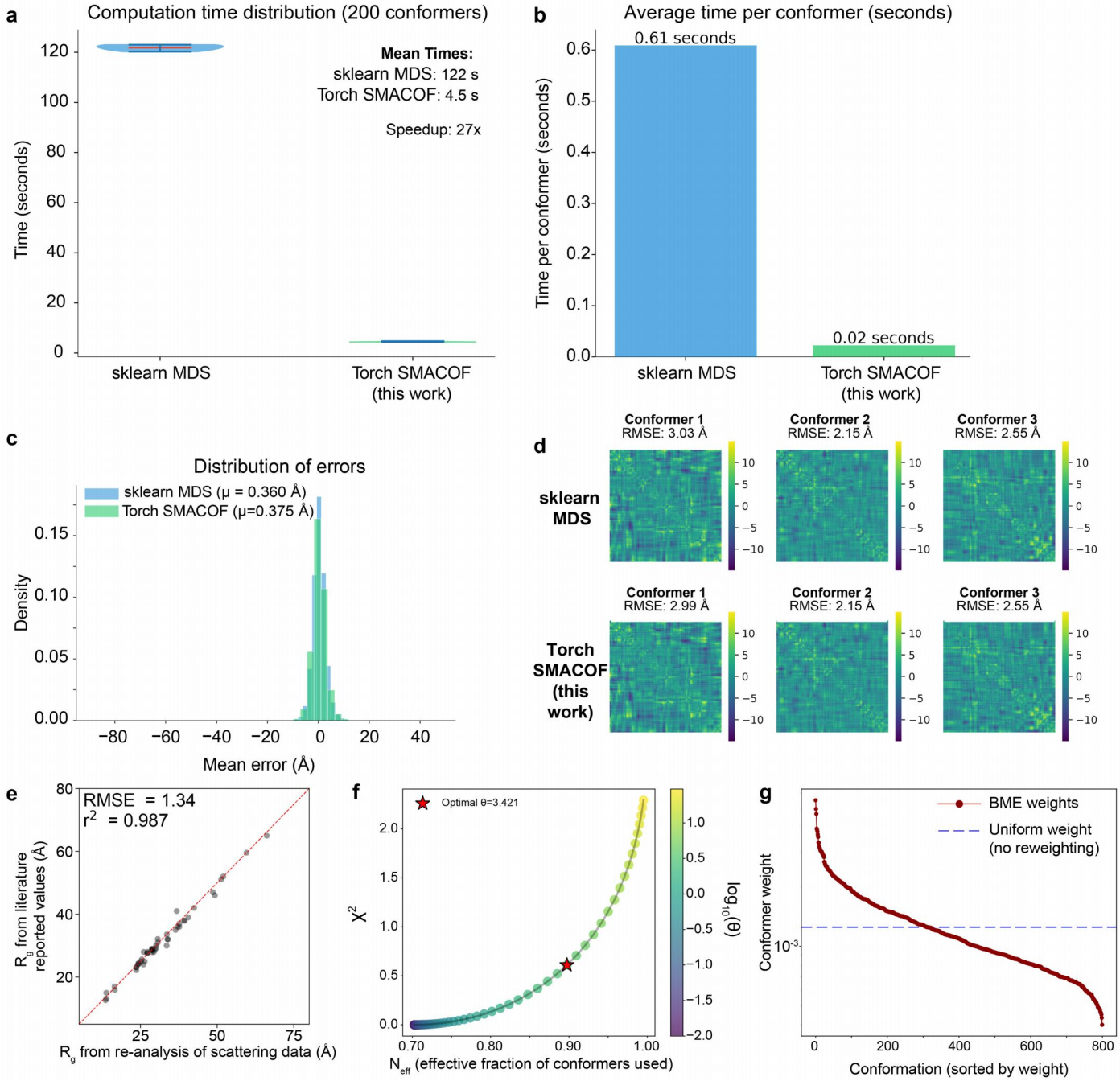
Extended Data Fig. 9 | See next page for caption.



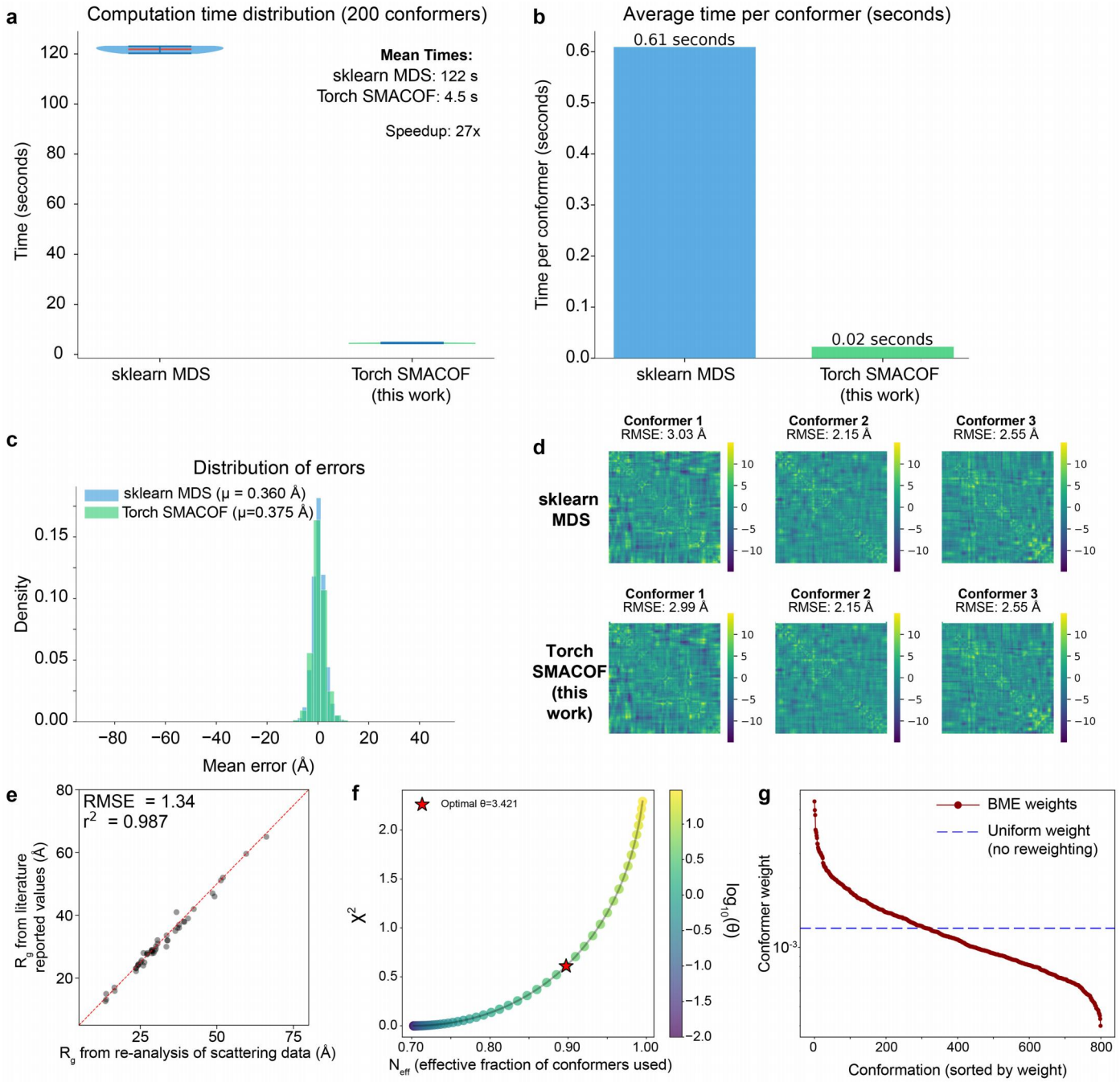
扩展数据图 9 | 图注请参阅下一页。

Extended Data Fig. 9 | Ensemble guidance and comparison with IDPFold and IDPForge. A, R_g biased ensemble generation. Ensembles of an 84-residue RNA POLII CTD construct biased towards the specified R_g values with different harmonic constraints. R_g values as a function of target (R_g) and guidance strength. The *x*-axis reports on value for harmonic constraint, while the *y*-axis reports the target R_g value. Elements in the matrix report on the ensemble-average R_g value under the combination of target and constraint. **B**, Equivalent analysis for R_e values as a function of target (R_g) and guidance strength. Equivalent biasing using end-to-end distance as the target is shown in Supplementary Fig. 5. **C**, 2D histograms of R_g (X-axis) and R_e (Y-axis) for ensembles generated under the specified guidance. For each sub-plot, the *x*-axis reports on the ensemble R_g while the *y*-axis reports on the ensemble R_e. For the matrix of subplots, the *x*-axis reports on the harmonic constraint applied and the *y*-axis reports on the target R_g value. An equivalent matrix of plots for biasing using end-to-end distance is shown in Supplementary Fig. 5. **D**, Comparison of idpSAM-derived ensembles vs. STARLING compared to experimental data. Focussing on sequences less than 60 amino acids in length (except Ash1, discussed below), we compared R_g values obtained from idpSAM ensembles vs. R_g values from STARLING ensembles. For idpSAM, 1000 conformers were used (idpSAM defaults), whereas for STARLING, 400 conformers were used (STARLING defaults). We note that the Analytical Flory Random Coil (AFRC) – a null model in which IDR sequence chemistry does not influence the overall dimensions and instead IDR dimensions depend solely on chain length and (weakly) the intrinsic dimensions of each amino acid – obtains an r^2 of 0.9 and an RMSE of 3.58 Å. We included Ash1 (83 amino acids) because, although it is larger than the ~60 amino acid cutoff used for training data in idpSAM, prior work has shown that SAXS data for Ash1 are well reproduced by CAMPARI/ABSINTH simulations. However, despite being trained on CAMPARI/ABSINTH simulations, idpSAM was unable to recapitulate this behaviour. **E**, Comparison of IDPFold-derived ensembles vs. STARLING compared to single-molecule FRET data reported by Holla et al.[34]. Correlation between experimentally-derived end-to-end distance (x-axis) and STARLING (black) or IDPFold (Blue) derived end-to-end distances. **F**, Comparison of SAXS scattering profile for the sNrich ensemble derived from IDPFold ensembles (red) vs. experiment (black). Inferred radii of gyration from IDPFold, STARLING, and SAXS are provided. **G**, Comparison of SAXS scattering profile for the dCh minus ensemble derived from IDPFold ensembles (red) vs. experiment (black). Inferred radii of gyration from IDPFold, STARLING, and SAXS are provided. **H**, Comparison of SAXS scattering profile for the dTRBP ensemble derived from IDPFold ensembles (red) vs. experiment (black). Inferred radii of gyration from IDPFold, STARLING, and SAXS are provided. **I**, Comparison of SAXS scattering profile for the sNh plus ensemble derived from IDPFold ensembles (red) vs. experiment (black). Inferred radii of gyration from IDPFold, STARLING, and SAXS are provided. **J**-**Q**, Comparison of IDPFold and IDPForge vs. other approaches compared to radii of gyration derived from experimental SAXS data. Agreement between SAXS experimental data and: **J**, STARLING; **K**, IDPFold; **L**, AFRC - a null model where only sequence length contributes to chain dimensions; and **M**, CALVADOS2 for the set of sequences that were predictable using IDPFold. IDPFold performs relatively poorly at reproducing experimental SAXS data. The sequences used here are the 104 for which IDPFold ensembles could be generated. Agreement between SAXS experimental data and: **N**, STARLING; **O**, IDPForge; **P**, AFRC; and **Q**, CALVADOS2 for the set of sequences that were predictable using IDPForge. IDPForge again does a relatively poor job of reproducing experimental SAXS data. Sequences used here are the 110 sequences for which IDPForge ensembles could be generated. **R**, Comparison of time per conformation for STARLING, IDFold, and IDPForge as a function of sequence length. The variability in IDPForge times comes from a stochastic relaxation step.

扩展数据图9 | 集成引导法与 IDPFold 及 IDPForge 的对比分析。A，R_g偏置集成生成法：针对具有不同调和约束条件、且偏向于指定R_g值的 84 个残基 RNA 构建体生成集成集合。R_g值随目标值（R_g）及引导强度的变化关系。*x*轴表示调和约束值，y 轴表示目标R_g值；矩阵中的元素表示在目标值与约束条件共同作用下的集成平均R_g值。**B**，针对analysis for R_e values as a function of target (R_g）及引导强度变化关系的等效偏置分析。以端到端距离作为目标值的等效偏置方法详见补充图 5。**C**，在指定引导条件下生成的集成集合的of R_g（X 轴）与R_e（Y 轴）二维直方图；每个子图中，X 轴表示集成R_g值，Y 轴表示集成R_e值。对于子图矩阵，x轴表示所施加的谐波约束，y轴表示目标R_g值。使用端到端距离进行偏置分析的等效图矩阵见补充图 5。**D**：对比 idpSAM 捕获的集合与 Starling 捕获的集合与实验数据。我们重点关注长度小于 60 个氨基酸的序列（除下文讨论的 Ash1 外），将从 idpSAM 集合中获得的R_g值与从 Starling 集合中获得的R_g值进行比较。对于 idpSAM，采用 1000 个构象（idpSAM 默认设置）；而对于 Starling，则采用 400 个构象（Starling 默认设置）。我们注意到，解析弗洛里无规卷曲模型（AFRC）——一种零模型，在该模型中 IDR 序列化学性质不影响整体尺寸，而 IDR 尺寸仅取决于链长及（微弱地）每个氨基酸的固有尺寸——其 r^2值为 0.9，其 RMSE 值为 3.58 Å。我们纳入了 Ash1（83 个氨基酸），因为尽管其长度大于 idpSAM 训练数据所采用的约 60 个氨基酸的截断值，但先前的研究表明，基于 Campari/Absinth 模拟得到的 Ash1 SAXS 数据能够被准确复现；然而，尽管 idpSAM 是基于 Campari/Absinth 模拟进行训练的，但它仍未能复现这一现象。**E**：对比 IDPFold 生成的集合数据与 Starling 方法所得结果与 Holla 等人报道的单分子 FRET 数据[34]。实验测得的端到端距离（x 轴）与 Starling（黑色）或 IDPFold（蓝色）结果之间的相关性。**F**：基于 IDPFold 系统生成的 sNrich 系综（红色）与实验数据（黑色）所得的 SAXS 散射轮廓对比图；同时提供了通过 IDPFold、Starling 以及 SAXS 方法推导出的回转半径。**G**：基于 IDPFold 系统生成的 dCh 减集合（红色）与实验数据（黑色）的 SAXS 散射曲线对比；同时给出了通过 IDPFold、Starling 和 SAXS 方法推导出的回转半径。**H**：基于 IDPFold 系统生成的 dTRBP 集合（红色）与实验数据（黑色）的 SAXS 散射曲线对比；同时给出了通过 IDPFold、Starling 和 SAXS 方法推导出的回转半径。**I**：基于 IDPFold 系统生成的 sNh 加集合（红色）与实验数据（黑色）的 SAXS 散射曲线对比；同时给出了通过 IDPFold、Starling 和 SAXS 方法推导出的回转半径。**J-Q**：将 IDPFold 和 IDPForge 与其他方法进行比较，并与基于实验 SAXS 数据推导出的回转半径进行对比。 SAXS 实验数据与以下方法的吻合度：**J**（Starling）；**K**（IDPFold）；**L**（AFRC ——一个仅序列长度影响链尺寸的零模型）；以及 M（CALVADOS2，适用于可通过 IDPFold 预测的序列集）。IDPFold 在重现实验 SAXS 数据方面表现相对较差。本文所用序列为可生成 IDPFold 等价集合的 104 条序列。 SAXS 实验数据与以下方法的吻合度：**N**（Starling）；**O**（IDPForge）；**P**（AFRC）；以及**Q**（CALVADOS2，适用于可通过 IDPForge 预测的序列集）。IDPForge 在重现实验 SAXS 数据方面同样表现相对较差。本文所用序列为可生成 IDPForge 等价集合的 110 条序列。**R**：Starling、IDPFold 和 IDPForge 的单构象计算时间随序列长度变化的对比图。IDPForge 计算时间的变异性源于随机松弛步骤。



Extended Data Fig. 10 | 3D reconstruction from distance maps, BME reweighting, and reanalyzed SAXS data. a, Comparison of runtimes for native sklearn Multidimensional Scaling (MDS) algorithm vs. our Torch-based implementation of the Scaling by MAjorizing a COmplicated Function (SMACOF). We note that internally, the sklearn MDS algorithm uses a SMACOF implementation by default. **b**, Time per conformer for sklearn MDS vs. torch-based SMACOF. **c**, Comparison of error distributions (3D conformer vs. reconstructed conformer) for sklearn MDS vs. Torch SMACOF. Despite being substantially faster, our torch-based SMACOF implementation shows no difference in accuracy compared to the sklearn MDS implementation. **d**, Example distance maps showing deviations from reconstruction in a 384-residue sequence. **e**, Comparison of literature-reported R_g values (y-axis) obtained using Guiner analysis in the original publications, vs. our re-analysis of buffer-subtracted scattering data (x-axis) done using the Molecular Form Factor (MFF) approach of ref. 63, confirming the high quality of the data (very similar R_g values are arrived at regardless of the approach used to determine the R_g values). **f**, Theta (θ) scan for BME (Bayesian/Maximum Entropy) scan. This figure is generated automatically by STARLING's BME implementation. X-axis reports on the effective fraction of conformers used, while the Y-axis reports on the χ^2 value between experiment and ensemble (lower value means better agreement between experiment and ensemble). **g**, Visual representation of (ordered) conformer weights. This figure is also generated automatically by STARLING's BME implementation.



延伸数据图 10 |基于距离图、 BME 重加权及经分析的 SAXS 数据进行的三维重构。a，原生 sklearn 多维尺度分析（MDS）算法与我们基于 Torch 的“通过主化复杂函数进行尺度分析”（SMACOF）算法的运行时间对比。需注意，sklearn MDS 算法默认采用 aSMACOF 实现。**b**，sklearn MDS 与基于 Torch 的 SMACOF 对每个构象体的计算时间对比。**c**，sklearn MDS 与 Torch SMACOF 的误差分布（三维构象体 vs. 重构构象体）对比。尽管我们的基于 Torch 的 SMACOF 实现速度显著更快，但其精度与 sklearn MDS 实现相比并无差异。**d**，示例性距离图，展示在含 384 个残基的序列中与重构结果之间的偏差。**e**，文献报道的R_g值（y 轴）对比原始文献中采用 Guiner 分析法获得的结果，与我们利用参考文献63中的分子形状因子（MFF）方法对缓冲液扣除后的散射数据进行的重新分析结果（x轴）进行对比，从而验证了数据的高质量（无论采用何种方法确定R_g值，所得出的R_g值均极为接近）。**f**，Θ(θ)扫描（用于 BME /贝叶斯/最大熵扫描）。本图由 Starling 的 BME 实现工具自动生成。X 轴报告所用构象体的有效分数，而 Y 轴报告实验值与系综值之间的 χ^2值（数值越低，表明实验结果与系综结果越吻合）。**g**，（有序）构象体权重的可视化呈现。本图同样由 Starling 的 BME 实现工具自动生成。

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Plants

Seed stocks	N/A
Novel plant genotypes	N/A
Authentication	N/A

植物

种子库存	N/A
新型植物基因型	N/A
身份验证	N/A